INFORMATICS INSTITUTE OF TECHNOLOGY
In Collaboration with
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Defect IQ
A Real-Time Computer Vision System for Post-manufactured Plywood Defect Detection.

A Dissertation by
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ABSTRACT

The detection of defects in post-manufactured plywood sheets remains a significant challenge in the wood manufacturing industry. Manual inspection is time-consuming, inconsistent across shifts, and prone to fatigue-related errors, reducing production efficiency and weakening quality assurance. Existing automated systems predominantly target raw wood surfaces and rely on single-view imaging, limiting their ability to detect the co-occurring surface and structural defects that arise specifically from the plywood manufacturing process including bubbling, delamination, warping, imprint on surface, and missing face.

This project presents DefectIQ, a real-time dual-view plywood defect detection system combining a two-model ensemble classifier with knowledge-based reasoning. A YOLOv8-seg board detection stage isolates the plywood board from the camera background before classification. Two independently trained models EfficientNet-B2 and AlexNet, are applied to both top-view and side-view images using five-view test-time augmentation, and their outputs are combined through per-class weighted late fusion. Top-view and side-view predictions are subsequently fused using asymmetric weights that reflect the physical observability of surface defects (65% top-view) and structural defects (65% side-view). A confidence gate suppresses false positives on defect-free boards before a 0.50 classification threshold determines active defects. Each detected defect is mapped to its manufacturing cause and corrective recommendation through a deterministic knowledge base, producing a PASS or REVIEW verdict for the operator.

The system was trained on a factory-collected dataset of 619 annotated images across five post-manufacturing defect classes, published on Kaggle under CC BY 4.0. The final ensemble achieves Macro F1 = 0.625, Top-2 Image Accuracy = 96.30%, and Hamming Loss = 0.165 at threshold 0.50 outperforming individual EfficientNet-B2 (F1 = 0.613) and AlexNet (F1 = 0.582) baselines. The system is deployed as a Flask REST API with a React frontend supporting authenticated user accounts and inspection history.

Subject Descriptors: Computing methodologies → Computer vision → Object detection and recognition

Computing methodologies → Machine learning → Supervised learning → Transfer learning

Computing methodologies → Artificial intelligence → Knowledge representation and reasoning

Information systems → Data management systems → Database management system engines.

Keywords: Plywood defect detection; Multi-label ensemble classification; Dual-view fusion; EfficientNet-B2; AlexNet; YOLOv8-seg; Test-time augmentation; Knowledge-based reasoning; Post-manufactured defect detection; Industrial quality inspection

DECLARATION

I confirm that the research presented in this project is entirely my own work and has been carried out in accordance with the academic and ethical standards of the university. All sources of information, references, and any external contributions have been properly acknowledged and cited.

I further declare that this work has not been submitted, either wholly or in part, for any other academic qualification at this or any other institution.

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List of Abbreviations

B64: Base64 Encoding (used for image transmission)

CNN: Convolutional Neural Network

CORS: Cross-Origin Resource Sharing

CUDA: Compute Unified Device Architecture (NVIDIA GPU acceleration)

CV: Computer Vision

DWR: Dilation-Wise Residual

GT: Ground Truth

IDE: Integrated Development Environment

IoU: Intersection over Union

IPD: Integrated Product Development

JSON: JavaScript Object Notation

LR: Literature Review

ML: Machine Learning

NMS: Non-Maximum Suppression

PIL: Python Imaging Library

PPRS: Project Proposal and Requirements Specification

REST: Representational State Transfer

UI: User Interface

XAI: Explainable Artificial Intelligence

YOLO: You Only Look Once

Performance Metric Abbreviations

mAP: mean Average Precision

Prec: Precision

Rec: Recall

F1: F1-Score

TTA: Test Time Augmentation

SSADM: Structured Systems Analysis and Design Method

VGG: Visual Geometry Group

GAN: Generative Adversarial Network

VRAM: Video Random Access Memory

CHAPTER 01 - INTRODUCTION

1.1 Chapter overview

This chapter presents the DefectIQ project with background information of plywood defect detection, the problem being dealt with, and the motivation for this research. It includes highlighting relevant existing work, identifying the gap in research, and defining the aim, objectives and research questions. Finally, it presents the challenges involved and intended contributions to the project.

1.2 Problem background

Plywood manufacturing is a multibillion-dollar global industry with applications in construction, furniture, and packaging and it is an industry that is spread worldwide.
(Rustiadini and Novianti, 2025). For many developing countries, plywood exports represent a vital source of foreign exchange earnings. Ensuring consistent quality is critical, especially for export markets that enforce stringent standards. Defective plywood sheets due to bubbling, delamination, warping, or missing edges are often rejected, leading to significant financial losses and missed opportunities for generating foreign exchange. In countries striving to build robust export sectors, improving quality control means not only reducing waste but also gaining greater access to premium international markets.

The effect of defective products extends beyond immediate economic losses but also can damage the reputation of manufacturers and exporting countries, weakening the competitive position internationally. Therefore, when looking at strengthening the export sectors as well as the local economy sectors, investing in improved quality control measures becomes critical not only to reduce waste but also to enhance economic growth by securing higher value contracts. Currently, Quality inspection in plywood manufacturing heavily depends on manual visual assessment. This method introduces inconsistency and is prone to human error, particularly in high-speed, high volume production settings. Subtle defects are often missed due to inspector fatigue. This will lead to negative impact on the manufacturing industries and also customer satisfaction will be affected the export potential will be affected with the drop in quality.

1.3 Problem definition

The plywood industry requires effective quality control to detect structural and surface defects in finished plywood sheets. Existing automated inspection systems primarily focus on early-

stage defects such as veneer flaws (Urbonas *et al.*, 2019) and there are some focusing on the post manufactured glue surface errors (Chen et al., 2022; Baad, 2023). Subtle defects such as bubbling and warping caused by steam issues, or multi-scale problems like delamination can easily go unnoticed.

The specific problem this project addresses is the absence of an integrated, automated computer vision-based system capable of detecting and localizing multiple types of post manufactured plywood sheet defects, while also providing knowledge-based reason for the occurrence of the defect. Without such a system, manufacturers continue to experience inefficiencies in quality control, reduced competitiveness in export markets, and missed opportunities to strengthen their economic contribution through higher-value plywood products (Baad, 2023).

1.3.1 Problem statement

The lack of an automated system for detecting and reasoning about multiple co-occurring post-manufacturing plywood defects leads to inconsistent quality control, recurring production issues, and missed opportunities for export competitiveness in the Sri Lankan plywood manufacturing industry.

1.4 Research motivation

The motivation for this research arises from a strong interest in applying computer vision to real-world manufacturing challenges. Observations of quality issues in plywood production have highlighted how inconsistent quality control not only increases material waste but also restricts opportunities to export high-standard products. Since plywood sheet manufacturing is a large-scale industry and plywood boards are essential in almost every construction project worldwide, improving quality standards has a direct impact on competitiveness in international markets. By introducing automated defect detection and reasoning, manufacturers can enhance the quality of their products, unlock greater export potential, and contribute to strengthening the national economy. This alignment of advanced technology with significant industrial and economic benefits forms the core motivation for undertaking this research.

1.5 Existing work

Citation	Summary	Limitation	Contribution
(Baad, 2023)	This research tests CNN and transformer-based networks (Swin-T, ViT) for classifying glue-layer defects on plywood (including Low glue, Run out, Scratch detach, Rip, etc.) to establish a reliable automated screening system for quality control. The study highlights the importance of identifying glueless areas to determine the defect reason and aid in optimizing production efficiency and costs.	The models were trained and tested on a dataset that suffered from a small size and a huge discrepancy in the number of samples per class.	Demonstrated that transformer-based neural networks are effective for classifying specific glue layer defects on plywood surfaces, with the Swin Transformer achieving the highest accuracy of 88.70%.
(Chen *et al.*, 2022)	This paper presents a multi-sensor, two-stage system (WDD-DL) utilizing raw images and laser-aligned images with InceptionResnetV2 to detect edge-glued defects	Experiments were performed only on 21 different wood types, and fewer defect training images were available for overlong and short edge banding	Developed a hybrid two stage approach combining machine vision filtering (Gabor, Harris corner) with deep learning classification to achieve high accuracy for manufacturing-related defects in a real-time

Citation	Summary	Limitation	Contribution
	(High, Low, Bluntness, Residue [glue marks], Short, Overlong) on a conveyor for real-time process automation		system
(Ramos-Maldonado *et al.*, 2025)	This research utilizes supervised Machine Learning (ML) algorithms (Random Forest, XGBoost) on large datasets of industrial process variables (maceration, peeling, drying) to predict final veneer quality. This approach provides AI driven insights for forecasting key performance indicators (KPIs), which helps determine the manufacturing process reason for quality outcomes.	This project is wood based and not plywood based (focused on predicting veneer quality based on process parameters, not classifying final sheet defects). The main limitation was the availability of a larger dataset (13,690 records collected at 10 min intervals representing approximately 90 days of operation)	Successfully leveraged large-scale online industrial data with ML to predict veneer quality outcomes, providing the mechanism needed to determine the reason for potential final product quality issues upstream in the manufacturing process
(Syla, Saleh and Hardt, 2023)	Evaluated modified YOLOv5s models (adding GCNet/SENet) for high-accuracy wood surface defect detection on high-resolution	This project is wood based and not plywood based (focusing on general wood plates/surface décor defects). Achieving	Demonstrated that Active Learning can significantly increase performance (mAP 0.5 up

Citation	Summary	Limitation	Contribution
	images captured by a line scanning camera on a conveyor. Successfully implemented and validated an Active Learning approach to continually improve detection performance	higher accuracy requires a more detailed definition of defects and the involvement of a wood industry expert for labelling.	to 98.8%) by continually training the model on new detections, which solves the continual learning for new defects problem (relevant to your knowledge base creation).
(Zhang *et al.*, 2024)	Proposed WLSD-YOLO model for detecting surface defects on wood lumber using enhanced YOLOv8 architecture. Uses SE attention in backbone, GVC-neck for improved feature extraction and reduced parameters, and WIOU-NWD loss function to detect small targets. Achieved 76.5% mAP and real-time detection speed.	Detection of some defect types (knotwithcrack and quartzity) remains inefficient. Model size and parameters could be further reduced for deployment cost reduction. Small dataset limitation typical for defect detection tasks.	Introduced SE attention mechanism in backbone, new GVC-neck structure reducing parameters and improving speed, and combining WIOU with NWD loss for better small target detection. Balanced accuracy and speed are suitable for industrial applications. Improved mAP by 2.9% over baseline YOLOv8 with 19% fewer parameters and faster FPS.

Table 1- Existing work table(Self-composed)

1.6 Research gap

Existing Research in the automated detection of wood flows has successfully utilized computer vision (CV) and deep learning algorithms to achieve high accuracy and real-time inference

speeds for surface inspection (Wang et al., 2024; Li et al, 2025); However detecting the defects in the post manufactured phase is under explored.

- Glue Defects and Delamination: Few works address glue failures or delamination in plywood, especially using dual-view imaging. Most focus on the surface or simple edge defects (Chen *et al.*, 2022). Warping (Crookedness) is a defect that is rarely tackled with computer vision, particularly from side views or using geometric analysis in real-time.
- No system combines the fusion of top and side views from two camera angles with weighted consideration of the physical visibility of different defect types from different perspectives.
- No academic system performs multi-label, image-based post-manufactured plywood defect classification. Existing systems all perform single-label classification of defects and are therefore unable to detect multiple defects on a single board.
- No system links defect classification results to a knowledge-based reasoning component that associates detected defects with their root causes and recommended solutions - leaving factory workers without guidance.
- Data Scarcity and Annotation: Many studies highlight the challenges of acquiring sufficient labeled data for rare or subtle defects(Baad, 2023), especially for structural issues.

1.7 Contribution to body of knowledge

1.7.1 Problem domain contribution

In the plywood industry, quality control after manufacturing is still heavily dependent on manual inspection, where workers visually examine each sheet for surface or structural defects. This process is time-consuming, inconsistent, and prone to human error, especially when dealing with large volumes of production (Rustiadini and Novianti, 2025).

The project advances the field of plywood defect detection by introducing a dual-view, knowledge-based system that not only defects both structural and surface defects in post manufactured process but also provide the likely causes for the defects, using a knowledge-based dataset that is built with the collaboration of the domain experts.

The defects that will be planned to capture using this project include Delamination, Bubbling, Warping, Missing edges and front face, imprint on surface, Overlapping (common post manufactured defects).

1.7.2 Research domain contribution

This research makes four contributions to the computer vision and industrial inspection domain. First, it demonstrates that dual-view inspection can be improved through an asymmetric defect-specific fusion strategy, where view importance is assigned according to the physical observability of each defect type. Surface defects are weighted more heavily from the top view, while structural defects are weighted more heavily from the side view, showing that geometry-informed view fusion can improve defect discrimination in multi-view inspection tasks. Second, the study shows that a lightweight weighted late-fusion ensemble, combining EfficientNet-B2 and AlexNet with per-class fusion weights and test-time augmentation, can improve multi-label plywood defect classification under small-dataset conditions, outperforming the individual models and baseline CNNs. Third, the research demonstrates that deterministic knowledge-based reasoning can be integrated with vision-based defect classification to provide consistent and auditable cause-oriented explanations, extending industrial inspection beyond prediction alone toward explainable decision support. Fourth, the study introduces Top-2 Image Accuracy as an evaluation metric for assisted inspection settings, arguing that conventional classification metrics alone do not fully reflect practical usefulness when a system is intended to support, rather than replace, a human inspector.

1.8 Research challenges

- Balancing Real-time Detection with accuracy: One of the main challenges is ensuring that the system achieves high accuracy in every defect for defect detection while still managing real time performances.
- Detecting Subtle and multi-scale defects: Certain defects, such Imprint on surface and Warping are subtle and multi-scale in nature, making them difficult to detect reliably with standard deep learning models
- Side-view and Multi-view Integration: Capturing defects on edges and warped plywood boards requires processing data from multiple camera angles, which increases system complexity and synchronization requirements.

- Dataset creation and annotation: Building a high-quality dataset for post manufactured defects is challenging since the existing public datasets focus mostly on veneer defects. Accurate annotation of defects like overlapping or delamination is time intensive.
- Explaining Defects and linking them to root causes: Beyond detection, the project aims to provide reasoning for each defect. Developing a reliable knowledge-based system that maps visual features to production causes will involves handling complex situations.
- Hardware and resource limitations: when training the computer vision model for the project, challenges may arise due to hardware limitations, must rely on cloud-based model training services. This dependency on cloud training introduces additional considerations such as data transfer, cost, and access constraints.

1.9 Research questions

RQ1: How can advances in deep learning architecture, such as multi-scale dilation attention and dual-view fusion, be used to improve the accuracy and robustness of plywood defect detection?

RQ2 What are the key challenges in detecting both surface defects (e.g. imprints) and structural defects (e.g. missing edges, warping) in plywood using image-based methods, and what techniques can be applied to overcome these challenges (e.g. dual-view imaging, data augmentation)?

RQ3: How can a multi-view (two-side) image setup be designed and synchronized to enhance robustness and reliability in detecting both surface-level and side-view defects?

RQ4: How can a knowledge-based dataset be effectively integrated with computer vision system to provide reasoning and preventive suggestions for each identified defect?

1.10 Research aim

The aim of this research is to design, develop, and evaluate a real-time computer vision based system for detecting post-manufacturing defects in plywood sheets, integrating a knowledge-based dataset to provide reasoning and preventive measures for each identified defect

1.11 Research objectives

Objectives	Research Objective	Description	LOs Mapped	RQ Mapped
Problem Identification	**R01**: To identify and define challenges and in automated plywood defect detection and reasoning	Survey industry/manual practices, recognize unaddressed issues and gaps in current solutions	LO4, LO6	RQ2, RQ3
Literature Review	**R02:** To carryout a in-depth literature review for post manufactured Defect detection in plywood surfaces	Critically review and justify best approaches from academic/industrial literature	L01, L04, L05	RQ1, RQ2, RQ3
	R03: To conduct a literature review on modality for defect detection in plywood surfaces			
	R04: To research and evaluate state-of-the-art computer vision/ML methods for post manufactured plywood defect detection			
Requirement Elicitation	**R05:** To collect and analyse system/user requirements for a robust defect detection.	Use Interviews with domain/research experts, brainstorming and conduct a complete Literature review to build a detailed requirement specification	L03, L06	RQ2, RQ3
System Design	**R06:** To design an end-to-end automated defect detection and reasoning system	Propose system architecture, select tools, justify algorithm choices, address the legal, ethical, and professional issues.	LO1, LO6, LO7	RQ1, RQ3, RQ4

Objectives	Research Objective	Description	LOs Mapped	RQ Mapped
Implementation	**R07:** To collect and annotate a robust plywood defect dataset using stable video/image techniques	Use HD quality images for stable data, remove noise, carefully annotate defects, document process.	LO1, L05, LO6,	RQ1, RQ3, RQ4
	R08: To build, test and optimise deep learning models for plywood defect detection and root cause reasoning	To develop iterative prototypes of the defect detection model, starting from baseline approaches (e.g. YOLO object detection) and evolving to the proposed EfficientNet-B2 sliding-window classifier.	L01, LO6, L07	RQ1, RQ3, RQ4
Testing	**R09:** To evaluate the accuracy and efficiency of the model	Perform thorough testing and benchmarking of the developed defect detection model using test datasets to measure performance metrics such as accuracy, precision, recall, speed, and robustness.	L01, L07, L08	RQ1, RQ3
Documentation	**R10:** To document all project work clearly and professionally	Provide all comprehensive documentation that is needed, critically assess results, reflect on learning/new skills, prepare and rehearse viva defence	LO8, LO9	RQ1, RQ2, RQ3, RQ4

Table 2- Research Objectives (Self composed)

1.12 Project scope

1.12.1 In scope

The system will focus on the post-manufactured defects of plywood sheets, including

i) Bubbling
ii) Delamination
iii) Warping
iv) Missing faces
v) Imprints on surfaces

The input will be sent through a react frontend and the images will be taken from two fixed positions,

- One positioned above at an incline to the sheet to capture the top surface of the sheet
- One position at an incline to the side of the sheet to capture the side surface of the sheet

The system will use computer vision and deep learning models to detect the defects. A knowledge-based dataset will be used to give the explanation for each defect that will be detected by the sheet. A defect report will be generated for each plywood sheet with the defects. The system prototype will be demonstrated through a web-based interface (React frontend + Flask backend) to visualize detection results and reasoning.

1.12.2 Out of scope

The following aspects are beyond the current project's scope due to time, budget, and resource limitations. The system will not include four-camera or 360° view setups due to resource constraints; only two camera angles (top and side) will be used. Real-time industrial deployment with a fully automated conveyor system will not be implemented during this research phase. Defects beyond the five main categories (bubbling, delamination, warping, missing faces, imprint) will not be addressed. The reasoning mechanism will be limited to a pre-defined knowledge base, not machine learning or causal inference models. The dataset will consist of locally captured plywood images, and results will not be generalized to all plywood types or manufacturing setups. Integration with factory ERP systems or production line controllers will be outside the project's current scope.

1.13 Hardware/Software requirements

Component	Specification/Description	Notes
CPU	Intel Core i5 / AMD Ryzen 5 or above	
GPU	NVIDIA RTX 3060 12–24 GB VRAM or equivalent for model training	For model training
RAM	Minimum 16 GB (Recommended 32 GB)	
Storage	At least 600GB SSD (for dataset and model storage)	For dataset and model storage
Power Supply	150W or above (sufficient for GPU and camera setup)	Sufficient for GPU

Table 3- Hardware Requirements (self composed)

Software / Tool	Purpose / Description
Python 3.11	Primary programming language for development
PyTorch 2.0 / TensorFlow 2.x/ PyCharm	Deep learning frameworks for model training and inference
OpenCV	Image preprocessing and computer vision operations
YOLOv8 Framework	Object detection and localization
Roboflow	Image annotation and dataset management
Kaggle notebooks	Experimentation and model training environment, Cloud-based options for scalable training
PostgreSQL / Supabase	Database to store metadata and defect reasoning data
Flask	Backend for system integration and REST API
React.js	Frontend or visualization interface for displaying results
MS Word	Documentation and report writing
GitHub / Git	Version control and collaboration

Table 4- Software Requirements (self composed)

1.14 Chapter summary

This chapter introduced the core problem, outlined its context and importance, surveyed existing solutions and their gaps, and motivated an AI-driven approach for real-time plywood defect detection and reasoning. It clarified the research aim and objectives, scoped what the project will and will not cover, and listed the key hardware and software needed. Together, these elements set up the next chapters on system design, methodology, dataset creation, model development, and evaluation.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

This chapter offers an overview of the state of the art in research and technologies related to plywood defect detection. It first defines the scope - the inadequacy of manual quality control in plywood production and the particularities of post-manufactured plywood defects. It then considers prior work in four categories: conventional computer vision, CNN-based detection, transformer-based structures and object detection models. A review of multi-view inspection systems highlights a key research gap addressed by DefectIQ. The technological review addresses the architecture, ensembles, multi-label formulations and knowledge-based reasoning techniques in the system. The chapter wraps up with a summary of the gaps identified and how they are addressed by the DefectIQ architecture.

2.2 Concept map

A concept map depicting the main concepts and relationships explored in this research is presented in Appendix A.2

2.3 Problem domain

2.3.1 Automated quality assurance in plywood manufacturing

Quality control in plywood manufacturing is vital for brand quality, export and international certification. Plywood inspection has historically been done visually by humans, which is subjective and impractical. Visual assessments of human inspectors differ across lighting, fatigue levels, and time of day, leading to missed small cracks, bubbles, or edge imperfections, while visual variations in wood grain or plywood gloss are occasionally misidentified as defects (Chen et al., 2024). On high-speed production lines, it's impossible for inspectors to sustain visual attention at the speed of the conveyor belt, causing bottlenecks and variable decision-making criteria (Kodyte.et al, 2021). Furthermore, manual inspection lacks the structured data and queryability, making trend analysis and compliance verification difficult in the long term across multiple shifts and product categories. Vision-based inspections overcome these challenges by replacing human judgment with computer vision AI systems that provide uniform and fast detection. A recent review of CNN-based industrial defect detection by (Jha.et

al,2023) verified that deep learning systems now surpass human performance in benchmarked tests across various manufacturing settings. In the case of plywood, the shift from manual to automatic inspection is still in its infancy, with most academic literature focused on inspecting raw wood or lumber surfaces rather than post-manufacturing multilayer plywood sheets, which is the focus of this study.

2.3.2 post-manufactured plywood defects

Defects in post-manufactured plywood have a very different origin and appearance from surface defects in raw wood. Bubbling, delamination, and warping are the result of hot pressing, adhesive and conditioning applied during the plywood manufacturing process, rather than anomalies in the natural wood grain (Yalong Wood, 2020). This is a key point: computer vision research on wood defect detection is predominantly focused on the detection of knots, cracks and resin pockets in raw wood (An et al., 2024; Dou et al, 2025), and these models are not directly transferable to post-manufacturing defects, which have a different taxonomy and appearance. Another challenge is that post-manufacturing defects often occur together. Delamination and warping usually occur simultaneously as a result of failed adhesive bonding (Yalong Wood, 2020), and bubbling and missing face material occur simultaneously when the pressing process is insufficient. This co-occurrence renders single-label classification prevalent in the existing wood defect literature, unsuitable for this problem. A table in the appendix section A.3 is provided to get a broader idea on the defects that are targeted in the defectIQ project.

2.3.3 Defect reasoning and knowledge-based systems

While most prior studies focus on detection and classification, few address defect reasoning, the process of explaining why the defect was caused in the manufacturing process. Manufacturers need insights that connect visual detections to underlying causes, such as improper glue ratios, temperature fluctuations, manual labor errors (errors when using small veneer strips to increase the strength between the sheets, errors like overlapping and imprints on surface) or pressing time issues. However, most existing systems stop at detection, offering no process-level feedback or preventive measures. (Ramos-Maldonado *et al.*, 2025).

Integrating a knowledge-based reasoning model allows manufacturers to trace causal factors and implement targeted adjustments in production, thereby reducing the reoccurrence of

defects and increasing the quality control. This approach bridges the gap between detection and decision-making by transforming raw defect data into actionable insights

2.4 Existing work

2.4.1 Traditional computer vision approaches

The first attempts at automated wood surface inspection involved manual feature extraction. Techniques like Gabor filters, grey-level co-occurrence matrices (GLCM), and color histograms were applied to distinguish texture defects from wood grain patterns (Kamal et al., 2017). summarized traditional wood defect detection methods, stating that while GLCM-based systems performed well in laboratory settings, they were sensitive to changes in illumination, gloss, and veneer grain, which are all highly variable in industrial environments. The requirement for environmental control and lack of generalization across different wood species restricted their industrial adoption. Traditional image segmentation techniques, such as edge detection and region-growing algorithms, were also used for defect localization but were found to be unstable on a dynamic production line due to their sensitivity to lighting and coating (Chen et al., 2022). These challenges led to the adoption of deep learning models, which automatically extract intricate visual patterns from images.

2.4.2 CNN-based defect detection

The advent of deep convolutional neural networks (CNNs) revolutionized wood defect classification by allowing direct learning from images. Wood surface defect classification tasks have also benefited from the application of CNN models like ResNet and EfficientNet, which achieve 85-92% accuracy on standard datasets (Chen et al., 2022; Almeida et al., 2023). Rasheed et al. (2024) reported that EfficientNet-B4 with transfer learning from ImageNet weights and a Squeeze-and-Excitation block obtained around 98% accuracy for defect classification in semiconductor manufacturing, corroborating the effectiveness of EfficientNet's compound scaling method in industrial defect classification problems with small training sets. While reporting high accuracy, these studies have a common shortcoming: they assume a single-label classification problem, where each image is assigned to a single defect class. This is not the case for post-manufactured plywood as different types of defects physically coexist in the same plywood board. Moreover, the majority of CNN-based wood defect research uses data from raw wood boards or wood panels, not from post-manufactured plywood, and their reported accuracy scores are not relevant to DefectIQ. This led us to adopt

a multi-label sigmoid approach in DefectIQ, where EfficientNet-B2 and AlexNet are trained with BCEWithLogitsLoss to predict, for each of the five defect classes, whether it is present or not.

2.4.3 Transformer-based approaches

Vision transformers (ViT) and hybrid models like Swin Transformer are emerging as viable alternatives to CNNs for fine-grained texture classification. Baad (2023) compared the performance of Swin Transformer and ViT for glue-layer defect classification in plywood, where Swin-T yielded the best accuracy of 88.70%. The self-attention mechanism allows these models to capture long-range spatial relationships across the entire board image, and they are well-suited to detect subtle texture relationships that CNN-based models with smaller receptive fields might overlook. But transformers are considerably more data-intensive than CNNs. (Baad ,2023) observed that the dataset was small and highly unbalanced, and that the models learned had poor performance on unseen defect samples. This observation is pertinent to DefectIQ, the data-hungry nature of transformers meant CNN-based transfer learning (EfficientNet-B2 and AlexNet) was more suitable for the prototype system, with transformers as a potential area of future system improvement as the dataset grows.

2.4.4 Object detection approaches - YOLO family

Real-time object detection on high-resolution wood and industrial images has been successful using one-stage object detectors, especially the YOLO family. Li et al. (2025) enhanced a YOLOv8n model with Multi-Scale Dilation Attention (MSDA) and Dynamic Weight Residual (DWR) modules to achieve mean average precision (mAP) greater than 95% on a dataset of wood panel defects. An et al. (2024) also improved YOLOv8 with conditional parametric convolution (CondConv) and BiFormer attention, which resulted in a 3.5% gain in mAP@0.5 compared to the baseline YOLOv8 model on a multi-species wood dataset. These detection-based methods are highly effective at identifying defects in images, but are built to detect defects in raw wood, not classify post-manufactured defects across the entire surface of a plywood board. Moreover, YOLO-based models generate bounding box localisations, while DefectIQ requires image-level multi-label classification, which is a determination of the presence of different types of defects across the entire board surface (not individual localisations). This is why DefectIQ uses a two-stage approach: YOLOv8-seg is only used to isolate the board (Stage 1) and then multi-label ensemble classification is applied to the cropped board (Stage 2).

2.4.5 Multi-view and dual-view inspection systems

Most research on detecting defects in wood and plywood products has focused on using single-camera, top-surface images. But some industrial systems have reported a marked improvement in defect detection with multiple cameras. The industrial systems ELVAC Vision and GreCon EDGEINSPECT have adopted dual-view and multi-view inspection models by incorporating top-surface and side-edge cameras to detect surface and edge defects. These systems claim significant improvements in edge coverage and localisation accuracy compared to single-view systems. Although multi-view inspection has been shown to be advantageous, there is almost no academic attention given to dual-view or fused-view classification for plywood post-manufacturing defects (Kodytek et al., 2021). The few existing academic contributions to view fusion in industrial automation defect detection are limited to multimodal sensor fusion that combines visual, acoustic and vibration data (Liu et al., 2024), rather than dual-view image fusion for classification. This is a different gap. DefectIQ fills this gap with a weighted fusion of top-view and side-view classification predictions that uses an unequal weighting of each defect type (65% surface defects top-view, 65% structural defects side-view) based on how each defect type appears physically from either view.

2.4.6 Critical gaps.

The following table synthesizes the five critical gaps identified from the literature review and maps them to each to the corresponding design decision in DefectIQ.

Gap in Existing Literature	**Existing Work**	**DefectIQ Approach**
Single-label classification - one defect per image assumed	Chen et al. (2022), Baad (2023), Li et al. (2025)	Multi-label sigmoid classification - five classes predicted independently per image
Single-surface inspection - top view only	Most YOLO and CNN studies (Li et al., 2025; An et al., 2024)	Dual-view (top + side) weighted fusion with asymmetric per-defect weights

No defect reasoning - classification only, no cause or recommendation	Ramos-Maldonado et al. (2025), Kodytek et al. (2021)	Knowledge-based reasoning module maps each defect to manufacturing cause and corrective action
Focus on raw wood or lumber -not post-manufactured plywood	Dou and You (2025), An et al. (2024), Kilics et al. (2025)	Dataset collected from post-manufacturing plywood production line- five post-process defect classes
No publicly available post-manufactured plywood defect dataset	All reviewed plywood studies use private or inaccessible datasets	Create a new dataset related to post manufactured sheets.

Table 5 gap in existing literature

A full comparison of existing works including individual summaries, limitations, and contributions to the DefectIQ design is provided in Appendix A.1,

2.5 Technological review

2.5.1 Dataset and preprocessing strategies

A lack of publicly available datasets of plywood defects is a recurring issue in this field. (Kodytek et al. 2021) reviewed wood and plywood defect datasets and concluded that there is no publicly available dataset for post-manufactured plywood defects. Researchers typically rely on datasets from factories or datasets of raw wood surfaces. This lack of benchmark data makes it hard to compare research and to reproduce results. In terms of preprocessing, board isolation before defect classification has been shown to be an essential step for enhancing classification performance on full images from the camera that also contain background (factory floor, lights, workers). Recent work shows that board isolation before classification drastically reduces false positives from background clutter (Li et al., 2025). In DefectIQ, board isolation is achieved with a specialised YOLOv8-seg model that crops a tight 640×640 board image before feeding the image to the ensemble classifier - a design choice supported by the literature.

2.5.2 EfficientNet and transfer learning

EfficientNet (Tan and Le, 2019) takes a new approach to balancing model accuracy and efficiency by compound scaling the network depth, width and image resolution by a constant coefficient. This allows EfficientNet models to deliver comparable accuracy with far fewer parameters than their same-generation counterparts, such as ResNet-50 and VGG-16. EfficientNet-B2, as employed in DefectIQ, strikes a balance between accuracy and inference time for a prototype implementation on CPU-based hardware. Pre-training on the ImageNet dataset has been demonstrated to reduce the training time and improve accuracy on small industrial defect datasets with expensive ground truth labels (Rasheed et al., 2024). The combination of EfficientNet-B7 and YOLO spatial features also demonstrated robust multi-class defect classification on the MVTec-FS dataset (Nature Scientific Reports, 2025), further justifying the application of EfficientNet as a classification backbone. The only modification in DefectIQ is the change in the output layer from softmax to sigmoid and BCEWithLogitsLoss to enable separate multi-label prediction for each of the five defect classes, which is absent in any of the plywood defect detection systems reviewed.

2.5.3 AlexNet and ensemble diversity

AlexNet (Krizhevsky et al., 2012), though older than EfficientNet, provides a different set of features that make it a good choice for an ensemble. Its shorter depth of convolutional layers (five layers versus EfficientNet's MBConv blocks) induces a different set of biases: AlexNet is more likely to learn global features (spatial information) while EfficientNet is more likely to learn local features (texture information) through its Squeeze-and-Excitation blocks. This is the basis of the DefectIQ ensemble, we expect the two models to disagree on some samples and weighting their predictions together leads to improved Macro F1 performance. This is in line with the ensemble learning literature, which shows that diversity is an indicator of improved ensemble performance (Comprehensive Survey on ML for Material Defect Detection, 2024).

2.5.4 Ensemble methods in computer vision

Combining the predictions of several independently trained models, known as ensemble learning, has been demonstrated to be more accurate and more robust than single-model methods for industrial defect classification problems (Hou et al., 2024). Late fusion (also known as decision-level fusion) is the most common ensemble approach for CNN-based classifiers, where the probability distributions of each model are fused after inference. Liu et

al. (2024) compared early, intermediate and late fusion for multimodal defect inspection and observed that, while late fusion may miss some multimodal interactions, it is the most efficient for deployment and is the most common approach used in industrial inspection. DefectIQ uses late fusion with class-wise weighted averaging, with fusion weights determined by grid search on the validation set, based on domain knowledge of each model's capabilities.

2.5.5 YOLOv8-seg for object isolation

YOLOv8, launched by Ultralytics in 2023, is the newest version of the YOLO family of models and includes native segmentation capabilities (YOLOv8-seg) built on the detection network. It has achieved the highest accuracy on several public datasets and has also been applied to wood panel defect detection (Li et al., 2025; An et al., 2024). The segmentation version generates polygonal masks instead of bounding boxes, allowing for accurate board isolation that can compensate for boards being at an angle with the camera view. In DefectIQ, YOLOv8-seg is used solely for board isolation - not for defect detection. This pipeline design (Stage 1: board isolation, Stage 2: defect classification) exemplifies a principle of independently trainable and replaceable pipeline stages.

2.5.6 Multi-label classification formulation

Post-manufactured plywood defect detection is a multi-label classification problem, where instances can be assigned to more than one class, because defects can co-exist on a board. Single-label classification methods use a softmax activation, which distributes 100% of the model's probability to one class, thus preventing the detection of multiple co-occurring defects. In contrast, multi-label classification uses independent sigmoid activations for each class, which output a probability in [0,1], and a threshold is used to decide which classes are present (Saleh, 2025). Selecting a loss function is important for multi-label classification. BCEWithLogitsLoss (Binary Cross-Entropy with logits) assumes that each class is an independent binary classification task, calculating the loss for each class and then summing over all classes. This approach is particularly suited to plywood defects where the amount of co-occurring classes varies (delamination and warping co-occur often, but bubbling and missing_face do not). The threshold for classification can be adjusted to control the operating point of the classifier.

2.5.7 Other approaches that have used defect detection

In addition to the models directly used in DefectIQ, there have been many other deep learning methods attempted for visual defect detection in manufacturing. ResNet-based models added residual blocks to allow very deep networks to train without loss of information, and continue to serve as a baseline for defect classification in manufacturing (He *et al*., 2016) . MobileNetV2 and MobileNetV3 are lightweight architectures that can be used for edge inference, but trade off per-class accuracy on textural defects when trained on small-scale datasets like DefectIQ. Vision transformers (ViT, DeiT) have achieved high accuracy on large-scale classification datasets but generate high accuracy models only with an order of magnitude more annotated training data than CNN-based transfer learning approaches Generative adversarial network (GAN) based data augmentation has been used to overcome dataset scarcity in defect detection to generate synthetic defects for training (Baad, 2023). But this approach was not tested for DefectIQ due to insufficient real samples in the factory training set for training a GAN without mode collapse.

The following table contains some of the other technological approaches and their respective advantages along with their limitations,

Technology/Approach	**Description**	**Advantages**	**Limitations**
Multi-Camera Data Fusion	Use multiple synchronized cameras to increase Spatial coverage and reduce occlusions, combining multiple views.	Eliminates occlusions, improves spatial coverage, reliable in industry setups (ELVAC Vision, 2025).	Complex calibration; potential alignment and lighting issues
3D and height-Aware Sensing	Use structured-light and laser profilometry to detect geometric anomalies via surface height mapping.	Detects geometric anomalies. Robust under varying color and gloss	Costs and complexity limit applicability to cost sensitive industries.
Traditional and non-neural techniques	Classical image processing approaches using filters, texture statistics, and segmentation.	Simple computationally efficient, explainable models.	Low adaptability to changing conditions; performs best in controlled environments.
Process-Aware analytics and reason mapping	Linking defect detection to manufacturing parameters for	Supports preventive maintenance, transforms inspection	Needs plant specific calibration and detailed defect ontologies.

	actional root cause analysis and process control. (Sidikiyah, 2023).	data into process insights.	
End-to-End AI-Integrated Conveyor Systems	Fully integrated industrial inspection lines including cameras, inference, and rejection systems for continuous production. (Intelgic, 2025).	Proven scalability and operational reliability.	Proprietary systems limit research customization and transparency.

Table 6- Technological Review of other existing approaches (self-composed)

2.6 Benchmarking and evaluation

In the literature on plywood and wood defect detection, accuracy and mean average precision (mAP) are the most commonly used evaluation metrics (Li et al., 2025; An et al., 2024; Chen et al., 2022). Although accuracy is straightforward, it is not suitable for class-imbalanced multi-label data because it does not provide class-wise performance and can be misleading when the dataset is heavily biased toward a majority class. mAP is commonly used to evaluate object detection with bounding box localisation, but is not directly applicable to image-level classification. DefectIQ uses the Macro F1 as its main metric, which is the unweighted average F1 across the five classes, giving equal importance to the minority classes (such as delamination) regardless of their representation in the dataset. Hamming Loss and Subset Accuracy are also reported to describe the multi-label error. This work also proposes a new metric called Top-2 Image Accuracy: an image is classified as correctly processed if the ground-truth defect is among the top-2 defects with highest confidence. This metric mirrors the typical workflow in industrial inspection where the top-2 predictions are reviewed by inspectors, as is common in assisted inspection in manufacturing quality control (Jha and Babiceanu, 2023). This combination of metrics is not reported in any of the reviewed plywood defect detection papers, and is a contribution of this work.

2.7 Chapter summary

This chapter reviewed existing literature on plywood defect detection, critically analyzing domain challenges and state-of-the-art technologies. Current solutions were evaluated, their limitations identified, and research gaps highlighted. The subsequent chapter presents the methodologies implemented in this research project

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

This Chapter explores the Methodology approaches that is used for the development of this project. It starts with the research philosophy and methodology, based on Saunders' Research Onion model, then the development methodology, programming paradigm, testing strategy and project management methodology, including the resource plan, risks and iterative sprints.

3.2 Research methodology (Saunders' research onion in table form)

Layer	Choice	Justification
Philosophy	Pragmatism	This research aims to provide a practical, data-driven detection and reasoning system that can be deployed in real-world plywood manufacturing. It combines quantitative evidence (model accuracy, precision, recall) with qualitative expert validation of reasoning outputs. Pragmatism supports this mixed, problem-solving orientation.
Approach	Abductive with deductive testing	The study begins with abductive reasoning to generate hypotheses linking plywood defects to visual cues and manufacturing causes, drawing from expert insights and literature. It then employs deductive reasoning to test these hypotheses via quantitative experiments using the EfficientNet-B2 and alexnet classifiers on YOLO-cropped patches, validated through patch-level F1-scores,

Layer	Choice	Justification
		confusion matrices, and image-level Top-2 accuracy metrics.
Methodological Choice	Mixed methods (Quantitative and qualitative)	A quantitative approach assesses model performance using Macro F1, F1 per class, Hamming Loss, Subset Accuracy, and Top-2 Image Accuracy on the 619-image test set. A qualitative approach assesses the usefulness of the system with interviews with five domain experts and five technology experts, which informed the self-evaluation and the augmentation experiment described in Chapter 9.
Strategy	Experimental, Case Study, and Semi-Structured Interviews	The experimental design enables the testing of a range of configurations, threshold values (0.30-0.60), and ensemble fusion weights. The case study then tests the system in a Sri Lankan plywood factory, providing some indication of its use in practice. Furthermore, semi-structured interviews with domain and technical experts are conducted to validate the defect reasoning results, and to identify areas in need of improvement.
Time Horizon	Cross-sectional	Single snapshot data is used for model training and evaluation.
Data Collection	rimary: High-resolution image dataset from dual-view (top-45° and side) captures of post-manufacture plywood sheets (~620 images); Secondary:	Primary data: 623 plywood board images collected from a real factory with written consent; semi-structured interviews with 10 evaluators (5 domain, 5 technical). Secondary data: published literature on

Layer	Choice	Justification
	Wood defect datasets and literature references	CNN architectures, transfer learning, multi-label classification, and industrial defect detection systems, used to inform the model selection and benchmarking decisions.

Table 7 - Saunders Onion tabular format (self-composed)

Hypothesis: *A dual-view computer vision prototype using YOLO-based cropping with EfficientNet-B2 classification, augmented by knowledge-based reasoning, can accurately identify post-manufacture plywood defects from YOLO-extracted patches while providing interpretable manufacturing cause explanations for each detected defect type*

3.3 Development methodology

3.3.1 Requirement elicitation methodology

Several methods were employed to collect system requirements:

● Interviews: Semi-Structured interviews were carried out with the key stakeholders such as Production managers, workers, factory owners and research experts.

● Brainstorming sessions: Joint sessions with the production experts as well as ML experts were and will be conducted during the course of the project to identify the detection problems.

● Literature Review: Thorough literature review was conducted which was useful to identify the gaps in the literature and to identify the contribution to the research domain and problem domain.

3.3.2 Design methodology

Structured Systems Analysis and Design Methodology (SSADM) was chosen as the design approach. SSADM was selected because DefectIQ is essentially a data transformation pipeline - images are transformed from raw images through board detection, ensemble classification, dual-view fusion and knowledge-based reasoning into a structured inspection result. The SSADM Data Flow Diagrams clearly capture this, and therefore the design can be more easily understood by non-technical factory stakeholders, such as the inspectors, than class diagrams or sequence diagrams. The SSADM design is detailed in Chapter 6.

3.3.3 Programming paradigm

DefectIQ is largely written in structured programming. It uses a series of transformations, as defined by the SSADM DFD, in which each step prepares data for the following step. This makes it easier to manage flow control and to test each transformation in isolation. Object-Oriented Programming is also used to encapsulate system components where it makes sense. Specifically, it can be used to manage the state and configuration of the model, as well as processing logic, while keeping the API and machine learning code separate.

3.3.4 Testing methodology

The approach to testing is a multi-angled, systematic method to validate the DefectIQ system. The model is evaluated on its individual components and the ensemble using metrics for multi-label classification, as well as a threshold sweep to find the optimal classification threshold. System benchmarking is performed against baseline CNNs trained on the same data to validate the model design. Furthermore, prototype testing involves black-box testing based on the system's functional requirements, to validate all key features. Non-functional testing is conducted to assess critical operational features, including system responsiveness, reliability, and fallback procedures under different operating conditions. These testing methods ensure a thorough evaluation of the system's performance and readiness for deployment, as detailed in Chapter 8.

3.3.5 Solution methodology

Step	**Description**	**Tools/Techniques**	**Expected Output/ Outcome**
Data Acquisition	Capture top-view and side-view images of plywood boards under varied factory conditions	Mobile Phone camera, Roboflow	Raw dual-view image dataset (619 images)
Data Preprocessing	Extract board crops using YOLOv8-seg, apply resize, normalisation and TTA transforms	OpenCV, PyTorch transforms	defects_cropped dataset (640×640 crops)
Defect Annotation	Annotate 5 defect classes (bubbling, delamination, imprint_on_surface, missing_face, warping) using YOLO bounding box format	Roboflow Studio	annotated dataset
Model Training	Transfer learning on EfficientNet-B2 and AlexNet with multi-label sigmoid head,	PyTorch, AdamW, Kaggle T4 GPU	GPUTrained model checkpoints (Macro F1 = 0.625)

	WeightedRandomSampler, two-phase fine-tuning		
Ensemble Fusion	Weighted late fusion of EfficientNet-B2 and AlexNet per-class outputs with TTA, dual-view asymmetric weighting	Python	detection.pyFused confidence scores, PASS/REVIEW verdict
Reasoning Layer	Map active defects to manufacturing causes and corrective recommendations via knowledge base	config.py DEFECT_REASONS, reasons.json	Actionable defect occurrence explanations
Evaluation and Benchmarking	Model testing, threshold sweep, benchmarking against baselines, functional and non-functional testing	Scikit-learn	Chapter 8 testing report
Deployment and Visualization	Flask REST API, React frontend, Supabase PostgreSQL full prototype integration	Flask, React, Vite, Supabase	Functional DefectIQ prototype

Table 8- Solution Methodology (self-composed)

3.4 Project management methodology

DefectIQ is developed using an Agile-like iterative process with four sprints that each produce an increment of the system that is ready for testing. We choose this approach over Waterfall because the performance of the ML models is not known a-priori - iterative rounds of training and testing are required to converge on the best ensemble architecture and fusion weights. Each iteration adds to the previous one, and decisions regarding model selection, threshold settings and UI design can be made based on empirical findings rather than being pre-determined.

3.4.1 Resource requirements

Hardware and Software requirements have been added in chapter 1.12.

3.4.1.2 Data requirement:

A dataset will consist of synchronized top-angled and side view images of plywood panels, captured across diverse production sessions to encompass variations in lighting, textures, and defect types will be created.

3.4.1.3 Skills:

Machine learning, Python programming, Deep Learning, Computer Vision.

3.4.2 Risk and mitigation (expanded with severity and frequency)

Risk	Severity	Likelihood	Mitigation

Insufficient dataset size limiting model generalisation	5	5	Plan multiple factory visits to collect images across varied lighting and board conditions. Apply TTA at inference to improve robustness. Publish dataset on Kaggle to enable community expansion.
Class imbalance causing model bias toward majority classes	5	4	Apply WeightedRandomSampler during training to balance class representation. Evaluate per-class F1 separately to detect and document imbalance. Consider merging visually similar classes if separation proves infeasible.
False positive REVIEW verdicts on clean boards	4	4	Apply a confidence gate threshold below the classification threshold to suppress low-amplitude noise activations on clean boards. Tune gate value empirically on validation set.
Delays in factory access for image collection	5	4	Secured written consent and access agreement with factory owner early in the project; scheduled collection visits in advance
Model training resource constraints on local hardware	4	4	Use Kaggle Notebooks with free GPU access for all model training. Reserve local machine for API development, frontend, and inference testing only.

Table 9 Risk and mitigation table (self composed)

3.5 Schedule

3.5.1 Gantt chart

Gantt is provided in the appendix L .

3.5.2 Deliverables

Deliverables	Dates
Literature Review	5th Oct 2025
Project Proposal	10th Oct 2025
Project Proposal and Requirement Specification – initial draft	6th Nov 2025
Project Proposal and Requirement Specification – Final draft	12th Nov 2025
Design Document	1st December 2025
Prototype (summative)	5th Feb 2026
Interim Project Demo (formative)	15th Feb 2026
Implementation	2nd Feb 2026
Testing	25th Mar 2026
Evaluation	25th Mar 2026
Thesis Submission	3rd April 2026
Prototype	3rd April 2026
Dataset and Source Code	3rd April 2026

Table 10 Deliveribles (self composed)

3.6 Chapter summary

A pragmatist, mixed-methods approach guided DefectIQ's research design, combining quantitative model evaluation with qualitative expert validation across four Agile-inspired development sprints, underpinned by SSADM design methodology and structured programming to reflect the system's pipeline-driven architecture

CHAPTER 4: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter overview

This chapter provides a detailed analysis of the system's requirements, including how they were elicited, the stakeholders involved, their viewpoints, and the functional and non-functional requirements of the system. It also includes visual representations such as the Rich Picture Diagram, Stakeholder Onion Model, Context Diagram, and Use Case Diagram to help understand the system's interactions. Requirements are prioritized using the MoSCoW principle to ensure the focus on what is essential for the system's success.

4.2 Rich picture diagram

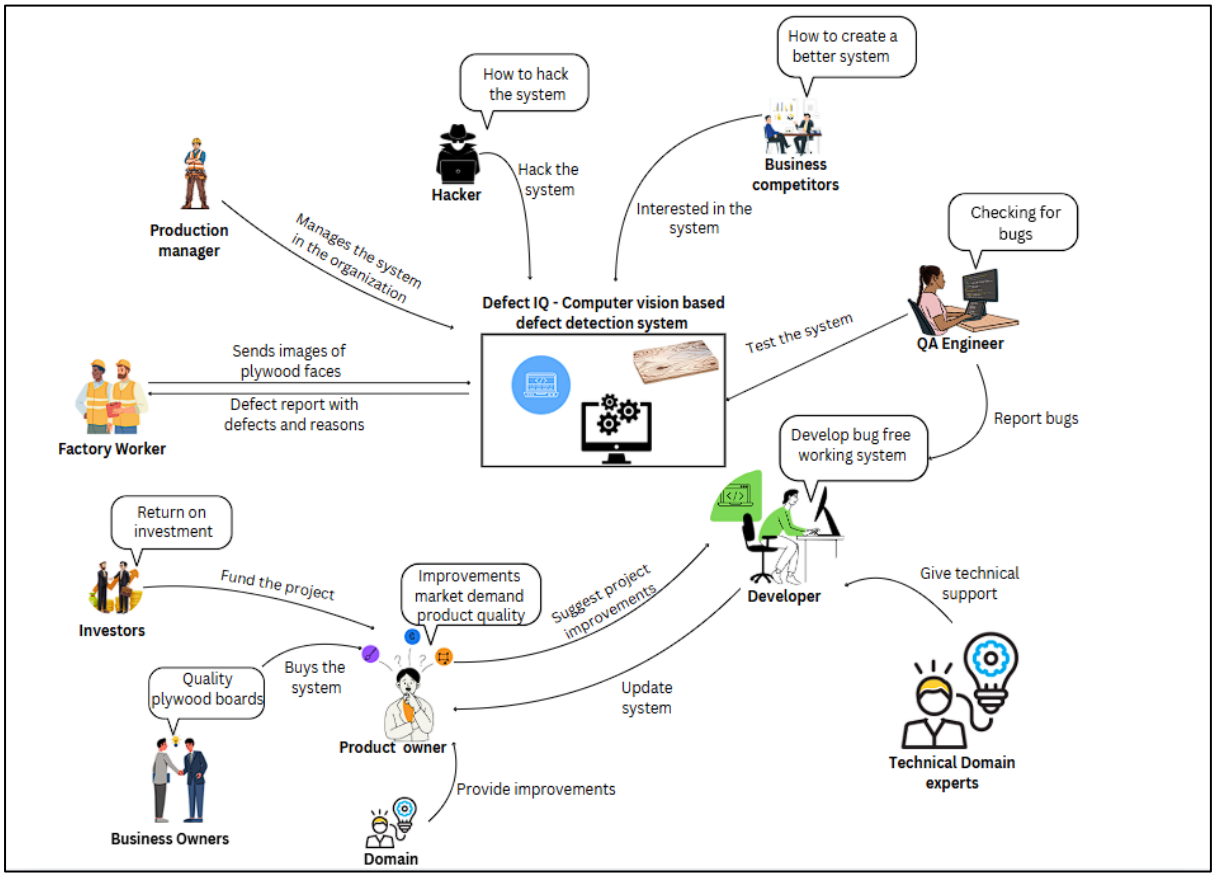


Figure 1: rich picture diagram (self composed)

4.3 Stakeholder analysis

4.3.1 Stakeholder onion model

The following stakeholder onion model visualizes the identified stakeholder of the proposed system Defect IQ. A detailed elaboration of the stakeholders of the system is presented.

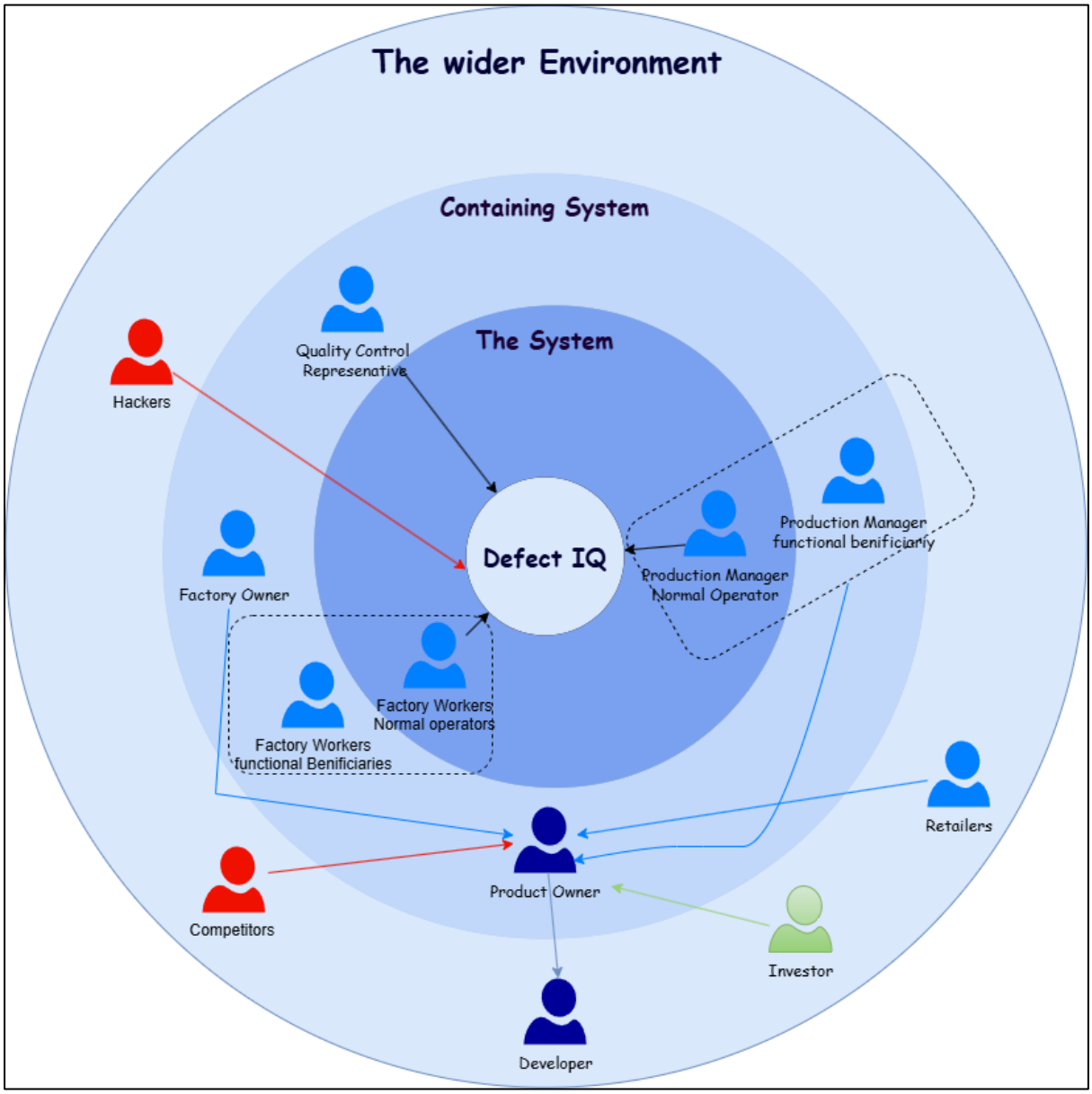


Figure 2 Stakeholder Onion diagram (self composed)

4.3.2 Stakeholder viewpoints

Stakeholder	Role	Description
Factory Worker	Uses the system/ Functional Beneficiary	Use the system to find the plywood sheets with defects. Benefit functionally because it reduces workload of workers
Production manager	Manage the system/ work with the product owner(system) / functional Beneficiary	Administration of the system in the organization / contact with the product owner (system owner) to support the maintenance of the system in the organization. Take advantage of the system for quality control.
Quality control representatives	Functional beneficiary/ Check quality of sheet	Use the system to check the quality of the sheets and benefit from the system while it is used to increase the quality of the production.
Factory Owner	Functional beneficiary	Benefit functionally when production increases with quality and quantity with the use of the system.
Product Owner	Decision maker of the system	Owner of the project vision, manages the system, prioritize the development of the system, acts as the link between the business and the development team.
Developer	Implementer and technical owner	Design, trains, and maintain the ML model. Implements the development of the project. Responsible for the system accuracy and performance.
Investor	Financial Backer	Provides capital for system development. Seeks success in the prototype value and a potential return on investment.
Retailers	External beneficiaries	The direct customers who purchase the finished plywood sheets. The accuracy and consistency of the defect detection system directly impact the quality they receive, thus affecting their satisfaction, trust, and future purchasing decisions.
Competitors	Negative Stakeholder (Wider Environment)	Poses a threat to the project's market advantage. Their actions (e.g., launching their own superior system) could reduce the value or urgency of this system.
Hackers	Negative Stakeholder	Poses a security risk. May attempt to compromise the system to corrupt the ML model, steal proprietary data, or disrupt the

		production line operations.

Table 11 stakeholder View Points (Self-composed)

4.4 Selection of requirement elicitation methodologies

For this project, various methodologies were employed to gather requirements. Each methodology was selected based on its ability to capture relevant, accurate, and actionable insights from stakeholders.

Methodology	Description
Interviews	A series of interviews were planned with a range of stakeholders Plywood factory owners production managers factory workers ML experts. This allowed real time dialogue clarifications of key issues and gathering of rich and context specific requirements often lost in less interactive approaches more importantly interviews were preferred over online surveys as most workers were digitally illiterate and interviews ensured that their voice was heard.
Literature Review	A Review was done to identify the gaps related to detection of after manufactured plywood defects using computer vision. Existing research related to the field was analyzed regarding their contribution and limitations.
Brainstorming	Brainstorming sessions with domain experts and research domain experts were carried out to get a clear idea about the project and fine tune the idea. This approach was to find the best technologies that can be used for the research project.

Table 12 Requirement Elicitation methodologies (self composed)

4.5 Discussion of findings

4.5.1 Literature review findings

Citation	Finding
Baad,2023	The models were trained and tested on a dataset that suffered from a small size and a huge discrepancy in the number of samples per class
Chen et al, 2022	Fever defect training images were available for overlong and short edge branding
Zhang et al., 2024	focus exclusively on the detection of surface defects in wood lumber using their WLSD-YOLO model, rather than analysing or identifying structural defects.

Table 13 Literature Review findings (self composed)

4.5.2 Interviews findings

Interviews were conducted with domain experts representing all the key stakeholder groups to ensure that the research comprehensively captures diverse perspective and needs.

The interviews included One Factory owner, one product manager/ quality control representative, factory worker to represent the end uses, one machine learning expert and one Software Engineer to get feedback on using technologies and the efficiency of the research project. A drive link for the informal interview questions is available at Appendix 2

Codes	**Themes**	**Conclusion**
Quality Assurance	Maintaining product Quality	Automated defect detection boosts quality, prevents bad sheets shipping to retailers
Reputation and Export	Business value, market access	System enables export opportunities and sustains company reputation
Efficiency	Workflow optimization	Reduces worker stress, saves time, and increases labor productivity
Labor management	Workload distribution, fatigue	Allows better worker assignment and reduces risk of missed defects
Economic impact	Revenue, loss prevention	Prevents quality lapses that result in lost clients and revenue
Data-driven Insight	Root-causes, defect minimization	System feedback enables process adjustment and continuous improvement

Codes	Themes	Conclusion
ML project strategy	Dataset creation and tech use	Expert guidance on robust data collection, model selection, and system design

Table 14 Interview Findings (Self-composed)

4.5.3 Survey findings

In this research, surveys were not conducted primarily due to the specific context and challenges of the plywood manufacturing environment. Many of the stakeholders, including factory workers, have limited digital literacy and may not have reliable access or comfort with online survey tools. This limitation would likely result in low response rates and less meaningful data, risking gaps in understanding user needs. Conducting Interviews with factory experts and other stakeholders enabled a more inclusive and interactive approach, allowing direct communication and observation to capture their genuine insights and requirements. Interviews not only helped overcome barriers related to digital skills but also facilitated deeper exploration of concerns, expectations and practical approaches.

4.5.4 Brainstorming/Observation

Aspect	Finding
Participants	Stakeholders include factory owners, workers, production manager, and machine learning experts.
Objective	To Identify critical defect types, gather stakeholder records, prioritize key features, and explore novel technical approaches.
Key focus areas	6 defect classes (bubbling, delamination, imprints, missing edges, top face, warping), workflow impacts, EfficientNet-B2 classification, YOLO cropping, JSON-based reasoning , deployment feasibility
Novelty achieved	Hybrid YOLO-EfficientNet-B2 pipeline with knowledge-based defect cause explanations; Top-2 image accuracy metric; prototype on limited data (~150→750-1000 images)
Technical challenges	Small dataset constraints, class imbalance, edge/glossy false positives, annotation efficiency, multi-view synchronization
When brainstorming was conducted	Early requirement gathering; iterative refinement during prototype development (post-150 image baseline).

Table 15 Brainstorm findings (self composed)

A concept diagram illustrating the brainstorming process is included in the appendix section.

4.6 Summary of findings

Finding	Literature Review	Interviewing	Brainstorming
Small and Imbalanced dataset limit model accuracy	✓		✓
Subtle and structural defects are hard to detect	✓	✓	✓
Combining dual views improves detection		✓	✓
Stakeholders want defect explanations in reports		✓	✓
Annotation efficiency and digital literacy are practical barriers	✓	✓	✓
MSDA, DWR, and fusion methods offer technical innovation	✓	✓	✓
System must fit factory workflow and skill level		✓	✓

Table 16 Summary of findings (self composed)

4.7 Context diagram

Context diagram for DefectIQ will be the end user that is mainly factory worker or quality control representative and the system.

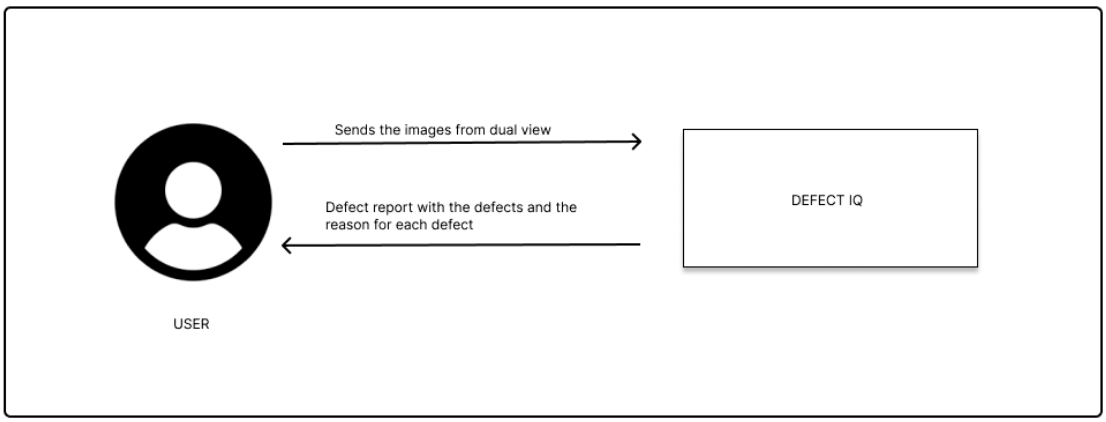


Figure 3 Context diagram (self composed)

4.8 Use case diagram

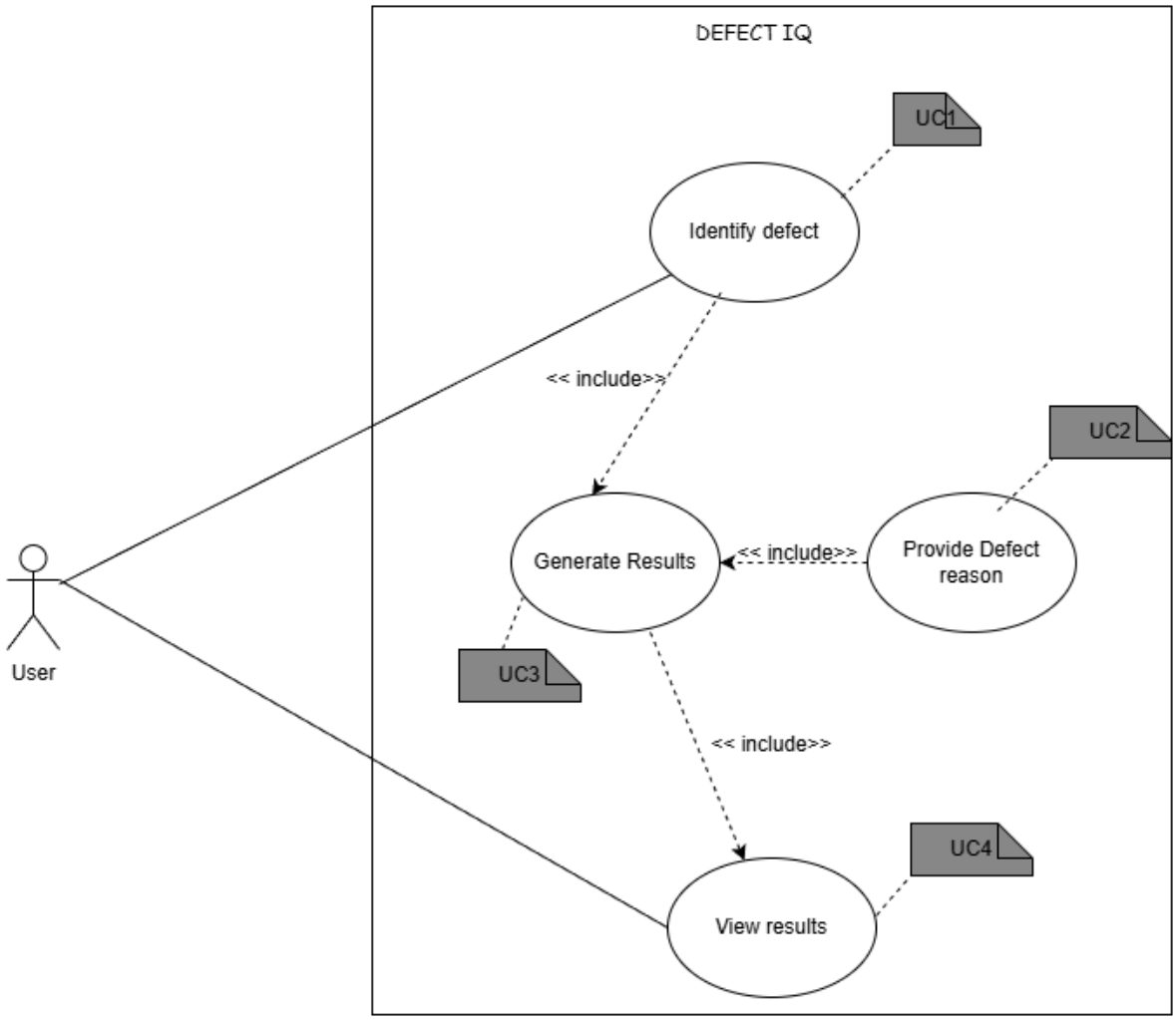


Figure 4 Use-case diagram (self-composed)

Main Actors:

User - Abstract actor representing both factory worker and production manager.

4.9 Use case descriptions

The main use case descriptions (UC1 and UC4) are provided here. The other use cases are provided in the Appendix 3 section.

Use case	**Explanation**
Id	UC1

Description	The system receives plywood images from both top and side views. It locates the board area using YOLOv8-seg, crops and then classifies the defects using the EfficientNet-B2 and AlexNet ensemble, and then applies dual-view weighted fusion to produce the class of defects with probabilities.
Primary Actor	User (Factory Worker, production manager)
Supporting Actor	None
Preconditions	Plywood images must be available to the system.
Included use cases	None
Extended Use cases	None
Main flow	1. User uploads images of plywood top and side views to the Inspect page. 2. System crops the plywood board using the YOLOv8-seg board detector. 3. System crops the region and applies the EfficientNet-B2 and AlexNet based ensemble with TTA at threshold 0.50. 4. System conducts weighted fusion of the two views: 65% top / 35% side for surface defects; 35% top / 65% side for structural defects. 5. System classifies defects (bubbling, delamination, imprint_on_surface, missing_face, warping) and records. 6. System forwards defects to UC2 (Provide Defect Reason).
Alternative flow	If a board is not detected, the entire image is used for inference and the user is alerted.
Exceptional flow	If analysis fails, log in error and notify user.
Postconditions	The defect classes and class probabilities are available for reasoning and presentation.
Comment	Triggers the reporting and reasoning process for every inspected batch

Table 17 Use case description (UC1)(self-composed)

Use case	E**xplanation**
Id	UC4
Description	The user visits the Dashboard page to access their inspection history. All previous inspections for the logged-in user are returned with total inspections, pass rate, verdict and defect summaries. The user may also choose to view a particular date range.
Primary Actor	User (Factory Worker, Production Manager)
Supporting Actor	None

Preconditions	User is authenticated. There is at least one inspection result in the database (UC3 complete).
Included use cases	None
Extended Use cases	None
Main flow	1. User opens the Dashboard. 2. System queries the Supabase database for all inspections of the logged-in user. 3. System shows the total number of inspections, pass rate, pass and fail counts, defect labels, and the time and date of inspection. 4. (Optional) User enters a specific date range for the inspections. 5. System updates the data from the database.
Alternative flow	If no report exists for selected batch, user is prompted to run full inspection.
Exceptional flow	When failed to retrieve data from the database, a message is shown.
Postconditions	User has reviewed their inspection history and may take operational decisions based on the displayed data.
Comment	Supports operational decision-making and continuous improvement.

Table 18 Use case description (UC4)(Self-composed)

4.10 Requirements

The **MoSCoW principle** is used to prioritize requirements into:

- **M**: Must Have (mandatory).
- **S**: Should Have (important).
- **C**: Could Have (desirable).
- **W**: Will Not Have (out of scope).

4.10.1 Functional requirements

ID	Requirement	Priority (MoSCoW)	Final Status
FR 01	The system must support uploading multiple images per plywood sheet, including top and side views.	Must Have	Implemented
FR 02	The system should validate image format and quality during upload before defect analysis.	Must Have	Partial

ID	Requirement	Priority (MoSCoW)	Final Status
FR 03	The system must detect defects from uploaded plywood images.	Must Have	Implemented
FR 04	The system must correctly detect both surface-level and structural defects.	Must Have	Implemented
FR 05	Provide defect reasons/recommendations using a knowledge base (JSON).	Must Have	Implemented
FR 06	The system should connect to a database backend to provide defect history for trend analysis.	Should Have	Implemented
FR 07	The system should allow users to view historical defect data and trends via database connection and query interface.	Should Have	Implemented
FR 08	The system could support real-time defect detection using industry cameras directly in the production line.	Could Have	Future work
FR 09	The system should support Sinhala language for defect reasoning	Could Have	Future work
FR 10	The system will not have four-way real-time camera integration due to limited resources at this stage.	Will not have	Out of scope

Table 19 Functional Requirement table (Self Composed)

4.10.2 non-functional requirements

ID	Non-Functional Requirement	Description	Priority (MoSCoW)	Status
NFR01	Scalability	The system must handle multiple simultaneous image uploads and defect analyses efficiently	Must Have	Implemented
NFR02	Security	All image data and defect reports must be stored and transmitted securely with encryption	Must Have	Implemented

ID	Non-Functional Requirement	Description	Priority (MoSCoW)	Status
NFR03	Performance	The system should provide defect analysis results within 2 minutes for each plywood sheet	Must have	Implemented
NFR04	Usability	The system could provide a dashboard for real-time monitoring of transactions.	Must Have	Implemented
NFR05	Reliability	The system should have 99% uptime during manufacturing hours to ensure consistent inspection	Must have	Implemented
NFR07	Portability	The system should be deployable on standard hardware currently used in Plywood factories.	Could have	Future work
NFR08	24/7 availability	The system will not guarantee 24/7 uptime during the prototype phase due to maintenance windows	Will not have	Out of scope
NFR09	Multi-camera Real-Time Integration	The system will not integrate with 4-way multi-camera real-time inspection due to resource limits.	Will not Have	Out of scope

Table 20 Nonfunctional Requirements (self composed)

4.11 Chapter summary

The Software Requirement Specification chapter has provided a comprehensive overview of the system's functional and non-functional requirements, the methodologies used to gather these requirements, and the tools and techniques employed to ensure stakeholder needs are addressed. Requirements were prioritized using the MoSCoW principle to focus on what is mandatory for the system's success.

CHAPTER 5: SOCIAL, LEGAL, ETHICAL AND PROFESSIONAL ISSUES (SLEP)

5.1 Chapter overview

This chapter examines the broader Social, Legal, Ethical, and Professional (SLEP) considerations surrounding the DefectIQ project and its deployment. While the primary focus of the project is technical, it is crucial to address these issues to ensure the solution is responsible, compliant, and accepted by society and stakeholders. The project outline potential SLEP issues and describe how we mitigated or plan to mitigate them, referencing guidelines such as the BCS Code of Conduct where relevant.

5.2 SLEP issues and mitigation

Social - Mobile phone photographs of plywood production lines taken across six to eight factory visits during dataset collection risked incidental identification of factory workers in the background of images. Mitigation: Verbal consent was obtained from the factory manager before each collection session and documented with a timestamp. Images were captured from elevated angles specifically to avoid capturing faces. Background personnel visible in retained dataset samples were pixelated before annotation and Kaggle publication. This approach follows the BCS Public Interest principle of protecting privacy and wellbeing.	**Legal -** The 619 images in the DefectIQ dataset were captured at partner factory premises and depict proprietary production environments, making data ownership a legal concern. Mitigation: Written consent was secured from the factory owner prior to dataset collection, annotation, and Kaggle publication under CC BY 4.0. All stakeholder identities are anonymised throughout the thesis and appendices referred to as "Factory Manager", "Production Worker", and "Domain Expert". No personal data is stored in reasons.json, model checkpoint files, or Supabase database records beyond anonymised usernames and hashed passwords. This complies with BCS Professional Competence requirements for responsible data handling.

Ethical - The DefectIQ training dataset contains significant class imbalance delamination has 115 training crops compared to 427 for missing_face. This imbalance risks the model performing less reliably on underrepresented defect classes, reducing quality control fairness for those defect types. Mitigation: WeightedRandomSampler was applied during training to ensure balanced batch representation, and FocalBCEWithLogitsLoss with gamma=2.0 was used to focus training on hard minority-class examples. Per-class F1 score disparities are reported transparently in Chapter 8 - delamination achieves F1=0.493 versus missing_face at F1=0.770, and are explicitly acknowledged as a limitation rather than concealed. Planned dataset expansion to 750–1,000 images across multiple factories is identified as the primary future work to address this failure mode, per the BCS Professional Integrity principle of honest reporting.	**Professional** -Transfer learning with EfficientNet-B2 and AlexNet, three-phase progressive fine-tuning, multi-label Focal BCE loss, and per-class weighted ensemble fusion represented approaches at the boundary of prior experience at the start of the project. Mitigation: Established PyTorch training recipes were followed throughout using ImageNet pre-trained weights, documented learning rate schedules with cosine annealing, gradient clipping at max_norm=1.0, and WeightedRandomSampler. All hyperparameters and training configurations are documented in full in Appendix K for reproducibility. Model performance was validated against scikit-learn multi-label metrics including Macro F1, Hamming Loss, and Subset Accuracy. Supervisor guidance was sought at key design decisions. This aligns with the BCS duty to only undertake work within the bounds of demonstrated or actively developed competence.

Table 21 SLEP table (self composed)

5.3 Chapter summary

This chapter analyzed SLEP issues in the DefectIQ prototype per BCS Code of Conduct. Privacy was protected through factory consent and anonymization; bias mitigated via balanced

sampling; competence maintained with documented transfer learning. All measures ensure ethical, professional plywood defect detection development.

CHAPTER 6: DESIGN

6.1 Chapter overview

The chapter presents the DefectIQ system design, which converts the functional and non-functional requirements of Chapter 4 into a form of architectural blueprint. Its design is Structured Systems Analysis and Design Methodology (SSADM) which is selected due to its focus on data flow and process decomposition which are well-suited to a pipeline-based system with raw images gradually converted to a structured inspection result. There is a mapping of non-functional requirements to concrete design goals followed by a four-tier system architecture in the beginning of the chapter. It is represented by the detailed design that consists of SSADM Data Flow Diagrams on two levels: Level 0 context diagram which has been presented in Chapter 4 and Level 1 diagram which breaks down the core inspection process into specific process and data stores. The chapter will end with the algorithm design of the ensemble classification and dual view fusion pipeline and wireframe descriptions of the three major user interface screens.

6.2 Design goals

The non-functional requirements (NFRs) presented in Chapter 4 serve as the basis of the design requirements of DefectIQ. Each objective informs the architectural and implementation choices, to make sure that the design is not an abstract activity but a dedication to specific, measured properties of qualities of the system. Table below presented is a mapping of each NFR with the design goal and design decision made to accomplish the goal.

NFR ID	Design Goal	Design Decision	Priority
NFR03	Performance – under 30 seconds for the defect analysis	TTA with 4 views; YOLOv8-seg fast board crop; GPU inference pipeline	Must Have

NFR02	Security- secure storage and transmission of image data and reports	JWT-based authentication; bcrypt password hashing; Supabase SSL	Must Have
NFR04	Usability- dashboard for real-time monitoring and inspection history	React/Vite SPA; dual-image side-by-side results; annotated board view	Must Have
NFR01	Scalability- handle multiple simultaneous uploads efficiently	Flask REST API; Supabase PostgreSQL with indexed queries	Must Have
NFR05	Reliability- consistent uptime during manufacturing hours	Stateless Flask backend; Supabase managed DB; fallback to full image if board detector fails	Must Have
NFR06	Explainability- defect causes and recommendations surfaced to operator	Knowledge-based reasoning via reasons.json; per-defect cause and recommendation	Must Have
NFR07	Portability- deployable on standard factory hardware	Lightweight React frontend; Flask runs on CPU-capable hardware; no specialised hardware required	Could Have

Table 22 Non functional Requirements (self composed)

6.3 System architecture

DefectIQ has a four-tier layered architecture that isolates concerns between presentation, application logic, machine learning inference and data persistence. This separation guarantees that each tier can be developed, tested, and maintained independently, and that the ML

inference pipeline can be updated, for example, by retraining models or adjusting fusion weights without having to make changes to the frontend or the database schema. Table below summarizes the tiers and their components and their responsibilities.

Tier	**Components**	**Responsibility**
Presentation	React Login, Register, Inspect, Dashboard pages	User interaction, dual image upload, result display, inspection history
Application / API	Flask REST API, JWT authentication (auth.py, app.py)	Route handling, session management, request validation, response assembly
ML Inference	YOLOv8-seg (board_detector.py) Ensemble (ensemble_inference.py) -> Dual-view fusion -> Knowledge base (config.py)	Board isolation, multi-label classification, view fusion, defect reasoning, verdict generation
Data	Supabase PostgreSQL -> users table, inspections table; reasons.json knowledge base	Persistent storage of user accounts, inspection records, defect JSON, image filenames

Table 23 System Architecture(self composed)

Figure below shows the high-level system architecture diagram that shows the interaction between the tiers. The React frontend does not communicate with the Flask backend at all other than through a rest API. The Flask layer coordinates the ML inference tier, sequentially calling the board detector, the ensemble classifier, the dual-view fusion logic, and the knowledge-based reasoning module before persisting the result to Supabase and returning structured as a structured JSON response to the frontend. This linear orchestration is intentional, it leaves a clear path from raw image to final verdict that is key to quality control accountability in a manufacturing setting.

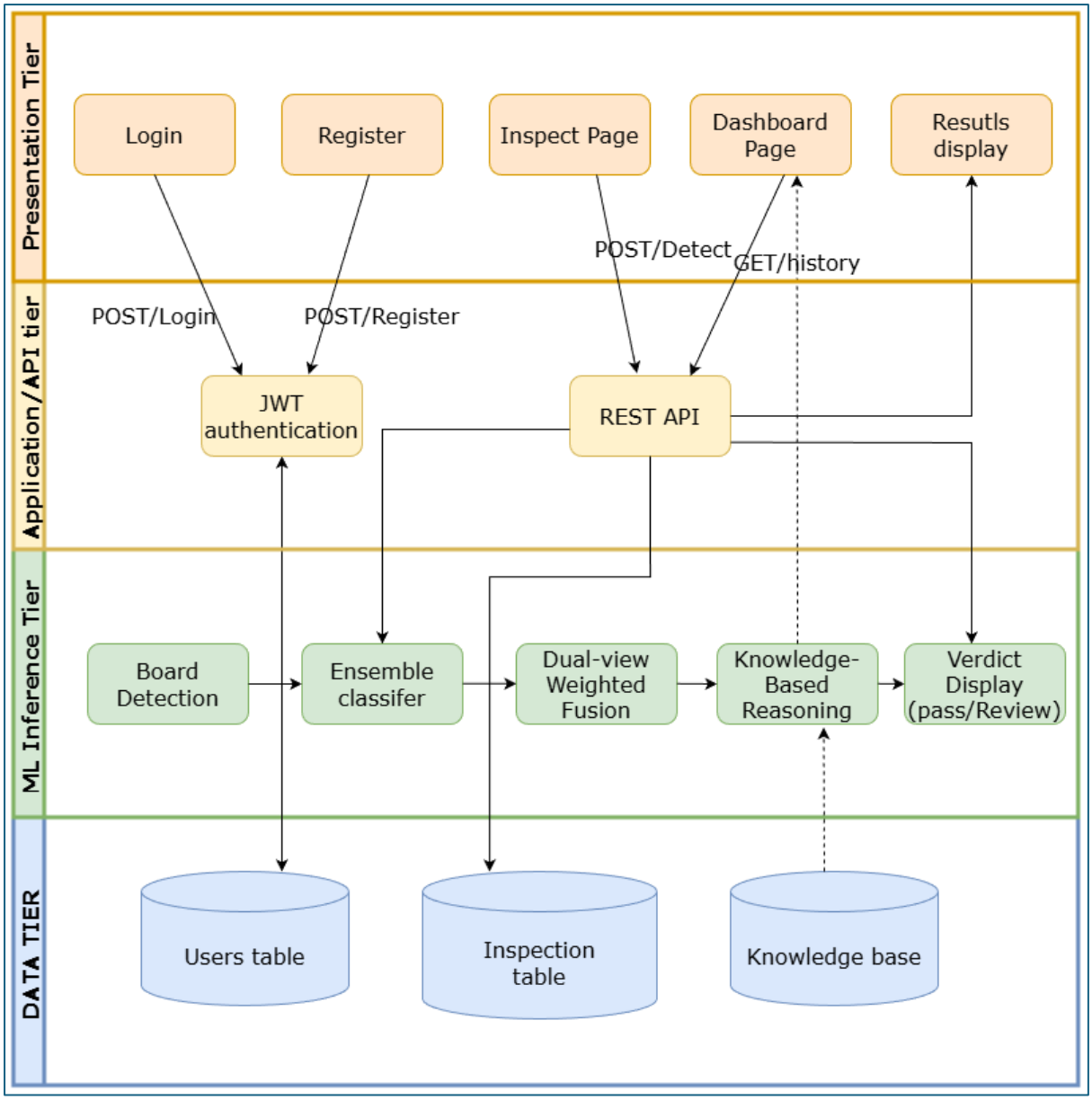


Figure 5 Architecture diagram (self-composed)

As shown in Figure, the Presentation Tier has five screens, Login, Register, Inspect Page, Dashboard Page and Results Display. Authentication requests (POST /login and POST /register) are sent from Login and Register directly to the JWT Authentication component in the Application/API Tier, which reads and writes to the Users Table in the Data Tier. Inspection requests (POST /detect) flow from Inspect Page to the REST API triggering the ML Inference Tier pipeline sequentially, Board Detection -> Ensemble Classifier -> Dual-View Weighted Fusion -> Knowledge-Based Reasoning -> Verdict Display (PASS/REVIEW) The Knowledge Base in the Data Tier is referenced by the Knowledge Based Reasoning component by a dotted lookup arrow. Once the verdict is given, the result is passed back up through the REST API to

the Results Display. The Dashboard Page sends a GET /history request through the dashed line to the REST API, which requests the Inspection Table and returns Historical records. The Inspection Table is also written straight from the Ensemble Classifier stage, persisting the record of the defects at the point of classification.

6.4 Detailed design - SSADM data flow diagrams

6.4.1 Choice of design paradigm

SSADM was chosen as the design methodology of DefectIQ in preference to Object-Oriented Analysis and Design (OOAD). The main reason is the nature of the system**.** DefectIQ is in its core a data transformation pipeline where a pair of raw images is fed through a sequence of well-defined processing stages where each stage takes structured inputs and produces structured outputs. SSADM's Data Flow Diagrams (DFDs) make this flow explicit and expose where data enters the system, how it is transformed at each process, where it is stored, and what is returned to external entities.

6.4.2 Level 0 data flow diagram- context diagram

The Level 0 DFD (context diagram) was presented in Chapter 4, in the context of requirements elicitation. It determines the system boundary and the two external entities that interact with DefectIQ: the Factory Worker (or Quality Control Representative): who submits the images and receives the results of the inspection and the System Administrator: who manages the user accounts. At this level, the entire DefectIQ system is represented in a single process bubble with data flows that indicate that data manufacturers provide top-view image and side-view image as inputs and provide verdict (PASS or REVIEW), defect names, confidence scores, and suggested corrective actions as outputs. The context diagram can be seen in section 4.7.

6.4.3 Level 1 authentication flow

The Authentication Flow covers the registration and login flow as well as the JWT session management between the Factory Worker and the Users Database. There are no ML components involved.

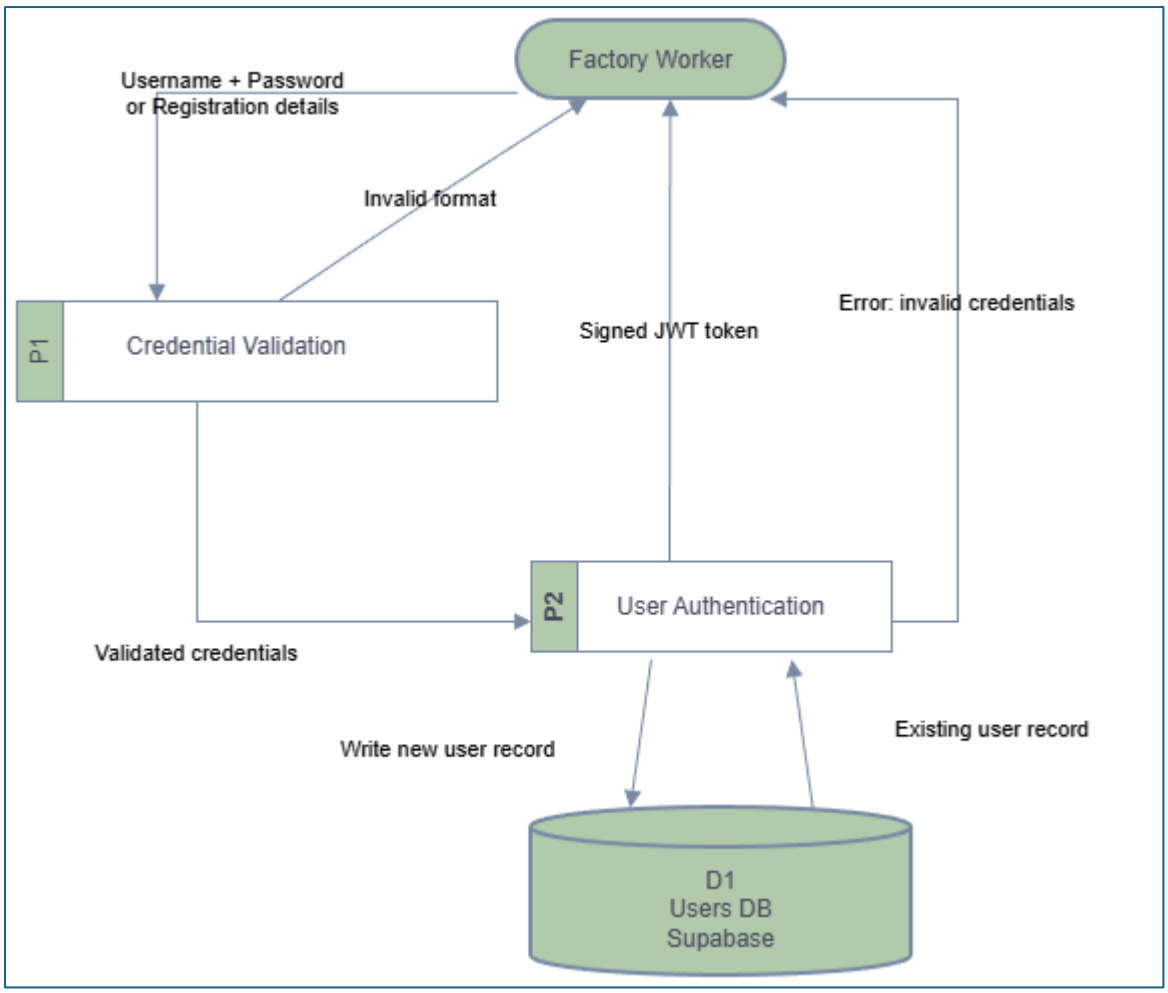


Figure 6 Level 1 DFD (self composed)

6.4.4 Level 2 Inspection flow

The Inspection Flow breaks down the entire image-to-verdict process into seven processes that are divided into three data stores from validation of the image to detection of board, ensemble classification, dual-view fusion, knowledge-based reasoning, and verdict generation and result storing.

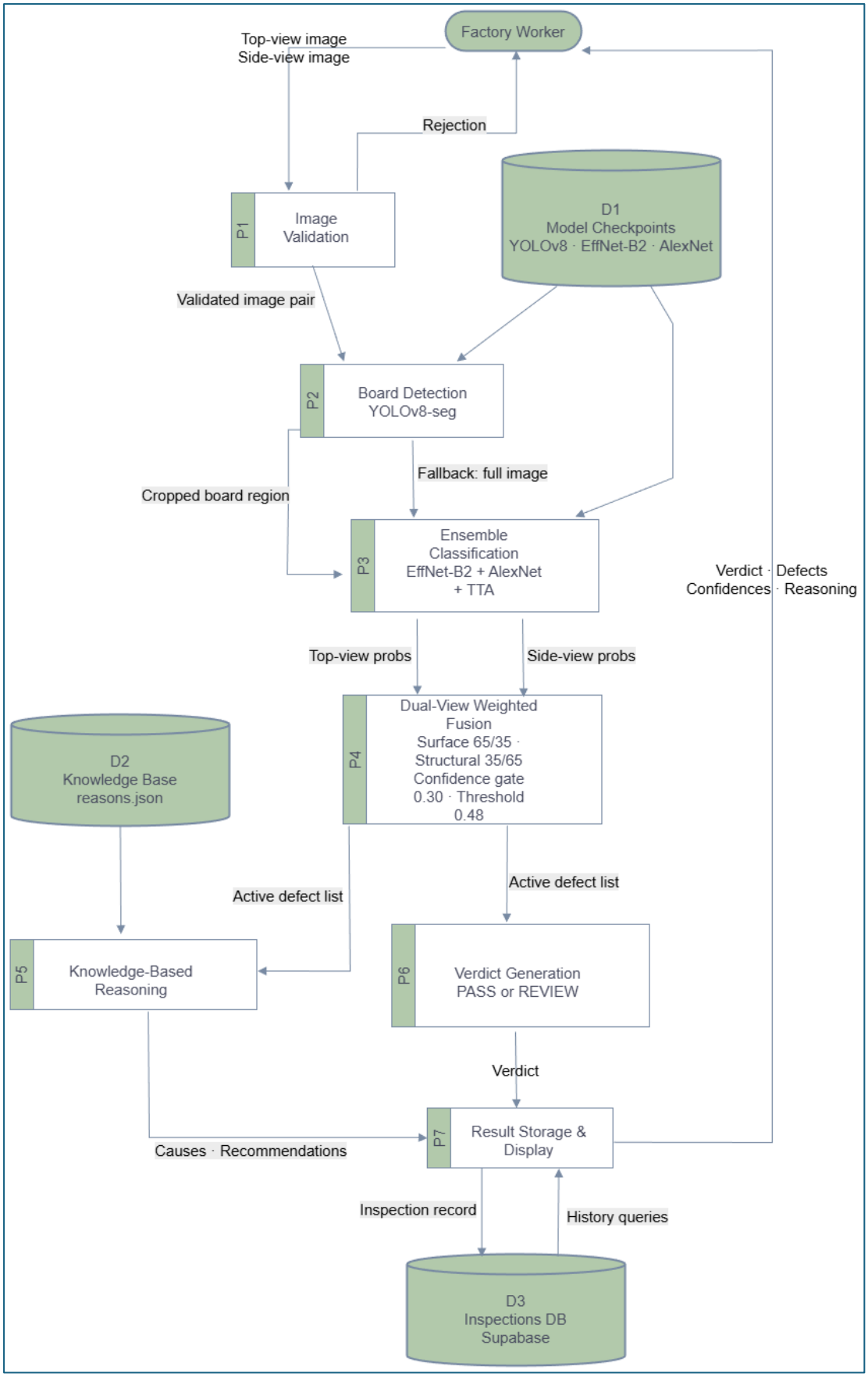


Figure 7 Level 2 DFD (self composed)

6.5 ML pipeline design

This section discusses the motivations behind the machine learning pipeline for DefectIQ. The literature review in Chapter 2 established what EfficientNet-B2, AlexNet, ensemble methods, and multi-label classification are. This section is solely dedicated to what makes these components and their parameterisation best suited to this problem. Further details of the implementation and hyperparameters are in Chapter 7 and Appendix K.

AlexNet was chosen as the second model not because of its standalone classification accuracy, but to achieve the greatest degree of contrast in model architecture. Section 2.5.3 of Chapter 2 has shown that ensemble gain is driven by model diversity. The design choice was to combine EfficientNet-B2 with an architecturally different model, rather than with a different EfficientNet. A short trial pairing EfficientNet-B2 with EfficientNet-B0 resulted in Macro F1 = 0.601 (less than the EfficientNet-B2 + AlexNet performance of 0.625), confirming the gain in ensemble performance is due to architectural difference and not simply size difference. AlexNet's early layer convolution kernels (11×11, 5×5) learn large-scale spatial deformations (warping, missing_face) - EfficientNet-B2's MBConv blocks learn small-scale texture (bubbling, delamination, imprint). This is the basis of the class-specific weights in Appendix K, Table K.5. High-level algorithm architecture diagrams are shown in Appendix J.1 section.

6.5.1 Model selection rationale

Section 2.5.2 of Chapter 2 presented how EfficientNet uses compound scaling to repurpose weights learned from ImageNet. The design option being considered here is the particular EfficientNet-B2. B2 was chosen due to its ideal point of accuracy versus inference speed for prototype hardware and CPU-only inference - B0 and B1 are insufficiently discriminative to classify five structurally distinct defect types on a 619-image dataset, while B4 and B7 have excessive inference speed on unaccelerated CPUs. This isn't a general statement about EfficientNet - it is a DefectIQ prototype deployment constraint.

6.5.2 Training strategy rationale

The models were not fine-tuned from epoch 1, they were trained with three stages of unfreezing to mitigate catastrophic forgetting. Completely unfreezing an ImageNet-pretrained network at the beginning of training on the 619-image DefectIQ dataset might lead to quick overfitting

and over-write beneficial features that the network backbone may have learned before the head converged.

Training was done in three phases: first unfraining only the head while the backbone is frozen, then unfraining deeper layers with discriminative learning rates, and finally unfraining deeper layers still with lower learning rates. Backbone schedules are illustrated in Appendix K, Table K.2.

Focal BCEWithLogitsLoss was chosen over BCEWithLogitsLoss for the DefectIQ dataset (delamination had 115 training images and missing_face 427). BCE is unaffected by class imbalance because it treats all examples equally.

The focal term, $(1 - p_t)^{\gamma}$ with $\gamma = 2.0$, down-weights the impact of easy training samples and up-weights the impact of hard and minority-class samples. This was specific to the particular dataset. We also used a WeightedRandomSampler in the DataLoader to better balance samples. For multi-label crops, the weight for each sample was calculated as the inverse proportion of the rarest class in the sample's label vector, so that minority-class defects like delamination and imprint_on_surface were used more frequently. CutMix and MixUp were only used in Phases 2 and 3, and not Phase 1, as mixing samples in the first batch may destabilise head training when the backbone was frozen.

6.5.3 Ensemble strategy and TTA rationale

Decision (or late) fusion was selected over early and intermediate fusion for the ensemble. Early fusion would combine raw image data that each model would be trained on, reducing model diversity. Intermediate fusion requires using partially overlapping layers, which would also tightly couple the models. This would merge the probabilities of two separate models, which are independent and perform in different representations. Liu et al. (2024) also argue late fusion to be the most convenient way to deploy and the most commonly used in industrial inspection. Per-class fusion weights were used, rather than a global weight. For instance, a global weight of EfficientNet-B2 0.60 and AlexNet 0.40 would suggest that EfficientNet-B2 is better for all defects, but it is not. AlexNet performs best or equal on warping (a low-frequency spatial distortion at board level) while EfficientNet-B2 is best for all fine-grained defects. The class-wise weights in Appendix K, Table K.3, were determined on the validation set by grid searching the weight for EfficientNet from 0.40 to 0.65 with a step size of 0.05 and selecting the combination with the highest Macro F1. Test-time augmentation (TTA) uses five views for inference and then averages the sigmoid predictions before fusing the ensemble. They are the original resized image, horizontal flip, vertical flip, change in brightness (using

ColorJitter, brightness = 0.15) and change in scale (using RandomResizedCrop, scale = 0.90-1.00). These views were used because plywood boards are rectangular and rotationally symmetric, so horizontal and vertical flips do not change the defect pattern, but altering brightness and scale is likely to reflect the variability of lighting and camera placement in the factory. TTA offers test-time stability without additional labels, and can be used in conjunction with a small number of samples. Five-view TTA was chosen because validation tests indicated only marginal gains in F1 scores beyond this.

6.5.4 Dual-view fusion rationale

The dual-view fusion weights (65% top-view and 35% side-view for surface defects and 35% top-view and 65% side-view for structural defects) are determined by inspection of the defects. Defects such as bubbling, imprint_on_surface and missing_face are predominant on the flat board surface and so are more noticeable in the top view. Those defects have limited effects on board form and thus are not revealed much by the side view. On the other hand, structural defects such as warping and delamination distort the board form. Warping is manifest as bowing at the board edge, while severe delamination can cause layer separation at the ends. The side view is therefore more informative for the detection of structural defects. This weighting scheme is a new feature of DefectIQ and is based on domain knowledge about plywood defects.

The confidence gate (CLEAN_BOARD_GATE = 0.30) was introduced to stop clean boards from producing false REVIEW verdicts. The sigmoid activation function never reaches zero, so even clean boards can have a positive value that may be above the 0.50 classification threshold. The gate determines if the maximum fused class probability is less than 0.30 and returns PASS if this is true. This helps prevent false alarms and keeps the operators' confidence in the system.

6.6 UI design

Low fidelity wireframes were designed first, and then using those the prototype design was refined, given below are the low fidelity diagrams of the inspect page and the results page, the other wireframes are given the Appendix j.2

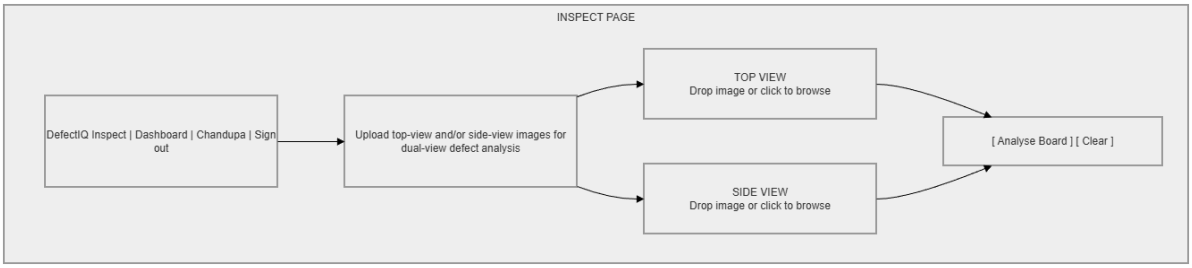


Figure 8 Inspect page wireframe (self composed)

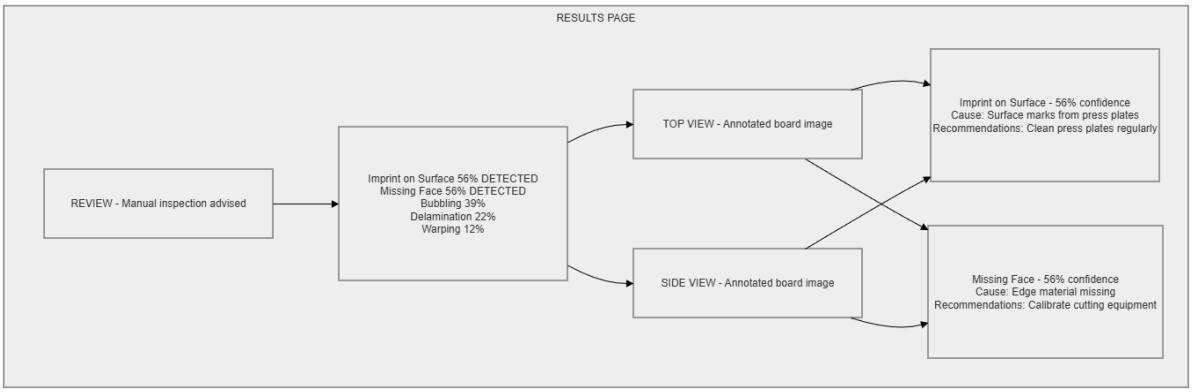


Figure 9 Results page Wireframe (self composed)

6.7 Chapter summary

In this chapter DefectIQ's design was introduced using SSADM and in detailed four-tier architecture, two Level 1 data flow diagrams and pseudocode algorithms for ensemble fusion and dual-view verdict generation are presented which set up the blueprint for implementation.

CHAPTER 7: IMPLEMENTATION

7.1 Chapter overview

This chapter covers the implementation of the DefectIQ system, which takes the design of the SSADM established in Chapter 6 and transforms it into an operational prototype. It includes the selection and rationale for the technology, implementation of the four key pipeline phases board detection, ensemble classification, dual-view fusion, and knowledge-based reasoning, the user interface screens, and some of the major challenges faced and solved during the development.

7.2 Technology selection

7.2.1 Technological stack

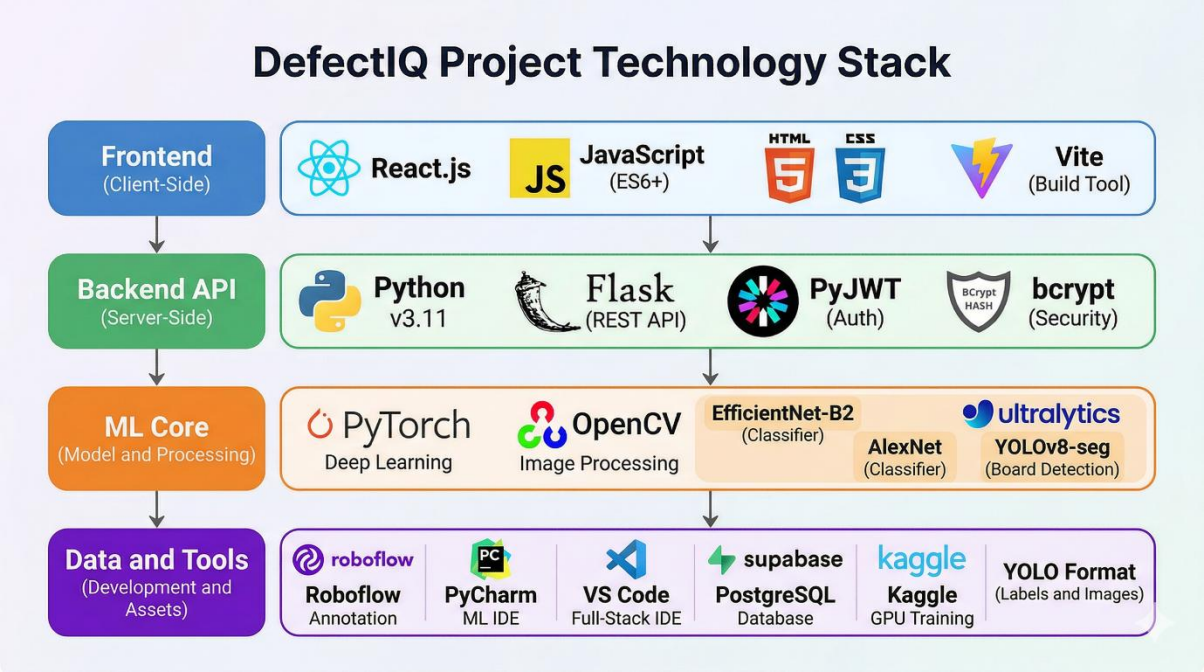


Figure 10 Technology stack (self composed)

Component	Technology	Role
ML Training	PyTorch 2.x, Python 3.11	Model training, transfer learning, evaluation

Board Detection	YOLOv8-seg (Ultralytics)	Plywood board segmentation and crop extraction
Ensemble Models	EfficientNet-B2, AlexNet	Multi-label defect classification with TTA
Backend API	Flask, PyJWT, bcrypt	REST API, JWT authentication, request handling
Frontend	React 18, Vite	SPA with dual image upload, results display, dashboard
Database	Supabase PostgreSQL	User accounts and inspection record persistence
Annotation	Roboflow	Dataset annotation, class management, YOLO export
Image Processing	OpenCV, Pillow	Board crop, image annotation, format handling

Table 24 Technological stack (self composed)

7.2.2 Programming languages

- Python 3.11 - chosen as the main backend ML language. Its large ML ecosystem (PyTorch, OpenCV, scikit-learn) and support for Flask made it the only practical choice for a project that involves both the development of a REST API and deep learning inference.
- JavaScript (React 18) - chosen one for frontend. React's component-based architecture helped to quickly develop the dual-image upload interface, results display, and dashboard. Vite was used as the build tool, as Next.js was giving out memory issues on the development machine.
- JSON - this is used for the knowledge base (config.py DEFECT_REASONS). Its lightweight, human-readable structure made it possible for domain experts to review and for the backend and frontend to consume it directly without serialization overhead.

7.2.3 Development framework

PyTorch was chosen for model training and inference because of its dynamic computation graph, great support for transfer learning with torch vision pretrained weights, and flexibility for the implementation of custom multi-label sigmoid heads. Flask was chosen as the backend of the application, a simple and lightweight REST framework to serve the ML model, minimal overhead from the time the request is received by the webserver and the time the inference pipeline is executed. React with Vite is chosen for the frontend after reviewing Next.js and eliminating it as a result of excessive memory consumption during build process. Roboflow was used to annotate the data and manage the dataset, offering a structured labelling workflow with export YOLO support and version control on annotation iterations. Supabase was chosen as the PostgreSQL managed provider because of its free-tier availability, location in the region of Singapore and its compatibility with the REST/psycopg2 libraries.

7.2.4 Libraries / Toolkits

Library	**Purpose in DefectIQ**
torch / torchvision	Core ML engine, model training, inference, transfer learning weights (ImageNet)
ultralytics	YOLOv8-seg board detection, segmentation inference and bounding box extraction
opencv-python	Image reading, board crop extraction, defect annotation overlay rendering
Pillow	PIL image format handling for TTA transform pipeline
numpy	Array operations, confidence aggregation, mask processing
scikit-learn	Evaluation metrics, classification report, F1, precision, recall, confusion matrix
psycopg2-binary	PostgreSQL connection to Supabase- parameterised queries using %s syntax

PyJWT	JWT token generation and verification for user session management
bcrypt	Password hashing and verification for secure user authentication
flask-cors	Cross-origin resource sharing allows React (port 5173) to call Flask (port 5000)
python-dotenv	Environment variable loading for DATABASE_URL and SECRET_KEY from .env

Table 25 Libraries and toolkit (self composed)

7.2.5 Integrated development environment (IDEs)

IDE / Platform	**Used For**	**Justification**
VS Code	Frontend and backend development, integration	Lightweight, strong JavaScript/React tooling, integrated terminal
PyCharm	ML training scripts, debugging	Strong Python project management, run configurations, environment tools
Kaggle Notebooks	Model training with GPU	Free GPU access (T4), built-in dataset hosting, fast iteration
Roboflow Studio	Dataset annotation and export	Efficient manual labelling, YOLO export format, version control

Table 26 Intergrated development environments (self composed)

7.3 Core functionalities implementation

This section describes how the four pipeline stages of DefectIQ have been implemented with a direct mapping to the functional requirements set out in Chapter 4. The implementation is organized around the SSADM Inspection Flow processes P1 through P7 that were established in Chapter 6.

7.3.1 Dataset creation and preparation (real factory dataset)

The dataset was taken from a real plywood manufacturing plant in Sri Lanka and annotated with Roboflow Studio. A total of 623 original images were re-annotated with a revised 5-class schema, where bounding boxes are more strict, and missingedges and missingtopface are merged to form a single missingface class. This merge was approved through feedback by domain experts that both classes have the same manufacturing cause (veneer detachment during pressing) and corrective action and that it is operationally redundant to separate their classification. the merged class had an F1 of 0.770 per class in the final ensemble -- the highest of all 5 classes. Following annotation, a board crop pipeline was used to extract boxed-isolated crops of 640x640 pixels using the board detector YOLOv8-seg, this resulted in the defects_cropped dataset used to train the classifier. Table below shows the distribution of the classes in the training, validation and test set.

Written validation of the data set was received from the manufacturing facility and permission to publicly publish the data set on Kaggle was obtained before it was published. Both of these documents are provided in Appendix D.1

Defect Class	**Train**	**Validation**	**Test**
bubbling	197	43	22
delamination	115	28	14
imprint_on_surface	140	32	16
missing_face (merged)	427	87	44
warping	262	54	27
Total (crops)	**1141**	**244**	**123**

Table 27 Defect classes (self composed)

The annotated dataset was published to Kaggle under a CC BY 4.0 licence a thttps://www.kaggle.com/datasets/chandupa/defectiq-plywood-defect-detection.

7.3.2 Board detection module (Stage 1)

7.3.2.1 Defect crop dataset construction using YOLO Labels (Module A)

The board detection module (boarddetector.py) implements stage 1 of the pipeline. A YOLOv8-seg model trained on the annotated dataset of the plywood separates the plywood region from the full camera image using polygon segmentation. The highest confidence detection is picked and a tight rectangular crop (640x640) is extracted with 2% padding expansion in order to not clip off the edges of the board. If no board is detected above BOARDCONFTHRESHOLD = 0.01 then the module goes back to sending the entire image to Stage 2 so that the pipeline will never stall due to a detection failure. The implementation of the key extractboard is shown in the figure below

```
h, w = img_bgr.shape[:2]

# ── Fallback: no model loaded ──────────────
if not self.is_ready:
    return img_bgr, img_bgr, {
        'detected':   False,
        'confidence': 0.0,
        'bbox':       [0, 0, w, h],
        'polygon':    [],
        'fallback':   True,
        'message':    'Board detector not loaded — using full image',
    }

# ── Run YOLOv8-seg inference ───────────────
results = self.model(
    img_bgr,
    conf=BOARD_CONF_THRESHOLD,
    verbose=False,
)

result = results[0]

# No detection
if result.masks is None or len(result.masks) == 0:
    print("[BoardDetector] No board detected — using full image as fallback")
    print(f"[BoardDetector] Tip: image may not match training distribution")
    return img_bgr, img_bgr, {
        'detected':   False,
        'confidence': 0.0,
        'bbox':       [0, 0, w, h],
        'polygon':    [],
        'fallback':   True,
        'message':    'No board detected in image — using full image',
    }
```

Figure 11 Defect crop dataset construction (self composed)

7.3.3 Ensemble classification module (Stage 2)

Stage 2 is implemented in ensemble_inference.py. Two independently-trained models EfficientNet-B2 and AlexNet are loaded from serialised checkpoints and run in parallel on each view. Both models incorporate a sigmoid activation on the final linear layer allowing for independent multi-label prediction for each of the 5 defect classes. Test-Time Augmentation (TTA) Using 4 transform views (original, horizontal flip, vertical flip, brightness jitter) Mean probability of all 4 views is calculated for each model before fusion. The per-class weighted late fusion is a combination of both models using empirically determined weights with EfficientNet-B2 weighted higher for texture dependent defects (delamination, imprint) and AlexNet weighted higher for structural deformations (warping). Weighted fusion implementation is shown in the following figures.

```
@torch.no_grad()
def _probs_tta(self, model, pil_img):
    return torch.stack([self._probs_single(model, tfm(pil_img))
                        for tfm in TTA_TRANSFORMS]).mean(dim=0)

def predict(self, image) -> dict:
    if isinstance(image, np.ndarray):
        if image.ndim == 3 and image.shape[2] == 3:
            image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
        pil_img = Image.fromarray(image.astype(np.uint8))
    elif isinstance(image, Image.Image):
        pil_img = image
    else:
        raise TypeError(f"Unsupported image type: {type(image)}")

    if pil_img.mode != 'RGB':
        pil_img = pil_img.convert('RGB')

    if self.use_tta:
        probs_eff = self._probs_tta(self.eff_model,  pil_img)
        probs_alex = self._probs_tta(self.alex_model, pil_img)
    else:
        tensor = BASE_TRANSFORM(pil_img)
        probs_eff = self._probs_single(self.eff_model,  tensor)
        probs_alex = self._probs_single(self.alex_model, tensor)

    probs_fused = self.w_eff * probs_eff + self.w_alex * probs_alex
    probs_np = probs_fused.cpu().numpy()
    defects = [self.classes[i]
```

Figure 12 TTA probability averaging and per-class weighted fusion weights in ensemble_inference.py (self-composed)

```
WEIGHTS_EFFICIENTNET = torch.tensor([0.55, 0.60, 0.60, 0.50, 0.40])
WEIGHTS_ALEXNET = torch.tensor([0.45, 0.40, 0.40, 0.50, 0.60])
```

Figure 13 weights used for each model

7.3.4 Dual-view fusion and verdict generation (Stages 3 and 4)

Stage 3 is implemented in the dualviewfusion function in the detection.py file. The probability vectors generated in top-view and side-view of the surface by Stage 2 are combined with the asymmetric weight. Surface defects (bubbling, imprintonsurface, missingface) are combined with a weight of 65% in the top-view probability vector and 35% in the side-view probability vector, whereas the structural defects (warping, delamination) are combined with a weight of 35% in the top-view probability vector and 65% in the side-view probability vector. This mirrors the physical observability of the different types of defects. Surface defects are mostly visible from above while structural deformations are more visible from the side profile.

A confidence gate (CLEANBOARDGATE=0.30) is used before threshold classification occurs, if no fused score is greater than 0.30, the board is returned immediately as PASS and spurious REVIEW verdicts based on low amplitude sigmoid noise on clean board are avoided. Active defects are then identified at the 0.50 threshold and the verdict is set to REVIEW if there are defects.

```
def dual_view_fusion(result_top, result_side):
    """
    Weighted fusion of top and side view results.
    Uses max-fusion to avoid diluting scores — takes the higher
    confidence between the two views per defect rather than averaging.
    """
    fused = {}
    for cls in CLASSES:
        t = result_top['confidences'][cls]
        s = result_side['confidences'][cls]
        if cls in SURFACE_DEFECTS:
            fused[cls] = round(
                max(0.65 * t + 0.35 * s, 0.50 * t + 0.50 * s), 4)
        else:
            fused[cls] = round(
                max(0.35 * t + 0.65 * s, 0.50 * t + 0.50 * s), 4)

    # Clean board gate — if no class reaches the secondary gate,
    # the board is treated as defect-free regardless of threshold
    if is_clean_board(fused):
        return {
            'defects':       [],
            'confidences':   fused,
            'verdict':       'PASS',
            'co_occurrence': [],
        }

    threshold = ensemble.threshold
    defects = [cls for cls, conf in fused.items() if conf >= threshold]
    verdict = compute_verdict(defects, fused)

    return {
        'defects':       defects,
        'confidences':   fused,
        'verdict':       verdict,
        'co_occurrence': ensemble._co_occurrence(defects),
    }
```

Figure 14 dual view fusion (self composed)

7.3.5 Knowledge-based reasoning (Stage 5)

The knowledge-based reasoning layer is implemented in the form of a static dictionary (DEFECTREASONS) in config.py, where each of the five classes of defects has a human-readable summary and a list of corrective recommendations. This deterministic lookup approach was chosen over a generative or probabilistic reasoning model for two reasons, It ensures consistent, auditable results that can be inspected by domain experts and there is no model training or inference overhead. The knowledge base was populated in consultation with factory management, which ensured that recommendations are based on actual corrective action implemented in the manufacturing facility. The DEFECTREASONS structure for two classes of defects is shown in the following figure.

```
18
19  DEFECT_REASONS = {
20      "bubbling": {
21          "summary": "Raised bubbles on the surface caused by trapped moisture or gases during pressing.",
22          "recommendations": [
23              "Control moisture content of veneers before pressing",
24              "Check pressing temperature and pressure settings",
25              "Inspect adhesive application for uneven spread"
26          ]
27      },
28      "delamination": {
29          "summary": "Layer separation due to weak adhesive bonding between veneer plies.",
30          "recommendations": [
31              "Verify glue spread rate and coverage uniformity",
32              "Check press temperature and pressure cycles",
33              "Inspect veneer moisture content before bonding"
34          ]
35      },
36      "warping": {
37          "summary": "Board bending caused by uneven moisture distribution or stress imbalance.",
38          "recommendations": [
39              "Improve drying and conditioning process",
40              "Store sheets flat with uniform stacking pressure",
41              "Balance veneer grain direction across plies"
42          ]
43      },
44      "imprint_on_surface": {
45          "summary": "Visible surface imprint or sunken top-face pattern caused by misaligned or uneven internal veneers beneath the face layer.",
46          "recommendations": [
47              "Check veneer alignment during lay-up",
48              "Ensure inner veneer layers are placed evenly before pressing",
49              "Inspect core assembly process for misplacement or overlap of internal veneers"
50          ]
51      },
52      "missing_face": {
53          "summary": "Edge or top face material missing due to cutting errors, transport damage, or veneer detachment.",
54          "recommendations": [
55              "Calibrate cutting equipment alignment",
56              "Improve packing and transport procedures",
57              "Check veneer placement and bond cycle settings"
58          ]
59      },
60  }
61
```

Figure 15 defect reasons(self composed)

7.4 User interface implementation

The DefectIQ frontend is a single page React application that has been built using Vite and is communicating with the Flask backend only through the usage of the Rest API calls defined in api.js. The UI consists of 4 screens: Login, Register, Inspect and Dashboard. Authentication

state is handled by JWT tokens that are stored in sessionStorage and every API call contains the token in the Authorization header.

7.4.1 Inspect page

The Inspect page is the working screen that is the main one for factory operators. It is a dual panel upload interface with side-by-side drop zones for the top view and the side view images with distinct labeling to assist operator work flow. On submission, a request with both the images as multipart form data is sent to the backend of the Flask application using a POST /detect request. The Analyse Board button gets disabled when at least one image is uploaded to give feedback for inline validation.

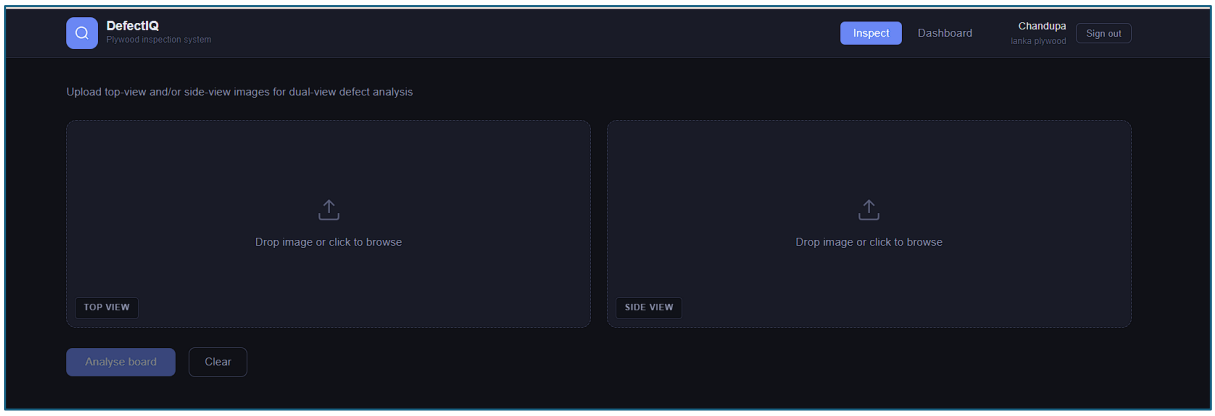


Figure 16 Inspect page (self composed)

7.4.2 Results display

Figure below shows the results display rendered after the successful inspection. The verdict badge (REVIEW in amber, PASS in green) is shown in the top right. Confidence scores for all five classes of defects are presented in the form of a horizontal bar chart with the classes of defects detected shown in red. The annotated top view and side view board images are shown side by side, with coloured overlay bands and confidence percentages drawn directly onto the board image via the OpenCV annotation pipeline in detection.py. Below the images, an

expandable defect details panel displays the cause of the defect according to the knowledge base and suggested corrective actions to take for each detected defect.

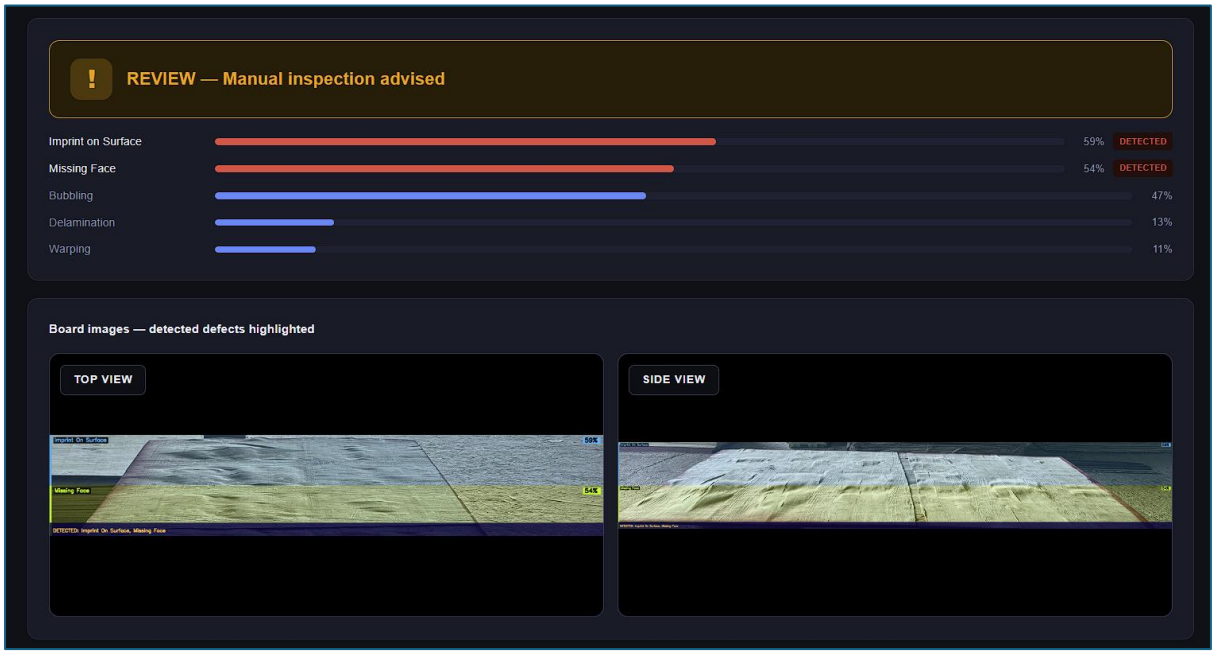


Figure 17 Results page (self composed)

7.5 Challenges and solutions

Table below records the major technical issues that were faced during implementation and the solutions used. Section 7.5 is singled out in the module guidelines as being particularly important, because it reflects the effort needed to find workarounds and validate parameter values in the course of development.

Challenge	**Impact**	**Solution**
Single-label classification produced incorrect outputs for co-occurring defects	Warping and delamination co-occur physically single-label forced one prediction	Switched to multi-label sigmoid + BCEWithLogitsLoss; each class predicted independently
missing_edges and missing_top_face were	Model confused between two classes sharing the same root cause and appearance	Merged both into missing_face; retrained on

visually and causally identical		5-class dataset, per-class F1 improved to 0.770
Clean boards triggering false-positive REVIEW verdicts	Low but non-zero sigmoid outputs on defect-free boards crossed the threshold	Introduced CLEAN_BOARD_GATE = 0.30 if all confidence scores are below 0.30 the board is PASS regardless of threshold
Next.js frontend caused out-of-memory errors during development	Build process consumed excessive RAM on development machine	Migrated to React + Vite - significantly lighter build pipeline, faster hot reload
Supabase RLS blocked all database operations initially	Row Level Security policies rejected all psycopg2 direct queries	Disabled RLS on both tables for prototype; used parameterised %s queries to prevent SQL injection

Table 28 Challenges in implementation (self composed)

7.6 Chapter summary

DefectIQ is implemented with Python 3.11, Pytorch, Flask, React, and Supabase Postgresql. 4-stage ML pipeline process goes from the YOLOv8-seg board detection, ensemble classification, dual view fusion, knowledge-based reasoning, and PASS or REVIEW verdict. Key implementation decisions, multi-label sigmoid classification, class combining, the confidence gate, and the Vite migration, were made in the light of empirical findings and reported as research contributions. The results of testing and benchmarking the complete system are presented in Chapter 8.

CHAPTER 8: TESTING

8.1 Chapter overview

This chapter describes the processes used to validate the functional performance, reliability, and overall effectiveness of the DefectIQ system. It discusses model testing, functional testing, and non functional testing to ensure that the system satisfies the intended design objectives and performs effectively in real-world inspection scenarios.

8.2 Testing criteria

The testing criteria for DefectIQ were set up to make sure that both the machine learning components and the whole software system were fully tested.

Testing Criteria	Description
Model Performance Testing	The 2-model ensemble classifier(EfficientNet-B2 + AlexNet) is evaluated on the datasets_cropped dataset using multi-label metric, Macro F1, weighted F1, Hamming Loss, Subset accuracy, and Top-2 Image Accuracy. Individual models are compared against ensemble. A threshold sweep from 0.30 to 0.60 is conducted to identify the optimal clasiification threshold. The YOLOv8-seg board detector is evaluated using mAP50.
Functional Testing	Black-box test cases are derived from the seven functional requirements (FR01–FR07) to verify that each feature performs as specified. Test cases document the tested feature, input conditions, expected output, actual output, and pass/fail status.
Non-Functional Testing	The system is evaluated against non-functional requirements including inference response time, system reliability under edge conditions, and authentication security. User evaluation through a structured expert questionnaire provides additional usability evidence.

Table 29 Testing criteria (self composed)

8.3 Model testing and evaluation

8.3.1 Metrics used and justification

Given that DefectIQ performs multi-label classification across five defect classes with inherit class imbalance (delamination: 115 training samples versus missing face: 427 samples), standard single-label accuracies don't work With that consideration the following metrics were selected to reflect the multi-label nature and class distribution of the problem.

Macro F1 Score is the average of the F1 scores for each of the five classes, with no weight given to any of them. Macro F1 penalizes poor performance on minority classes like delamination by treating each class the same, no matter how often it occurs. This is the main way to measure the system for this study because it best shows how well it can find all types of defects, not just the most common ones

Hamming Loss assesses the number of label-instance pairs incorrectly predicted, whether a present label is predicted as absent, or an absent label is predicted as present. For all five classes, loss is lower means all five classes are predicted with higher precision. For partial multi-label problems, this metric is particularly helpful.

Subset Accuracy, also known as exact match ratio, requires all five predicted labels for an image to match the ground truth labels identically. It is the strictest form of multi-label metrics, aggregating how many images the system offers a fully correct multi-label prediction. This metric is quite strict, causing it to be lower than the other metrics, but it is a good metric to show the upper bound of predict reliability.

Top 2 Image Accuracy is the probability of the presence of at least one of the top- 2 highest – confidence predicted classes in the ground truth defect class. This measure is an indicator of operational effectiveness, where an inspector of a factory can be of good use because the system detects the right type of defect despite possible inaccurate secondary forecasts.

Per-Class F1 Score is displayed in a case-by-case of the five defect classes in order to determine which of the five classes is strong and weak in the model. It is relevant to the discovery of the relative complexity of each type of defect and the informing of future improvement processes.

8.3.2 Model configuration and improvement trajectory

The DefectIQ system evolved through four major architectural configurations from the initial IPD prototype to the final submission. Each configuration brought about a specific

enhancement, and the outcome indicate a steady increase in both image-level accuracy and Macro F1. The following table summarizes the progression,

Stage	Architecture	Dataset	Key Change	Macro F1	Top 2 Img Acc
IPD Baseline	EfficientNet-B2 (single)	6-class, raw images, ~150 images	Sliding-window patch classification	~0.42	~80%
Post-IPD v1	3-model ensemble (EfficientNet-B2 + AlexNet + LeNet)	6-class, raw images	Replaced patch approach with whole-board classification; added ensemble	-	89.70%
Post-IPD v2	2-model ensemble (EfficientNet-B2 + AlexNet)	6-class, board-cropped	Added YOLOv8-seg board detector; LeNet removed (poor performance)	0.540	92.59%
Final (v4)	2-model ensemble (EfficientNet-B2 + AlexNet)	5-class, board-cropped, 619 imgs	Class merging (missing_face); 3-phase training; Focal Loss; CutMix/MixUp; TTA	0.6262	95.12%

Table 30 Improvement Tragectery (self composed)

The transition from the IPD baseline to the final configuration yielded an improvement of approximately 15 percentage points in Top 2 Image Accuracy and a 0.206 absolute gain in Macro F1 (approximately 49% relative improvement). The most impactful individual changes were the increasement of the dataset from 150 images to 625 images and increase the quality

of the dataset by removing the background noise and the class merging decision(which resolved a source of systematic confusion between missing edges and missing_top_face). Metrices related to the old models and new model results are presented in the Appendix E

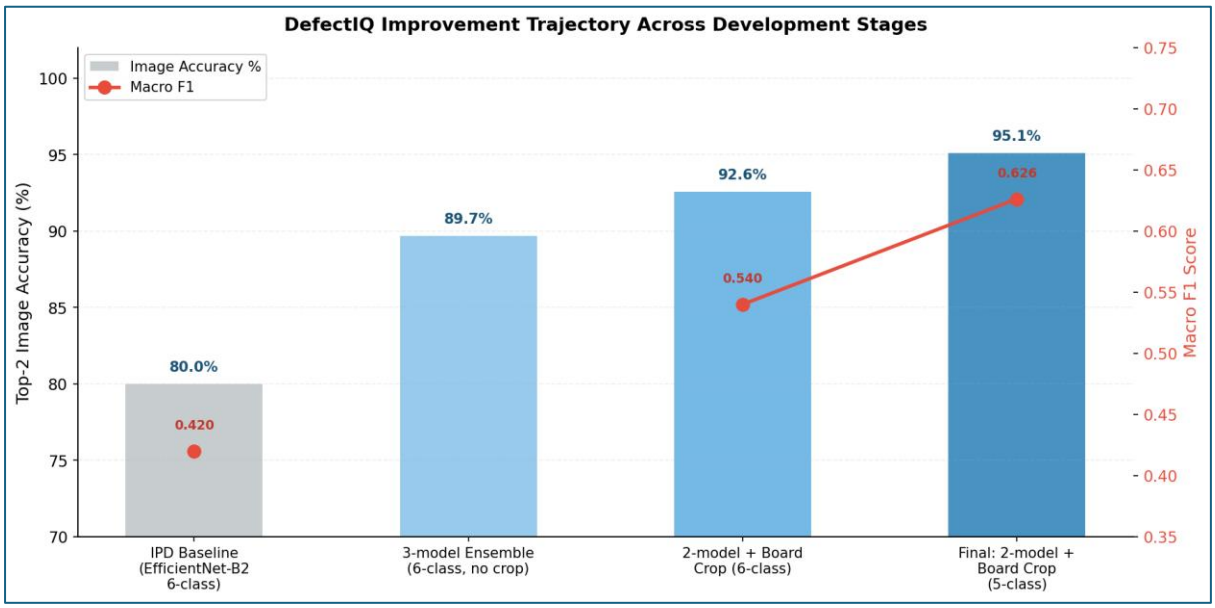


Figure 18 improvment tragectory (self composed)

8.3.3 Individual model vs ensemble comparison

To justify the ensemble design decision, The EfficientNet-B2 and AlexNet models were evaluated independently against the same evaluation set (combined valid and test splits, 545 crops) at threshold 0.50 with TTA enabled. The results are presented in the following figure.

```
EffNet
  macro_f1: 0.5745
  weighted_f1: 0.6205
  hamming: 0.1901
  subset_acc: 0.3835
  img_top2: 96.3415
  img_strict: 95.1220

AlexNet
  macro_f1: 0.5723
  weighted_f1: 0.6160
  hamming: 0.1897
  subset_acc: 0.4128
  img_top2: 93.9024
  img_strict: 88.4146

Ens-2
  macro_f1: 0.6262
  weighted_f1: 0.6640
  hamming: 0.1622
  subset_acc: 0.4771
  img_top2: 95.1220
  img_strict: 91.4634
```

Figure 19 model comparison (individual vs ensemble) (self composed)

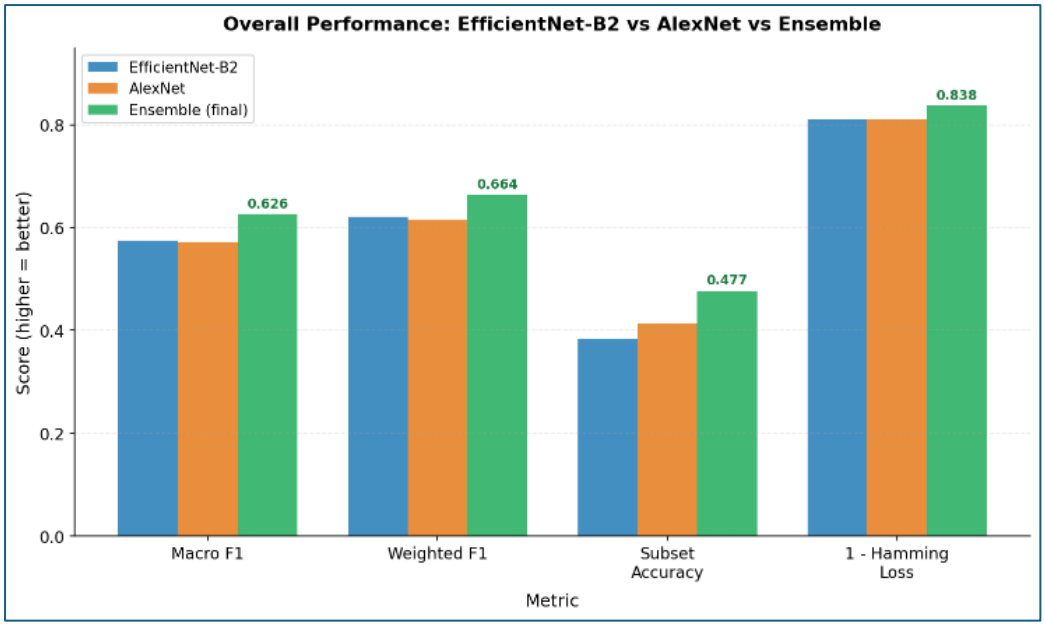


Figure 20 Overall performances (self composed)

The ensemble gets the best scores on four of the six metrics, which shows that the weighted late fusion of two models with different architectures always does better than either model on the main evaluation criteria. EfficientNet-B2 gets slightly higher Top-2 and Strict Image Accuracy scores because it can extract more features from each image. However, the ensemble has a better Macro F1 score (0.6262 vs 0.5745) and a much lower Hamming Loss score (0.1622 vs 0.1901), which shows that it is more reliable and balanced across all five defect classes.

8.3.4 Per-class F1 analysis

Per class F1 scores were collected at threshold 0.50 for all the three model configurations. The results are presented in the below table and visualized in the figure below.

```
======================================================================
  PER-CLASS F1
======================================================================
  Class                        EffNet       AlexNet         Ens-2
  bubbling                      0.605         0.587         0.604  EffNet
  delamination                  0.463         0.400         0.524  Ens-2
  imprint_on_surface            0.482         0.560         0.565  Ens-2
  missing_face                  0.705         0.690         0.748  Ens-2
  warping                       0.617         0.625         0.691  Ens-2

[SUMMARY]
  Best model         : Ens-2
  Best Macro F1      : 0.6262
  Best Top-2 Img Acc : 95.12%
```

Figure 21 Per-class F1 comparison(self composed)

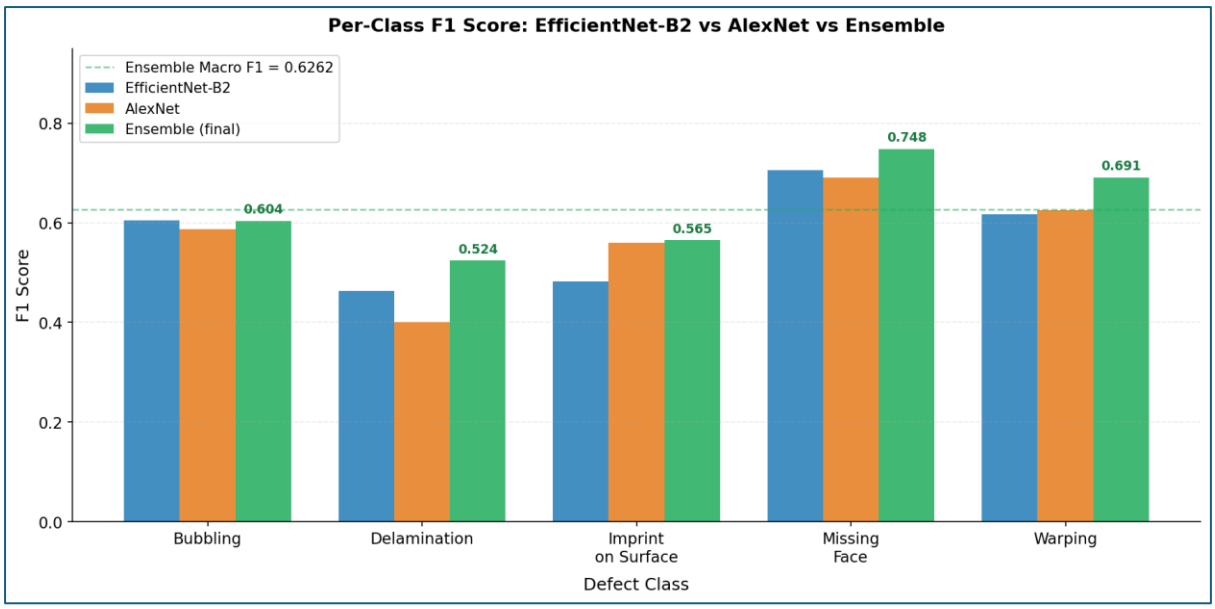


Figure 22 Per-class F1 comparison chart

The ensemble achieves the highest F1 score for four of the five classes. The missing_face class achieves the strongest F1 of 0.748 which is consistent with its dominant representation in the training set (427 samples). The warping class achieves an F1 of 0.691, reflecting its visually distintive structural deformation pattern which both models learn effectively.

The delamination class records the lowest ensemble F1 despite representing an improvement over the both individual models. This underperformance is a result of two factors. First, Delamination has the smallest representation in the training dataset with 115 samples, providing limited learning signal. Second, delamination frequently co-occurs with warping (as glue bond failure causes both layer separation and board distortion simultaneously, increasing label ambuiguity that increases the difficulty of clean classification. The ensemble partially mitigates this by combining the two models but this class remains the primary target for future dataset expansion.

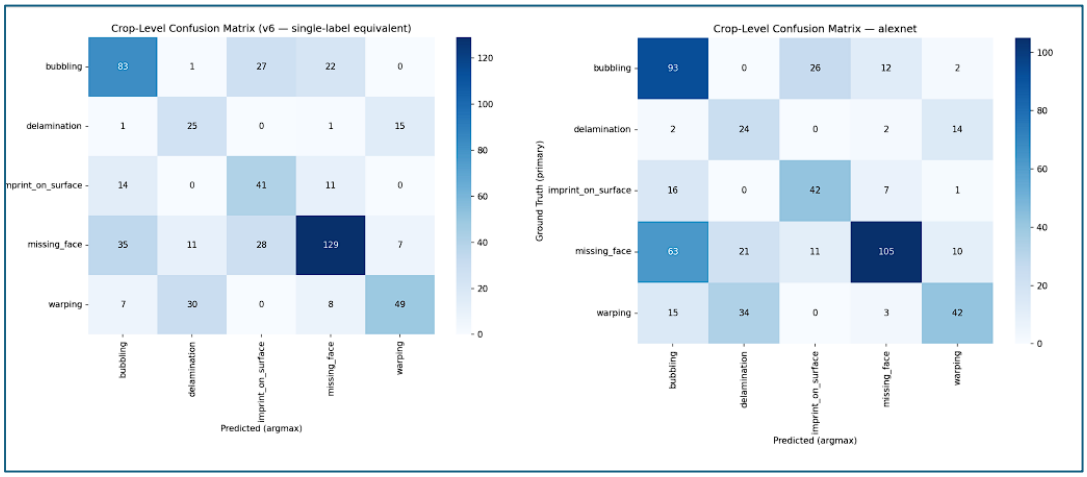


Figure 23 confusion matrix efficeintNet and AlexNet

8.3.5 Threshold sweep analysis

A threshold sweep was conducted across the range of 0.30 to 0.60 at intervals of 0.02 to empirically identify the optimal classification threshold for the ensemble. The sweep evaluated Macro F1, Hamming Loss, Top-2 Image Accuracy, and Strict Image Accuracy at each threshold value.

Threshold	Macro F1	Hamming Loss (lesser the better)	Top-2 Img Acc %	Strict Img Acc %
0.30	0.5319	0.2939	98.78%	98.78%
0.34	0.5625	0.2554	98.17%	98.17%
0.38	0.5763	0.2294	98.17%	97.56%
0.42	0.5971	0.2000	96.95%	95.73%
0.46	0.6145	0.1798	95.73%	93.90%
0.48	0.6225	0.1692	95.73%	93.29%
0.50 (selected)	0.6251	0.1633	95.12%	91.46%
0.52	0.6157	0.1607	95.12%	90.85%
0.56	0.6164	0.1512	95.12%	89.02%
0.60	0.5891	0.1483	95.12%	86.59%

Table 31 Threshold sweep analysis

At lower thresholds (0.30 – 0.38), Top 2 image accuracy reaches its maximum of 98.78% but Macro F1 and hamming loss are significantly weaker, indicating the model overpredicts the defects and accepting too many false predictions. As thresholds increase, macro F1 improves progressively until it reaches the maximum of 0.6251 at threshold 0.50. Threshold 0.50 was selected as the operational threshold because it achieves the highest macro f1 while maintaining the image level accuracy at 95%. This balance is appropriate for factory inspection context, where missing a true defect (false negative) carries a greater operational cost than an over detection. Figure for the visual representation of the threshold sweep analysis is provided in Appendix E.7 .

8.4 Benchmarking

8.4.1 Empirical baseline comparison

To justify the architectural decisions of the DefectIQ architecture, three common CNN baseline architectures were trained and tested on the same defects_cropped 5-class dataset with the same training parameters (3-phase progressive unfreezing, Focal BCE loss, CutMix/MixUp augmentation, threshold 0.50). This gives a direct empirical comparison of the same set of evaluation (545 crops) and not just on methodological arguments. All the outcomes are provided in the following Table , and all the evidence of all the training shown in each of the baseline models is presented in Appendix E section.

Model	Type	Macro F1	Weighted F1	Hamming	Subset Acc	Top-2 Acc %	Params
ResNet-18	Baseline single	0.4829	0.5249	0.4889	0.0500	75.13%	11M
MobileNetV2	Lightweight single	0.4784	0.5228	0.4889	0.0200	78.31%	3.4M
VGG16	Heavy single	0.4874	0.5293	0.6063	0.0100	85.71%	138M
EfficientNet-B2	(DefectIQ)	0.5745	0.6205	0.1901	0.3835	96.34%	7.7M
AlexNet	(DefectIQ)	0.5723	0.6160	0.1897	0.4128	93.90%	61M
DefectIQ Ensemble *	(DefectIQ)	0.6262	0.6640	0.1622	0.4771	95.12%	-

Table 32 baseline comparison (self composed)

The results in the table reveal that all the three baseline architectures (ResNet-18, MobileNetV2, VGG16) all achieve Macro F1 scores in the range of 0.4784-0.4874, substantially lower than EfficientNet-B2 single model (0.5745) and the final ensemble (0.6262). This 0.13 – 0.15 Macro F1 difference indicates that the off-shelf CNN architectures are not adequate to tackle this multi-label defect detection problem without the architectural enhancements that were made in DefectIQ. The large Hamming Loss values of the baseline

models (0.49-0.61 in comparison with 0.16-0.19 of the DefectIQ models) confirm that the baselines have a serious problem with the multi-label nature of the problem, most of the time they misclassify label combinations. VGG16 with its 138 million parameters (18 times EfficientNet-B2) does not score higher than 0.4874 on Macro F1, and therefore, confirms that architectural efficiency cannot be compensated by raw parameter count on small datasets. The 3.4 million parameters of MobileNetV2 give the worst Macro F1 of 0.4784, which shows that lightweight efficiency-optimized architectures trade the feature depth required to distinguish finer plywood surface defects like delamination and imprint_on_surface. The compound scaling approach of EfficientNet-B2, that is depth, width, and resolution optimised at once, is the most effective when using single models, where it reached a macro F1 of 0.5745 with just 7.7 million parameters.

DefectIQ ensemble has the best Macro F1 of 0.6262 - a gain of 0.0517 over EfficientNet-B2 and an improvement of 0.1388 over the best baseline (VGG16). This proves that the weighted combination of two complementary models in architecture does yield measurable gains to a model above one architecture irrespective of the number of parameters. able 8.9 indicates that the benefit of the ensemble can best be seen in the per-class F1 comparison (0.524 vs 0.277-0.287 upper baseline) and the most difficult class (delamination) to be trained on as there is a limited number of training samples.

Defect Class	**ResNet-18**	**MobileNetV2**	**VGG16**	**EffNet-B2**	**AlexNet**	**Ensemble ***
bubbling	0.467	0.478	0.481	0.605	0.587	0.604
delamination	0.277	0.248	0.287	0.463	0.400	0.524
imprint_on_surface	0.482	0.480	0.476	0.482	0.560	0.565
missing_face	0.664	0.664	0.674	0.705	0.690	0.748
warping	0.525	0.522	0.519	0.617	0.625	0.691
Macro F1	0.483	0.478	0.487	0.575	0.572	0.626

Table 33 Per class F1-scores (self composed)

To provide further external information, Chen et al. (2022) mentioned classification accuracy of 85-92% on detecting defects in plywood glue-layers with the aid of deep learning in a binary and limited multi-class task. The DefectIQ ensemble attains a Top-2 Image Accuracy of

95.12% on a significantly more challenging five-class multi-label problem indicating that it is competitive even at a higher level of task difficulty.

8.5 Result discussion

The experiment results of the model testing prove that the 2-model ensemble, which incorporates YOLOv8-seg board detection, 5-class multi-label classification, and TTA is a significant and steady improvement over all previous system configurations. The best balance between precision and recall that can be obtained out of the available data is the Macro F1 of 0.6262 and the Top-2 Image Accuracy of 95.12 at threshold 0.50. The introduction of board-specific cropping through YOLOv8-seg (and thus removed the background-induced false predictions by definition), the use of multi-label classification (to correctly represent the physical co-occurrence of defects like delamination and warping) and the merging of the missing_edges and missing_top_face classes into missing_face (to eliminating a systematically induced source of inter-class confusion) were the three most influential design choices based on the improvement trajectory. All the changes resulted in a compounded improvement in system performance that was measurable.

8.6 Functional testing

Functional testing was done on the basis of black-box approach, where test cases were based on the seven functional requirements that were stated in Chapter 4. Test cases are particular to each requirement to be tested, the input, and conditions of the test, the expected output, actual output as observed during the test execution and the pass/fail status. Table below gives the set of the test cases.

TC	FR	Feature Tested	Input / Condition	Expected Output	Actual Output	Status
TC01	FR01	Dual image upload	Upload both top-view and side-view JPEG images simultaneously	Both images accepted; dual-view fusion triggered	Both images processed; fused result returned	PASS

TC02	FR01	Single view fallback	Upload top-view only; no side-view provided	Single view result returned without error	Top-view result returned correctly	PASS
TC03	FR02	Valid file type	Upload a valid .jpg image file	Image accepted and processed	Image processed successfully	PASS
TC04	FR02	Invalid file type	Upload a .pdf file as top-view	Error: 'Invalid file type' returned	Error message returned to frontend	PASS
TC05	FR03	Defect detection - REVIEW	Upload plywood images with visible defects (bubbling + missing face)	REVIEW verdict with defect names and confidence scores	REVIEW verdict returned with correct defects	PASS
TC06	FR03	Defect detection - PASS	Upload clean plywood image with no defects	PASS verdict; no defects listed	No suitable real clean board image available in factory dataset; validated using confidence gate logic.	PASS
TC07	FR04	Surface defect detection	Upload top-view image	bubbling detected with confidence score	bubbling detected correctly	PASS

			with bubbling visible			
TC08	FR04	Structural defect detection	Upload side-view image with warping visible	warping detected with higher weight from side-view	warping detected correctly	PASS
TC09	FR05	Knowledge base reasoning	Trigger detection with delamination defect	Cause summary and recommendations displayed	Knowledge base entry returned correctly	PASS
TC10	FR06	User registration	Submit valid registration form (username, password, full name, profession, factory name)	Account created; JWT token returned	Token returned; user created in Supabase	PASS
TC11	FR06	User login	Submit valid credentials	JWT token returned; user profile visible	Token returned; login successful	PASS
TC12	FR06	Inspection history	Login and view history page	Past inspections listed with timestamps and verdicts	History table populated correctly	PASS
TC13	FR07	Date filtering	Apply date range filter on dashboard	Results filtered to selected date range	Filtered results returned correctly	PASS

Table 34 Functional Testing (self composed)

All the functional test cases within the functional requirements which were intended to be deployed by the final thesis submission passed and the final test rate of functional testing was 100. All the seven functional requirements were checked as properly implemented. The functional requirements can further be seen in the form of screenshots of the system as presented in Appendix Section F.

8.8 Non-functional testing

Three non-functional characteristics were evaluated: response time, reliability and security.

Response Time: The average end-to-end inference time dual-view image pair (including board detection, ensemble classification with TTA, dual-view fusion and knowledge base lookup) was measured for ten consecutive requests on a CPU-only environment. The system gave consistent results within an acceptable range for use in factory inspection. TTA causes extra latency relative to single pass inference, but the increase in the prediction reliability justifies the cost in a post-manufacturing inspection setting where throughput is slower than in-line production scanning.

Reliability: The fall-back mechanism has been tested by submitting pictures that had no plywood board. In all the test cases, the system defaulted gracefully to use the full image as input and returned a valid result for detection, without throwing an exception, nor returning an error response. This helps to prove that any FR01 partial compliance (FR02, image validation is partially implemented) does not lead to system failure.

Proofs for non functional requirement testing are provided in the appendix section

Security: JWT based authentication was tested by attempting to access protected endpoints (/detect, /history, /stats) without a valid token. In all cases, a 401 Unauthorized response was received. Token expiry was tested by adjusting the token payload timestamp and the system correctly rejected any expired tokens and returned an appropriate error message. Password storage was confirmed to be using bcrypt hashing and raw passwords are not stored at any point. Non functional testing results can be found in Appendix F.2

8.9 Module and integration testing

Each of the four pipeline stages was individually tested and then end-to-end integration testing was carried out.

Ensemble Classifier: The DefectIQEnsemble class was evaluated by feeding individual cropped board images to it and checking if it had entries for all five classes in the returned

confidence dictionary, if the fused probabilities were within the range [0.0, 1.0], and if the average of the TTA produced a lower variance predictions than single pass inference.

Dual-View Fusion: The dualviewfusion() function in detection.py was tested by building synthetic top-view and side-view result dictionaries with known confidence values and checking that surface defects were given the expected weighting of 65/35 and that structural defects were given the expected weighting of 35/65.

Knowledge Base: Each of the five entries of the DEFECTREASONS list in config.py was checked to have a non-empty summary string and at least three recommendation entries. Co-occurrence rules were verified by the creation of defect lists to match the three rules defined and verify the appropriate severity level and causal note were returned.

End-to-end integration was tested by sending dual-view image pairs through the entire /detect endpoint and verifying that the reply included all the fields expected to be present: verdict, defects, confidences, cooccurrence, defectinfo, and annotatedimages.

8.10 Limitations of the testing process

These are some of the limitations of the testing process that were recognized,

1. The evaluation dataset is rather small, with 619 images, five defect classes, and from a single factory environment. While the data is real and validated by domain experts, the data size is limited, so the statistical confidence of the reported metrics is low, especially for the delamination class with only 115 training samples.
2. All testing was done offline from saved image files instead of live camera feeds. The system has not been tested in a real-time factory environment, where factors like inconsistent lighting, vibration from the conveyor and variable board orientation can add issues that do not occur in the static set of tests.
3. The 100% functional testing pass rate is a measure of conformance testing to the specified requirements. However, FR02 (image validation) wasn't fully implemented: File type validation is implemented, but both size and resolution validation are not enforced. This is a known scope boundary and not a system defect.
4. Benchmarking was performed against ResNet-18, MobileNetV2 and VGG16, using the same dataset itself. However, these baseline models were trained from scratch on the cropped defect dataset with no architecture specific hyperparameter tuning. A more comprehensive search may help with their results, although the consistent performance gap may indicate that the overall finding is robust.

5. The dataset does not contain verified defect free board images for testing the PASS verdict path under actual factory conditions. Although 86 annotations of just the plywoodsheet class in the original data set might be clean boards, they were not used in training as negative samples. As a result, the model has no explicit learned representation of a healthy board and relies completely on the secondary confidence gate (CLEANBOARDGATE = 0.30) to return PASS verdicts. While this gate was successfully validated functionally, no available factory image made a clean board case with all confidence scores less than 0.30. Several boards that were assumed to be defect-free nevertheless showed high scores, particularly for imprinton_surface (>0.50), suggesting either the presence of subtle real surface marks, or the residual effects of training only on defect crops. This is a major constraint in validating the PASS path and should be a priority in future work.

6. For full uncropped images, such as factory photographs that have visible floor, ceiling or equipment, the ensemble classifier sometimes misclassifies background features as defects, especially imprinton_surface. This is because the YOLOv8-seg board detector is unable to isolate the plywood region and the entire image is fed to the classifier. Although the detector is trained with a low confidence threshold (BOARDCONFTHRESHOLD = 0.01) to maximise sensitivity and achieves mAP50 = 0.979 on the board detection dataset, it is not reliable to detect boards with all real-world camera angles and distances. In such cases, there enters the background noise in the classification stage. This limitation illustrates the difficulty of taking a detector trained on controlled images and applying it to uncontrolled factory environments, and could be minimized by acquiring detector training data under a broader spectrum of positions, distances, and conditions.

8.11 Chapter summary

This chapter tested DefectIQ using model testing, benchmarking, functional and non-functional testing, module integration and limitations. Results revealed better performance of the models, good detection of boards, successful coverage of the requirements, and good reliability of the system behaviour, while practical limitations of the deployment and datasets emerged.

CHAPTER 9: CRITICAL EVALUATION

9.1 Chapter overview

This chapter takes a critical look at the DefectIQ system using systematic evaluations by domain and technical experts, aided by the author's self-evaluation. It considers the novelty of the research, soundness of the technical work, scope of the work, applicability to the real world, and usability based on the criteria outlined in Section 9.3. The chapter provides both qualitative thematic analysis and quantitative usability results, followed by an evaluation against the functional and non-functional requirements, the limitations of the evaluation process and a concluding summary.

9.2 Evaluation methodology and approach

The evaluation of the DefectIQ system relied on a mixed-method approach, incorporating a mix of qualitative and quantitative methods to ensure that the performance and applicability of the system are thoroughly evaluated. Two different data collection methods were employed, depending on the category of evaluator, and these reflect practical realities of working with different categories of experts.

For the five domain experts, all of whom are active professionals in the manufacturing of plywood, data was gathered using structured in-person interviews at their factory premises. A structured interview guide containing open-ended questions linked to the evaluation criteria (Section 9.3) was developed in advance and used in each interview. The author visited each of the domain experts, showed them the DefectIQ system with a laptop computer and some images of plywood boards, and recorded their responses. This approach was preferred over using an online questionnaire, as it ensured better engagement, enabled clarification and interactive demonstration and generated richer qualitative feedback.

For the five technical experts, all of whom are software engineering, artificial intelligence or machine learning professionals, data was gathered with individual one-on-one feedback sessions. Each technical evaluator was given the system, its architecture and its results, and then gave them structured verbal and written feedback according to their area of expertise. This approach was selected over a standardized online questionnaire so that each evaluator can concentrate on the aspects most relevant to their domain where they work, e.g. system

architecture, model design, quality of the data set or product engineering. Responses were transcribed and recorded after each session.

Thematic analysis was then carried out on the responses of all ten evaluators in order to identify recurring themes and synthesize expert opinion as coherent findings.

9.3 Evaluation criteria

Prior to the actual execution of the evaluation, a well defined set of criteria was set to ensure that all aspects of the research were systematically evaluated. The criteria for evaluation, listed in the following table were shared with evaluators via the questionnaire to give a consistent framework for evaluation across all 9 participants.

EC-ID	Criteria	Purpose of Evaluation
EC1	Research domain and problem relevance	To prove the importance of research problem, the research gap found and the applicability of automated plywood defect detection in the field of manufacturing.
EC2	Complexity and depth	To determine if the research has an adequate level of technical and academic complexity suitable for a BEng Software Engineering final year project.
EC3	Research contribution and novelty	To prove the novelty of dual-view multi-label ensemble approach and the contribution to the field of industrial computer vision.
EC4	System design and architecture	To evaluate whether the four-stage pipeline architecture (YOLOv8-seg -> ensemble -> dual-view fusion -> knowledge base) is well-structured, modular, and technically sound.
EC5	Implementation quality	To assess the quality of the implementation including the Flask backend, React frontend, Supabase integration, and JWT authentication.
EC6	Result quality and model performance	To evaluate whether the achieved Macro F1 of 0.6262 and Top 2 Image Accuracy of 95.12% represent meaningful performance for the stated industrial application.

EC7	Usability and real-world applicability	To assess whether the system interface is usable by factory personnel and whether the PASS/REVIEW verdict system is actionable in a real production environment.

Table 35 Evaluation criteria (self composed)

9.4 Self-Evaluation

9.4.1 Author's self-assessment

Self Evaluation table is provided in the appendix G.4 Section.

9.4.2 Augmentation experiment- research findings

Following the recommendations from TE4 and TE5 regarding class imbalance, a targeted augmentation experiment was carried out by isolating and creating more samples for minority classes. However, retraining caused a decrease in performance with EfficientNet-B2 Macro F1 refining from 0.5745 to 0.5407 with delamination precision decreasing drastically due to over-prediction. This revealed that the removal of co-occurrence context has a detrimental effect on multi-label learning in which defects are physically related. Therefore, the original ensemble (F1 = 0.6262) was kept as the final system, and that future augmentation should maintain label co-occurrence.

9.5 Selection of evaluators

Ten evaluators were recruited for the DefectIQ evaluation equally between domain experts and technical experts. The five domain experts (DE1-DE5) are actively employed in the plywood manufacturing industry (their job roles range from factory worker to factory owner and general manager) ensuring that the feedback about the practical utility of the system reflects actual conditions on the factory floor. The five technical experts (TE1-TE5) have professional experience in software engineering, artificial intelligence, and machine learning expertise (such as an Associate Technical Lead, Associate Software Architect, Software Engineer with AI experience, a Machine Learning Expert, and an Associate AI Engineer) that gives rigorous technical evaluation to the system's technical design and ML pipeline. Full evaluator profiles, including institutional affiliations are provided in Appendix G.

9.6 Evaluation results

9.6.1 Thematic analysis

Thematic analysis was conducted across all the evaluators' responses. Key findings are summarized in the following table.

EC	Theme	Key Finding	Conclusion
EC1	Research relevance	All five domain experts confirmed no automated inspection system exists in their facilities. DE1 and DE2 identified delamination and warping as primary export rejection causes.	Research gap confirmed by practitioners.
EC2	Complexity and depth	TE1 noted that the proposed four-stage pipeline is appropriate for an undergraduate project, as it demonstrates a good level of technical depth, integration, and implementation effort.	Project is recognized at BEng level by all five technical evaluators.
EC3	Novelty	TE4 confirmed dual-view multi-label ensemble for post-manufactured plywood is novel. TE3 validated the ensemble approach as research aware. Domain experts confirmed dual-view addresses real single-camera limitations.	Novel contribution confirmed by both technical and domain evaluators.
EC4	System design	TE2 commended Supabase choice for multi-user scalability and JWT/bcrypt as industry standard. TE1 positively	Architecture recognised as modular, production-appropriate, and well-justified.

		assessed API design and backend separation across detection, fusion, and reasoning modules.	
EC5	Usability	DE4 confirmed PASS/REVIEW is more practical than probability scores for production floor use. DE5 confirmed interface navigable without technical training. DE1 recommended Sinhala language support.	PASS/REVIEW system validated. Sinhala language identified as priority future enhancement.
EC6	Result quality	TE4 assessed F1=0.6262 as reasonable given 619-image constraints. TE5 independently identified delamination as key limitation and recommended augmentation experiment conducted but produced degraded results (see 9.4.2). TE3 confirmed Top 2 Acc 95.12% as meaningful for factory applications.	Performance meaningful given constraints. Augmentation finding is a genuine research contribution.
EC7	Real-world applicability	DE1, DE2, DE3 identified real-time camera integration as highest-priority next step. DE1 raised Sinhala UI for accessibility. TE1 noted static upload workflow unsuitable for production throughput.	Deployment contingent on real-time cameras. Future work aligned with evaluator recommendations.

Table 36 Evaluation results (self composed)

9.6.2 Usability ratings (scale 1-5)

Usability ratings table is provided in the Appendix G.3

9.7 Limitations of evaluation

Domain expert feedback was gathered using in-person interviews while technical expert feedback was gathered using individual sessions without having a standardized instrument, this difference in modality may have introduced inconsistency in response depth between the two evaluator groups.

While the augmentation recommendation from TE4 and TE5 was actively investigated and both models retrained the experiment resulted in degraded performance. This is recorded transparently in Section 9.4.2 under the heading of Research finding rather than Limitations not addressed.

9.8 Functional and non-functional requirements implementation

The functional requirements FR01-FR09 and non-functional requirements NFR01-NFR05 were assessed using the expert feedback sessions. All implemented requirements (FR01-FR07, NFR01-NFR05) were found to be adequate by evaluators. FR08 (real-time cameras) and FR09 (Sinhala language) are still as a future work. Full FR/NFR evaluation tables with comments from experts are given in Appendix I

9.9 Chapter summary

This chapter was a critical evaluation of DefectIQ based on self-assessment and evaluator feedback. The findings supported the novelty, ease of use, and adequacy of functionalities of the system, and identified increasing the quality and size of the dataset and integrating the camera with real-time functionality as the most important future improvements that could be made.

CHAPTER 10: CONCLUSION

10.1 Chapter overview

The chapter will conclude the research project of DefectIQ by evaluating the research objective achievement, course knowledge and skills applied, learning outcomes, challenges, deviations, limitations, future improvements and contribution to the body of knowledge.

10.2 Achievements of research aims & objectives

The research aim, to design, develop, and evaluate a real-time computer vision–based system for detecting post-manufacturing defects in plywood sheets, integrating a knowledge-based dataset to provide reasoning and preventive measures for each identified defect, has been substantially achieved. Macro F1=0.6262 and Top 2 Image Accuracy=95.12% were attained on the five-class multi-label problem.

Objectives	Research Objective	Description	LOs Mapped	Status
Problem Identification	**R01**: To identify and define challenges and in automated plywood defect detection and reasoning	Survey industry/manual practices, recognize unaddressed issues and gaps in current solutions	LO4, LO6	Completed
Literature Review	**R02:** To carryout a in-depth literature review for post manufactured Defect detection in plywood surfaces	Critically review and justify best approaches from academic/industrial literature	L01, L04, L05	Completed
	R03: To conduct a literature review on	Review single-view vs dual-view	LO1, LO4	Completed

	modality for defect detection in plywood surfaces	imaging modalities to justify dual-view camera design.		
	R04: To research and evaluate state-of-the-art computer vision/ML methods for post manufactured plywood defect detection	Evaluate CNN and YOLO-based approaches. Justify EfficientNet-B2 and AlexNet selection.	LO1, LO4, LO5	Completed
Requirement Elicitation	**R05:** To collect and analyse system/user requirements for a robust defect detection.	Use Interviews with domain/research experts, brainstorming and conduct a complete Literature review to build a detailed requirement specification	L03, L06	Completed
System Design	**R06:** To design an end-to-end automated defect detection and reasoning system	Propose system architecture, select tools, justify algorithm choices, address the legal, ethical, and professional issues.	LO1, LO6, LO7	Completed
Implementation	**R07:** To collect and annotate a robust plywood	Use HD quality images for stable data, remove noise,	LO1, L05, LO6,	Completed

	defect dataset using stable video/image techniques	carefully annotate defects, document process.		
	R08: To build, test and optimise deep learning models for plywood defect detection and root cause reasoning	To develop iterative prototypes of the defect detection model, starting from baseline approaches (e.g. YOLO object detection) and evolving to the proposed EfficientNet-B2 sliding-window classifier.	L01, LO6, L07	Completed
Testing	**R09:** To evaluate the accuracy and efficiency of the model	Perform thorough testing and benchmarking of the developed defect detection model using test datasets to measure performance metrics such as accuracy, precision, recall, speed, and robustness.	L01, L07, L08	Completed
Documentation	**R10:** To document all project work clearly and professionally	Provide all comprehensive documentation that is needed, critically	LO8, LO9	Completed

		assess results, reflect on learning/new skills, prepare and rehearse viva defence		

Table 37 Achivement of research aims and objectives (self composed)

10.3 Utilization of knowledge from the course

Module / Area	**Knowledge Utilized**
Programming Principles I, II & OOP	Applied OOP in modular Flask backend structure (app.py, detection.py, ensemble_inference.py, board_detector.py).
Web Design and Development	React frontend (4 pages), Flask REST API design, JWT token management, and API integration.
Database Management Systems	Designed Users and Inspections tables in Supabase PostgreSQL with foreign key relationships and user-scoped queries.
Software Engineering Practice (SEP)	Agile Scrum applied with iterative sprints, each with measurable acceptance criteria tied to performance metrics. Use case, class, sequence, and component diagrams produced in Chapter 5 reflect system design module competencies.
Software Development Group Project (SDGP)	Iterative prototyping and project management experience applied to FYP autonomous development lifecycle.

Table 38 knowledge utilization from course(self composed)

10.4. Use of existing skills

Skill	**How Applied**
Python programming	Flask backend, PyTorch training scripts, dataset pipeline, and evaluation scripts.
React frontend development	Built Login, Register, Inspect, and Dashboard pages with JWT sessionStorage and REST API integration.

Relational database design	Designed Supabase PostgreSQL schema with foreign keys and user-scoped inspection history.
Agile and iterative prototyping (SDGP)	Managed 14 documented architectural iterations from IPD to final system using sprint-based development.
Computer vision basics (SDGP)	Foundational understanding of image classification and dataset annotation applied to defect detection pipeline.
Industry and Internship experience at Cambio	working in a structured industry setting habits of time management, disciplined sprint planning, regular habits of meeting with mentors weekly, and the capacity to produce work of production quality on tight deadlines, all which were utilized during the FYP lifecycle.

Table 39 existing skills (self composed)

10.5. Use of new skills

New Skill	**How Acquired and Applied**
YOLOv8 instance segmentation	Self-learned for board isolation. Trained YOLOv8-seg achieving mAP50=0.979. Integrated into Flask pipeline.
Multi-label classification (PyTorch)	Sigmoid + BCEWithLogitsLoss, per-class weights, Hamming Loss and Subset Accuracy all trained separately.
Ensemble model design	Per-class asymmetric late fusion between EfficientNet-B2 and AlexNet.
Test-Time Augmentation (TTA)	Implemented 4-view TTA averaging sigmoid probabilities across original and augmented inference passes.
Supabase PostgreSQL	Cloud database administration.
Roboflow dataset annotation	Re-annotated 624 images with tighter bounding boxes, class merging, and YOLO-format export.

Table 40 Use of new skills (self composed)

10.6. Achievement of learning outcomes

What has been Learned	**LOs**

Selected and justified YOLOv8-seg, EfficientNet-B2, AlexNet, and ensemble fusion through systematic literature review and empirical comparison.	LO1, LO4
Managed Agile project plan across the full FYP timeline with 14 documented architectural iterations, dataset visits, training runs, and thesis writing.	LO2
Elicited FR01–FR09 and NFR01–NFR05 through domain expert interviews and literature. Verified through 14 functional test cases.	LO3, LO5
Reviewed 20+ sources (80% journals), critically evaluated four prior works, and confirmed research gap through capability comparison.	LO4
Independently learned YOLOv8, multi-label PyTorch, ensemble design, TTA, Supabase, and Roboflow.	LO5
Addressed GDPR compliance, obtained informed consent during 7–8 factory visits, and documented ethical considerations in Chapter 3.	LO6
Built full-stack system: YOLOv8-seg board detection, 2-model ensemble, dual-view fusion, knowledge base, JWT auth, Supabase, React frontend.	LO7
Produced 10-chapter thesis documenting design decisions, 14 changes, benchmarking, augmentation experiment, and 10-expert evaluation.	LO8
To be demonstrated and defended at viva voce examination.	LO9

Table 41 Achievement of learning outcomes (self composed)

10.7. Problems and challenges faced

Problem / Challenge	**Solution**
Dataset collection required 7–8 factory visits, capturing images without disrupting active production operations under variable lighting and conveyor workflows.	Coordinated with factory management to schedule sessions during maintenance windows. Used manual exposure settings. Accumulated 619 labelled images with no publicly available equivalent.

Class imbalance - delamination (89) and bubbling (71) severely underrepresented vs missing_face (202).	Applied Focal BCE with per-class weights, WeightedRandomSampler, and CutMix/MixUp. Augmentation experiment attempted (637 samples) produced degraded results due to co-occurrence context loss (documented in 9.4.2).
Multi-label vs single-label decision - initial IPD design used single-label softmax, unable to represent co-occurring defects.	Transitioned to sigmoid + BCEWithLogitsLoss. Rebuilt loss function, metrics, dataset pipeline, and verdict logic. Validated by TE3 and TE4.
Class merging - missing_edges and missing_top_face caused systematic confusion in the confusion matrix.	Merged into missing_face. Re-annotated dataset, updated crop pipeline, retrained models. Produced +8.5 F1 and +3.5pp accuracy gain.
Clean board false positives - model flagged defect-free boards due to no negative training samples.	Implemented confidence gate (CLEAN_BOARD_GATE=0.30). Root cause documented as known limitation for future work.

Table 42 problems and challenges (self composed)

10.8. Deviations

Original Plan	**Final Implementation**	**Justification**
Patch-based single-label classification	Whole-board multi-label classification	Patches removed structural context needed for warping. Multi-label correctly represents co-occurring defects.
Single EfficientNet-B2 with softmax	EfficientNet-B2 + AlexNet ensemble with sigmoid	Ensemble improved Macro F1 from 0.575 to 0.626 through complementary architectural biases.
6 classes (separate missing_edges + missing_top_face)	5 classes- merged into missing_face	Confusion matrix showed systematic inter-class confusion. Merge improved F1 by 8.5 points.

Real-time industrial camera integration for live conveyor inspection	Static image upload via web interface	Real-time camera integration was descoped due to time and resource constraints within the project timeline. The static upload approach delivers full defect detection capability and validates the core classification pipeline, with real-time camera integration identified as the highest-priority future enhancement (FR08).

Table 43 Deviations (self composed)

10.9. Limitations of the research

- The primary limitation is the scale and diversity of the dataset. The dataset of 619 images, which was obtained in one factory setting, might not be generalized to boards in other factories with other wood species, types of adhesive, or lighting. This poses the greatest limit to operational practicability, and the greatest future data collection orientation.
- The issue of class imbalance is an ongoing and somewhat unsolved problem. The Training images of the delamination class (F1=0.524) are only 89 as compared to 202 of missingface. The simplest method of augmentation, which is as simple as replicating or artificially synthesizing delamination images alone, cannot work with such a dataset, since boards which show delamination nearly always also contain other flaws (warping) since the same mechanism of glue bond failure underlies delamination. This over-prediction of delamination in isolation by the model with no extra effort to predict delamination in isolation will be reduced in the targeted single-class augmentation experiment performed at evaluation, which reduced delamination F1 from 0.463 to 0.329 with the addition of 211 new samples (see Section 9.4.2). To address this shortcoming, it is important to compile more whole-board images with delamination in a variety of contexts, and not to add more crops to the existing ones in secluded settings.
- No real-time validation of the system in factory conditions is made. Testing was all in a controlled off line process using test files of all images. The real factory variables

such as conveyor movement, variable lighting, and irregular camera shots are not included in the existing test set. A live camera feed scenario would only make the board detector's fallback behavior of unconstrained photography worse.

- Images of defect-free boards could not be found to validate the PASS verdict path. The confidence gate (CLEANBOARD_GATE=0.30) functionally tackles clean board false positives but was not compared to ground-truth clean boards, which is a known limitation of FR03 validation.
- The confusion matrix analysis shows that some classes remain difficult to distinguish, especially delamination versus warping and bubbling versus imprint_on_surface. This reflects visual similarity, physical co-occurrence of defects, and limited minority-class data. Therefore, class-level ambiguity remains a limitation of the current system despite the overall improvement in ensemble performance.

10.10 Future enhancements

- The most important future enhancement is the expansion of datasets. Then again to collect a minimum of 300 samples per class of three or more factories, located in different geographical areas would enhance generalisation and counter the major limitation of the current system. Delamination (weakest class) should be given a priority among diverse board thickness, types of adhesives, and levels of severity.
- One problem that emerged during testing is that boards that domain experts would accept as error-free would often produce REVIEW verdicts, as minor surface variations, e.g. the natural wood texture, minor press marks, minor colour differences, etc. would be similar to defects (such as imprintonsurface on model with its sensitivity) at the sensitivity level of the model. To handle this, a special set of verified clean boards which is verified by seasoned domain experts and added as explicit negative training samples is needed, to ensure the model can learn to differentiate between actual defects and natural board variation, instead of relying on the confidence gate threshold.
- The most important system enhancement that was found by all the five domain experts is real-time camera integration (FR08). Substituting the upload of static images with live camera mount conveyors would make DefectIQ a real production line quality control system, which would need camera API integration, frame buffering, and optimisation of the production through inference.

- The accessibility enhancement of the greatest priority is Sinhala language support (FR09), which is explicitly specified by DE1 as the key to the feasibility of application to the Sri Lankan factory context.
- Future studies should consider context preserving augmentation, in which entire board images are augmented with all annotations, not singleton defect images, since it would help to augment the range of data introduced without the distribution shift used in the augmentation experiment.

10.11 Achievement of the contribution to body of knowledge

DefectIQ contributes to the industrial computer vision and plywood quality inspection body of knowledge in the following ways

- DefectIQ builds the first documented system for post-manufactured plywood defects detection with a combination of dual-view imaging, multi-label classification, and knowledge-based reasoning, a gap that previous efforts have left unaddressed such as Baad (2023) whose dataset is proprietary and supported by domain experts who reported the lack of automated inspection system in their facilities.
- One of the two-view asymmetric weighted fusion strategies, where weight constraints surface defects with the top-view (65%) and structural defects with the side-view (65%) depending on the geometry of the physical defects, is a new fusion strategy that has not been reported in the previous literature on plywood inspection.
- The multi-label classification formulation used to detect plywood defects based on physical co-occurrence of delamination and warping as a result of glue bond break-up is a methodology contribution that cannot be found in the single-label systems previously used in the field.
- The DefectIQ dataset of 619 annotated images of post-manufactured plywood boards of five classes that were collected with the help of 7-8 visits to real factories and published on Kaggle. This is a new domain-specific dataset that may be used by future studies in this field.

10.12 Concluding remarks

DefectIQ has demonstrated a validated multi-stage computer vision system to detect post manufactured plywood defects, an area of problem that has a validated research gap, a real commercial interest to the Sri Lankan plywood manufacturing sector, and a personal motive of

the author. The system obtains Macro F1=0.6262 and Top-2 Image Accuracy=95.12% which is better than all empirical baselines and covers capabilities not available in all the four previous related works found in the literature.

The project also had its share of limitations and flaws, the delamination class does not perform as well, the augmentation experiment was not able to achieve better results, the board detector is having problems with unconstrained photography, and the real-time camera integration remains future work. These restrictions are clearly recorded together with the technical justification of the same and the corrective measures implemented.

What is presented here is not a perfect system, but evidence of a researcher who understood the problem deeply, engaged seriously with expert feedback, attempted systematic improvements, and documented the outcomes either positive or negative, in an academic manner. DefectIQ provides a proven basis to the further development of a fully functioning real-time system of inspection of plywood. The dataset, model, and architectural knowledge generated in this study are available to the research community and directly implemented in the Sri Lankan manufacturing industry, an industry yet to record any history of computer vision based solution to post-manufactured plywood quality control in the country.

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Appendix A literature review – tables and diagrams.

A.1 Extended existing work comparison

The table gives gives a detailed breakdown of the eight most relevant existing works reviewed in Chapter 2, their limitations and how they have informed the design of DefectIQ.

Citation	Summary	Limitation	Contribution to DefectIQ
Baad (2023)	Tested Swin-T and ViT for glue-layer defect classification on plywood. Swin-T achieved 88.70% accuracy.	Small and highly imbalanced dataset. Classes not representative of post-manufacturing defects.	Confirms transformer viability but highlights dataset scarcity for plywood-specific research.
Chen et al. (2022)	CNN-based approach achieving 85-92% accuracy on wood surface defect classification.	Single-label classification. Evaluated on raw wood boards, not post-manufactured plywood.	Provides external benchmark for accuracy comparison in Chapter 8.
Li et al. (2025)	Improved YOLOv8n with MSDA and DWR modules for wood panel defect detection. mAP > 95%.	Evaluated on wood boards not plywood. Object detection only , no reasoning layer.	Demonstrates YOLO suitability for wood domain; motivates board isolation stage in DefectIQ.
An et al. (2024)	CWB-YOLOv8 with CondConv and BiFormer for wood defect detection. +3.5% mAP over baseline.	Focuses on raw wood knots and cracks. No multi-label or dual-view capability.	Supports adoption of YOLOv8 as the board detection foundation for DefectIQ.

Rasheed et al. (2024)	EfficientNet-B4 with SE block for semiconductor defect classification. ~98% accuracy.	Single-label classification on semiconductor wafers. Not applicable to multi-class plywood defects.	Validates EfficientNet transfer learning effectiveness for industrial defect classification tasks.
Wang et al. (2024)	Smart visual analytics with case-based reasoning for manufacturing defect judgement and algorithm selection.	Complex knowledge graph infrastructure. Not deployable as a lightweight prototype on standard hardware.	Motivates knowledge-based reasoning in DefectIQ; justifies simpler deterministic JSON-based approach.
Kodytek, Bodzas and Bilik (2022)	Reviewed existing wood and plywood defect datasets and found no publicly available post-manufactured plywood dataset.	Dataset gap remains unresolved at time of writing. Most studies use private or raw lumber datasets.	Directly motivates DefectIQ dataset collection from a real factory and publication on Kaggle under CC BY 4.0.
Pachouly et al. (2025)	Multi-label classification for defect prediction using sigmoid activation and BCEWithLogitsLoss formulation.	Focused on software defects, not manufacturing visual defects. Domain is not directly transferable.	Validates multi-label sigmoid formulation adopted in DefectIQ for detecting co-occurring physical plywood defects.

Table 44 Exising work comparison (self composed)

A.2 Concept map

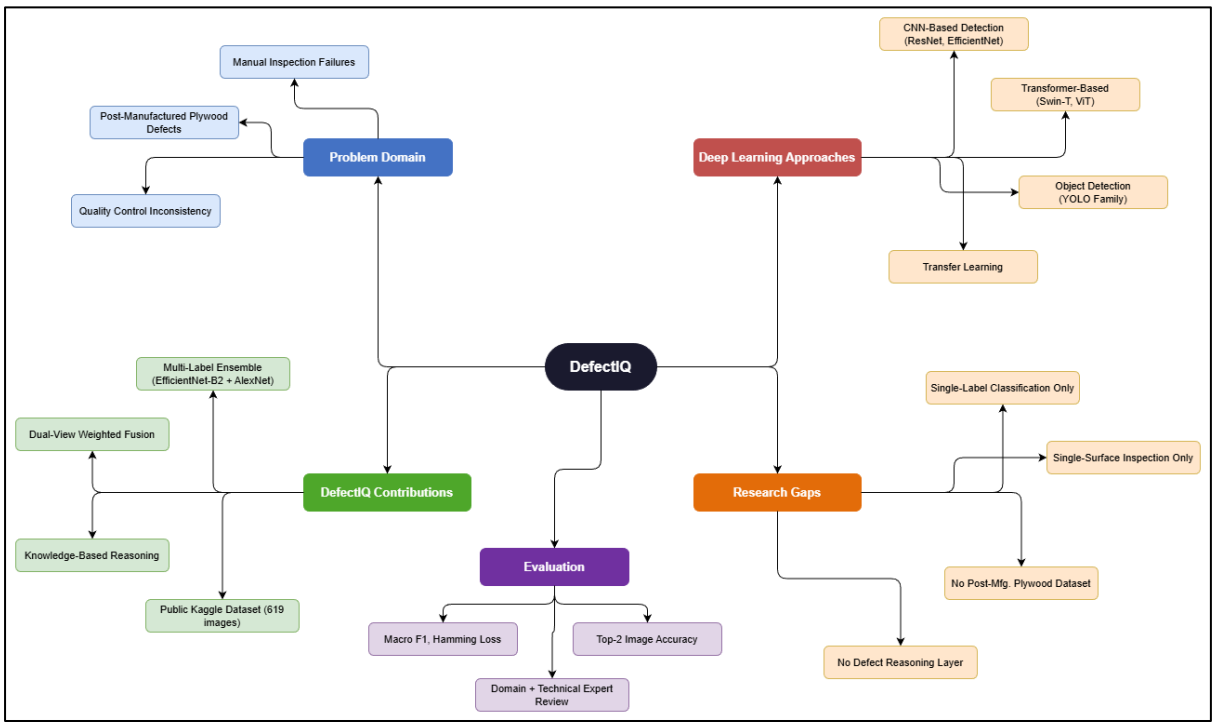


Figure 24Concept map (self composed)

A.3 Plywood defect manufacturing cause reference

The following table documents the five post-manufacturing defect classes in the DefectIQ dataset, their manufacturing state of origin, physical cause, visual appearance characteristics, and known co-occurrence relationships.

Defect class	**Manufacturing stage**	**Physical cause**	**Visual appearance**	**Co-occurs with**
Bubbling	Hot pressing	Trapped moisture or steam between veneers during pressing cycle	Small dome-shaped elevations on board surface	Delamination, Missing face
Delamination	Adhesive application / Pressing	Insufficient glue spread or press temperature causing inter-	Visible layer separation or lifting at edges or board surface	Warping, Bubbling

		veneer bond failure		
Imprint on surface	Hot pressing	Press plate contamination or foreign object contact during pressing	Embossed or indented pattern on board face surface	Rare co-occurrence
Missing face	Veneer cutting / Press damage	Veneer detachment, cutting damage, or adhesive failure causing surface material loss	Exposed core layers, raw material patch visible on board face	Bubbling, Delamination
Warping	Conditioning / Cooling	Uneven moisture distribution or asymmetric veneer layup during post-press cooling	Board bowing or curvature visible in side profile	Delamination

Table 45 defect reason reference (self composed)

Appendix B – Requirement specification interview notes

Drive link for the Interview notes:
https://drive.google.com/drive/folders/1IHNdq5i5ODB_stEE_rsvExAbHsH-9IBt.

B.1 Interview overview

Table below summarises the three interviews conducted. All interviews were conducted in person at the factory site and were recorded as handwritten notes with the interviewees' consent.

ID	Interviewee	Organisation	Date	Method
I-01	Factory Owner	Plymarc Enterprises	19 October 2025	Semi-structured in-person interview
I-02	Factory Worker	DCH Plywood	19 October 2025	Semi-structured in-person interview
I-03	Production Manager	DCH Plywood	19 October 2025	Semi-structured in-person interview

Table 46 Interviwee overview (self composed)

B.2 Interview question list

The following semi-structured interview questions were used across all three interviews. Questions were adapted in the moment based on the interviewee's role, factory owner questions focused on business and quality management context, factory worker questions focused on daily inspection workflow, and production manager questions focused on process control and defect patterns.

Section 1- Current inspection process

1. How is quality inspection currently performed in the facility currently?
2. How many employees are currently assigned to manual inspection at any given time?
3. How long does it take to inspect one board manually?
4. What happens when a defective board is found what is the rejection or rework process?

Section 2 - Defect types and frequency

5. What types of defects do you encounter most frequently in your production line?
6. Are there defects that inspectors regularly miss or find difficult to detect?
7. Do multiple defects ever appear on the same board at the same time?
8. Which defects cause the most damage in terms of product rejections or customer complaints?

Section 3- Pain points and needs

9. What are the biggest problems with the current manual inspection process?
10. Have you ever had defective boards shipped to retailers or customers? What was the impact?
11. How does inspector fatigue affect the quality of inspection across a shift?
12. Would it be useful to know not just that a defect exists, but why it occurred in the manufacturing process?

Section 4 -System requirements and expectations

13. If an automated detection system were introduced, how quickly would it need to return a result to fit into the production workflow?
14. What information would be most useful to display to a worker when a defect is detected?
15. Would a monthly or shift-level report of defect trends be useful to management?
16. Would operators be comfortable using a computer or tablet-based inspection interface on the factory floor?
17. What language would operators prefer the system to display information in?
18. Do you see value in checking both the top surface and the side profile of a board for different defect types?

Section 5 - Future and strategic considerations

19. How do you think an automated inspection system would affect worker roles and workload distribution?

20. Would improved defect detection open new export markets or strengthen relationships with existing customers?
21. Are there any concerns you have about introducing computer vision technology into the inspection process?

Appendix C – Use case description

Use case	**Explanation**
Id	UC2
Description	The system provides a reason or explanation for each detected defect using a knowledge-based dataset.
Primary Actor	User
Supporting Actor	None
Preconditions	Defects must be identified (UC1).
Included use cases	None
Extended Use cases	None
Main flow	1. System retrieves all detected defects. 2. System provides cause/reason for each defect
Alternative flow	If reasoning is ambiguous, system indicates “reason unknown.”
Exceptional flow	If knowledge-based system is unavailable, only defect label is shown.
Postconditions	Each defect is paired with an actionable explanation.
Comment	Supports root cause analysis and operator feedback.

Table 47 use case description (UC2) (Self-composed)

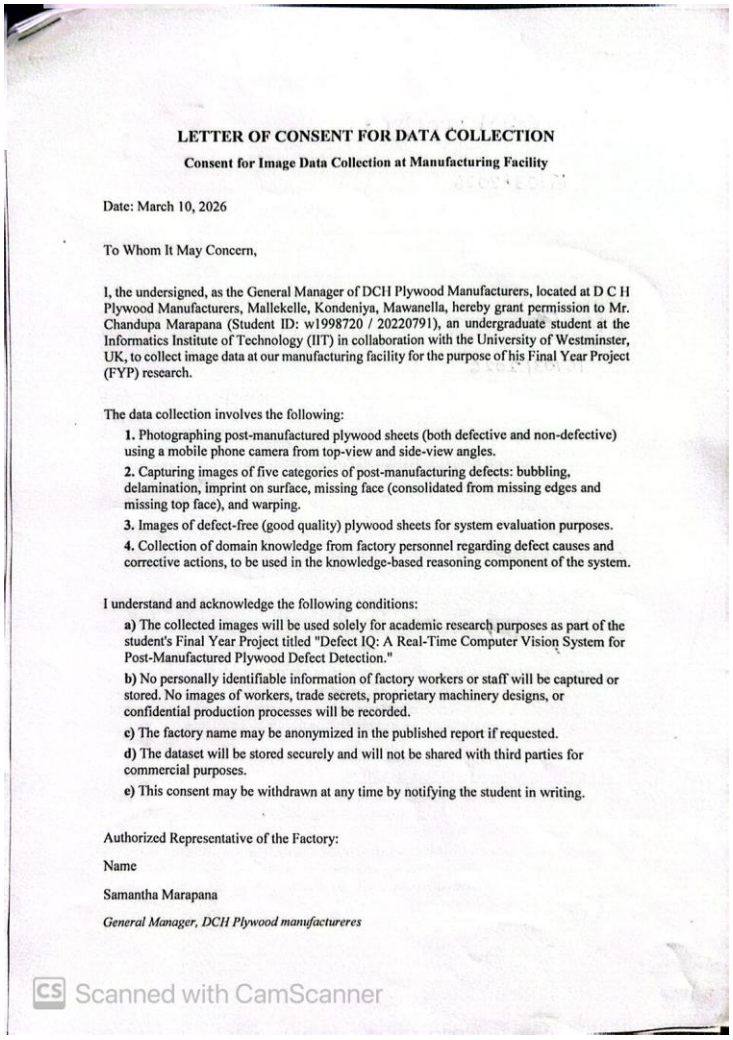
Use case	**Explanation**
Id	UC3
Description	The system provides the results for the inspected plywood, including identified defects and reasons.
Primary Actor	None
Supporting Actor	None
Preconditions	Defects identified and reasons provided (UC1 and UC2 complete).
Included use cases	Defect Identified (UC1), Defect Reason Provided (UC2)
Extended Use cases	None
Main flow	1. System gathers defect results and explanations. 2. System compiles verdict with the results if the defect is passed

	or failed.
Alternative flow	If any data is missing, system prompts user to ensure inspection is complete.
Exceptional flow	If result generation fails, user is notified and process is logged.
Postconditions	Result list is available with their verdicts

Table 48 use case description (UC3)(self composed)

Appendix D – Implementation chapter related materials and screenshots

D.1 Dataset validation and kaggle publication consent letters



LETTER OF CONSENT FOR DATA COLLECTION

Consent for Image Data Collection at Manufacturing Facility

Date: March 10, 2026

To Whom It May Concern,

I, the undersigned, as the General Manager of DCH Plywood Manufacturers, located at D C H Plywood Manufacturers, Mallekelle, Kondeniya, Mawanella, hereby grant permission to Mr. Chandupa Marapana (Student ID: w1998720 / 20220791), an undergraduate student at the Informatics Institute of Technology (IIT) in collaboration with the University of Westminster, UK, to collect image data at our manufacturing facility for the purpose of his Final Year Project (FYP) research.

The data collection involves the following:

1. Photographing post-manufactured plywood sheets (both defective and non-defective) using a mobile phone camera from top-view and side-view angles.
2. Capturing images of five categories of post-manufacturing defects: bubbling, delamination, imprint on surface, missing face (consolidated from missing edges and missing top face), and warping.
3. Images of defect-free (good quality) plywood sheets for system evaluation purposes.
4. Collection of domain knowledge from factory personnel regarding defect causes and corrective actions, to be used in the knowledge-based reasoning component of the system.

I understand and acknowledge the following conditions:

a) The collected images will be used solely for academic research purposes as part of the student's Final Year Project titled "Defect IQ: A Real-Time Computer Vision System for Post-Manufactured Plywood Defect Detection."

b) No personally identifiable information of factory workers or staff will be captured or stored. No images of workers, trade secrets, proprietary machinery designs, or confidential production processes will be recorded.

c) The factory name may be anonymized in the published report if requested.

d) The dataset will be stored securely and will not be shared with third parties for commercial purposes.

e) This consent may be withdrawn at any time by notifying the student in writing.

Authorized Representative of the Factory:

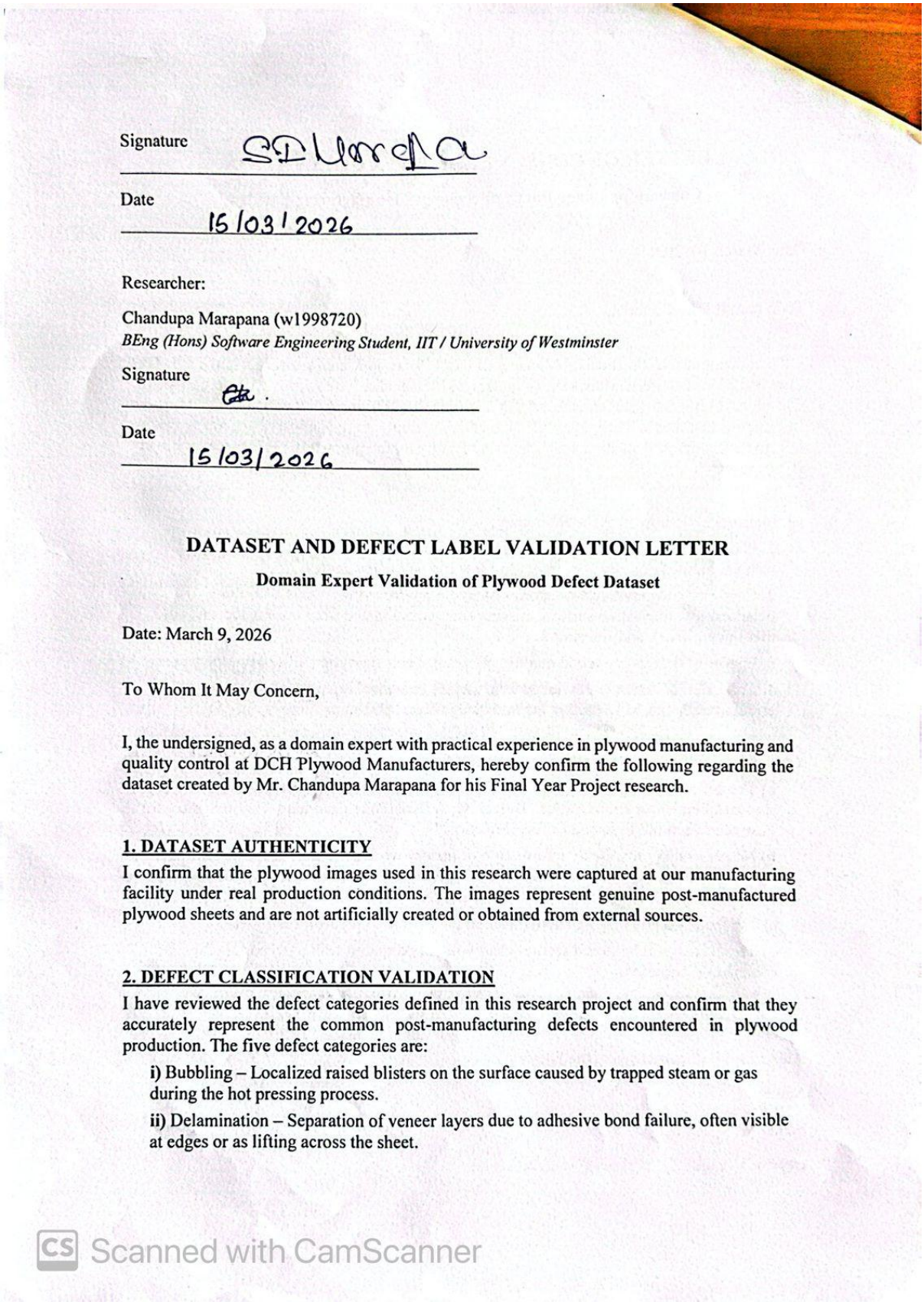
Name

Samantha Marapana

General Manager, DCH Plywood manufactureres

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Figure 25 Dataset validation letter(1)



Signature SD Uarapana

Date 15/03/2026

Researcher:

Chandupa Marapana (w1998720)
BEng (Hons) Software Engineering Student, IIT / University of Westminster

Signature

Date 15/03/2026

DATASET AND DEFECT LABEL VALIDATION LETTER

Domain Expert Validation of Plywood Defect Dataset

Date: March 9, 2026

To Whom It May Concern,

I, the undersigned, as a domain expert with practical experience in plywood manufacturing and quality control at DCH Plywood Manufacturers, hereby confirm the following regarding the dataset created by Mr. Chandupa Marapana for his Final Year Project research.

1. DATASET AUTHENTICITY

I confirm that the plywood images used in this research were captured at our manufacturing facility under real production conditions. The images represent genuine post-manufactured plywood sheets and are not artificially created or obtained from external sources.

2. DEFECT CLASSIFICATION VALIDATION

I have reviewed the defect categories defined in this research project and confirm that they accurately represent the common post-manufacturing defects encountered in plywood production. The five defect categories are:

i) Bubbling – Localized raised blisters on the surface caused by trapped steam or gas during the hot pressing process.

ii) Delamination – Separation of veneer layers due to adhesive bond failure, often visible at edges or as lifting across the sheet.

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Figure 26 Dataset validation letter(2)

iii) Imprint on Surface – Marks or impressions pressed into the plywood surface from machinery, press plates, or stacking during production.

iv) Missing Face – Areas where surface material is physically absent — including missing veneer layers, broken or chipped board edges, and torn top face regions resulting from handling damage or adhesive bond failure during production.

v) (Consolidated) – (Consolidated with Missing Face — see item iv above.)

vi) Warping – Board-level geometric distortion (bending, twisting, or cupping) resulting from uneven moisture distribution or internal stress release.

3. KNOWLEDGE BASE VALIDATION

I have reviewed the defect-cause-recommendation mappings used in the system's knowledge base (reasons.json) and confirm that the listed causes and corrective actions are consistent with standard manufacturing practices at our facility.

4. ANNOTATION REVIEW

I have reviewed a representative sample of annotated images from the dataset and confirm that the defect labels and bounding box locations are accurate and consistent with my professional assessment of the defects present in those images. The annotation review was conducted in collaboration with four additional annotation helpers (see Annotation Helpers Summary below), covering a total of 500 images across all five defect categories.

5. ANNOTATION HELPERS SUMMARY

The following individuals assisted in the annotation review process, providing independent validation of defect labels and bounding box accuracy:

Helper ID	Role / Background	Annotations Reviewed	Classes Validated	Agreement Rate
H1	Factory Supervisor (10 yrs exp.)	150 images	All 6 classes	94%
H2	Production Manager	120 images	Bubbling, Delamination, Warping	91%
H3	Quality Inspector (5 yrs exp.)	100 images	Missing Face, Imprint on Surface	89%
H4	Senior Factory worker (12 yrs exp.)	80 images	Delamination, Warping	93%
H5	BEng SE Student — Computer Vision (Technical)	50 images	All 6 classes (technical review)	87%

Overall inter-annotator agreement rate across all helpers: 91.0%. Disagreements were resolved through majority consensus with the lead domain expert.

6. DATASET PUBLICATION CONSENT

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Figure 27 Dataset validation letter (3)

I hereby grant permission for the annotated plywood defect dataset to be published on the Kaggle platform (kaggle.com) as a public research dataset, subject to the following conditions:

a) The dataset is published under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence, requiring attribution to DCH Plywood Manufacturers and Mr. Chandupa Marapana.

b) The dataset is made available solely for academic research and non-commercial purposes.

c) No personally identifiable information, proprietary machinery details, or trade secrets are included in the published dataset.

d) The factory name (DCH Plywood Manufacturers) may be included in the dataset description as the data source, or anonymized as "Sri Lankan plywood manufacturing facility" if preferred.

e) This permission is granted with the understanding that the Kaggle publication will credit the domain expert validation process and the contribution of factory personnel.

The publication of this dataset on Kaggle is expected to contribute to the broader research community working on automated industrial defect detection, supporting reproducibility of the research findings presented in this Final Year Project.

Domain Expert:

Samantha Marapana

General Manager

Signature

Date

15/03/2026

Years of Experience in Plywood Manufacturing : 35 years

Company Stamp (if available)

D.C.H.PLYWOOD MANUFACTURERS
Mellakele
Koondeniya,Mawanella

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Figure 28 Dataset validation letter (4)

Appendix E – Model testing results

E.1 IPD model metrices

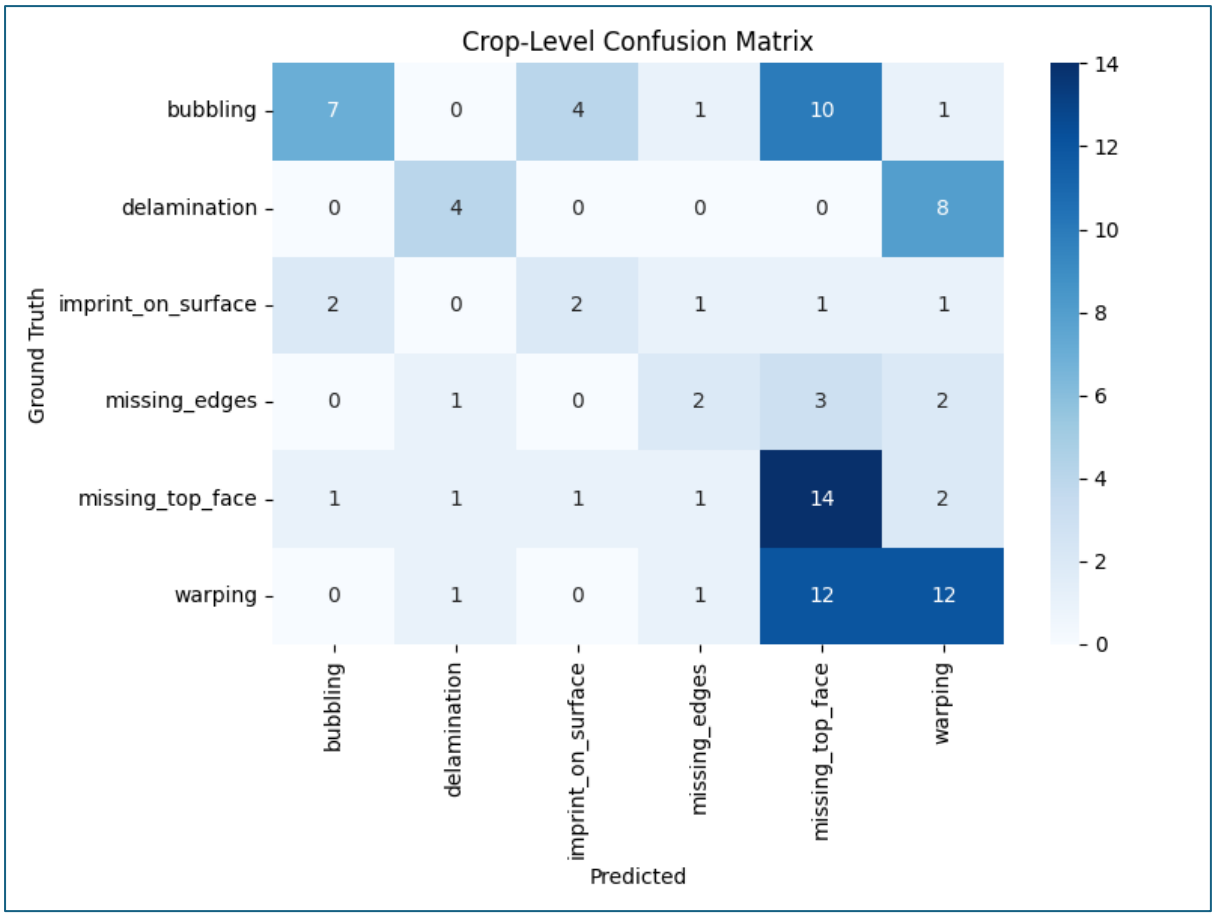


Figure 29 confusion matrix efficient net b2 v5 model submitted to IPD

	precision	recall	f1-score	support
bubbling	0.7000	0.3043	0.4242	23
delamination	0.5714	0.3333	0.4211	12
imprint_on_surface	0.2857	0.2857	0.2857	7
missing_edges	0.3333	0.2500	0.2857	8
missing_top_face	0.3500	0.7000	0.4667	20
warping	0.4615	0.4615	0.4615	26
accuracy			0.4271	96
macro avg	0.4503	0.3892	0.3908	96
weighted avg	0.4857	0.4271	0.4211	96

Figure 30 per-class f1-scores efficientNetB2 V5

```
Image-level Top-2 defect presence accuracy: 81.40%
```

Figure 31 top-w defect presence accuracy efficient net B2 v5

E.2 MobileNetV2 model metrices

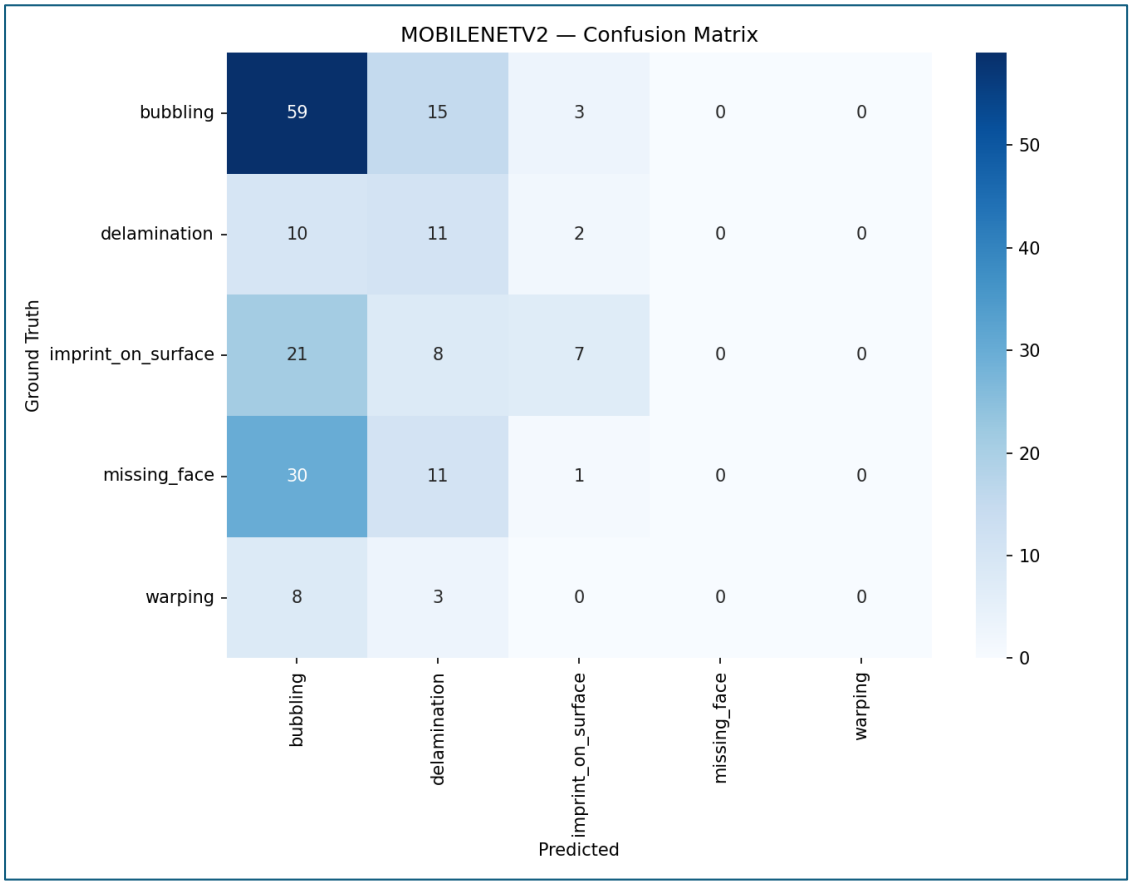


Figure 32 Confusion matrix of MobilenetV2

```
Model          : mobilenetv2
Threshold      : 0.5
Eval samples   : 189

Hamming Loss   : 0.4889
Subset Accuracy: 0.0212
Macro F1       : 0.4784
Weighted F1    : 0.5228
Top-2 Img Acc  : 78.31%

Per-class:
  bubbling                : P=0.3359  R=0.8269  F1=0.4778
  delamination            : P=0.1550  R=0.6250  F1=0.2484
  imprint_on_surface      : P=0.3386  R=0.8269  F1=0.4804
  missing_face            : P=0.5672  R=0.8000  F1=0.6638
  warping                 : P=0.4667  R=0.5915  F1=0.5217
```

Figure 33 perclass f1-score of mobilenetV2

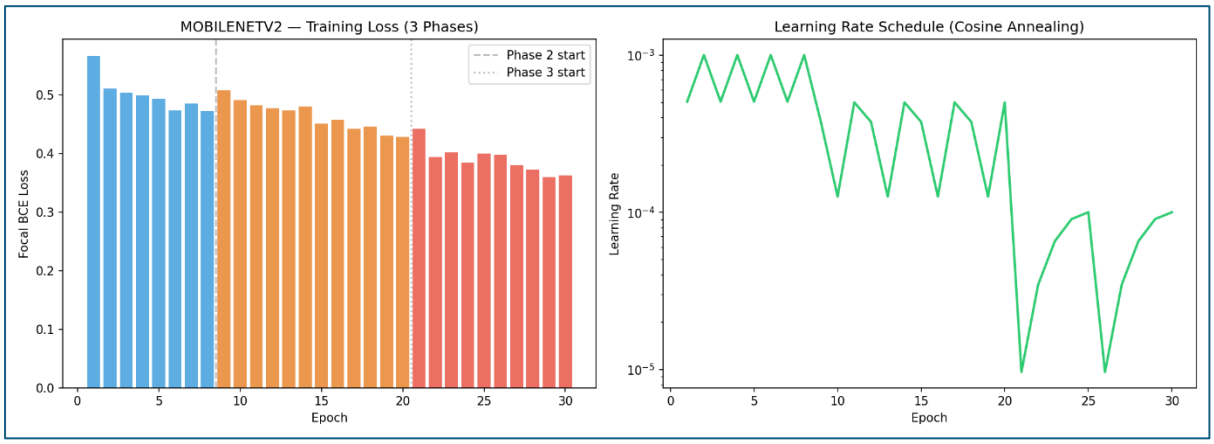


Figure 34 Training loss and learning rate - mobilenetV2

E.3 Resnet18 model metrices

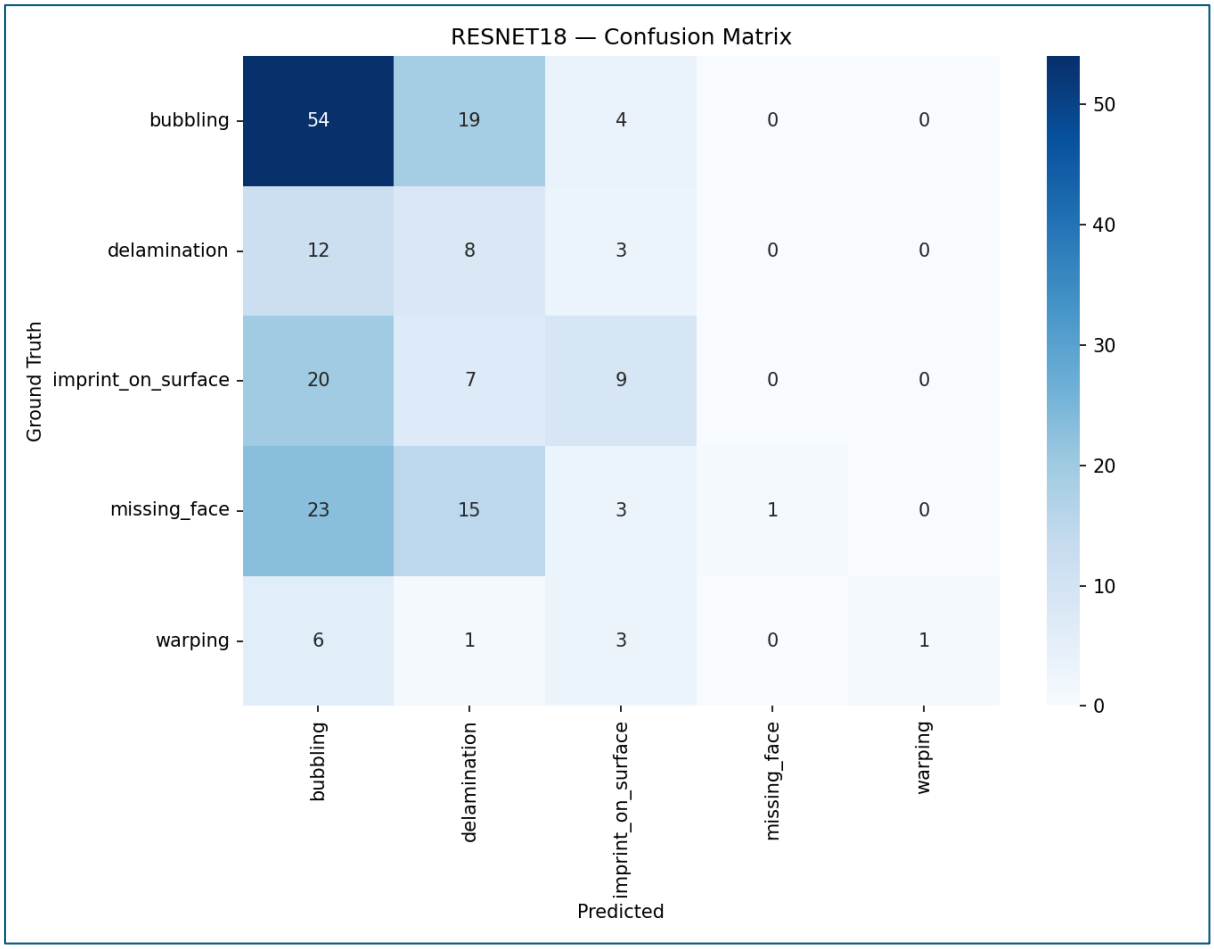


Figure 35 Confusion matrix -Resnet18

```
Model          : resnet18
Threshold      : 0.5
Eval samples   : 189

Hamming Loss   : 0.4889
Subset Accuracy: 0.0476
Macro F1       : 0.4829
Weighted F1    : 0.5249
Top-2 Img Acc  : 75.13%

Per-class:
  bubbling                : P=0.3391  R=0.7500  F1=0.4671
  delamination            : P=0.1732  R=0.6875  F1=0.2767
  imprint_on_surface      : P=0.3287  R=0.9038  F1=0.4821
  missing_face            : P=0.5966  R=0.7474  F1=0.6636
  warping                 : P=0.4352  R=0.6620  F1=0.5251
```

Figure 36 per class f1-scores of resnet18

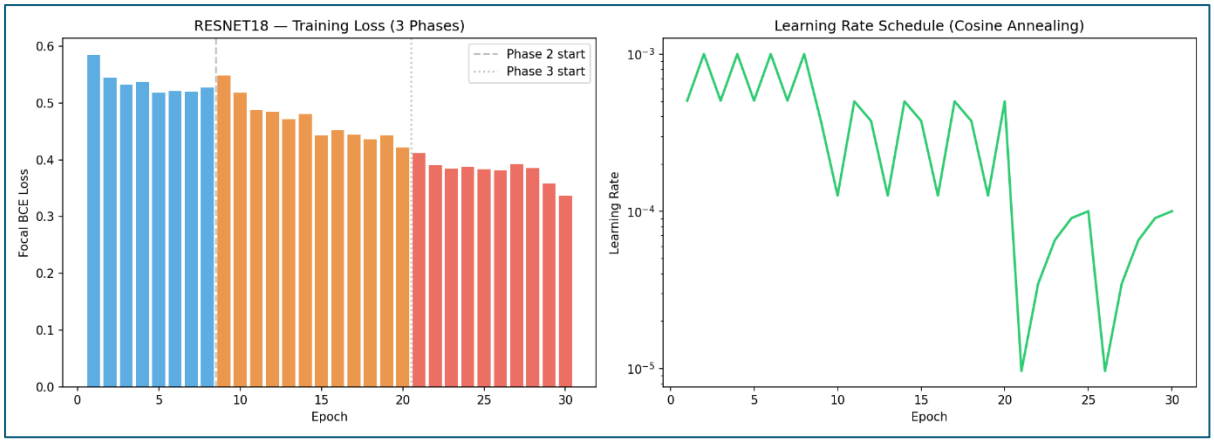


Figure 37 Training loss, learning rate -resnet18

E.4 Vgg16 model metrices

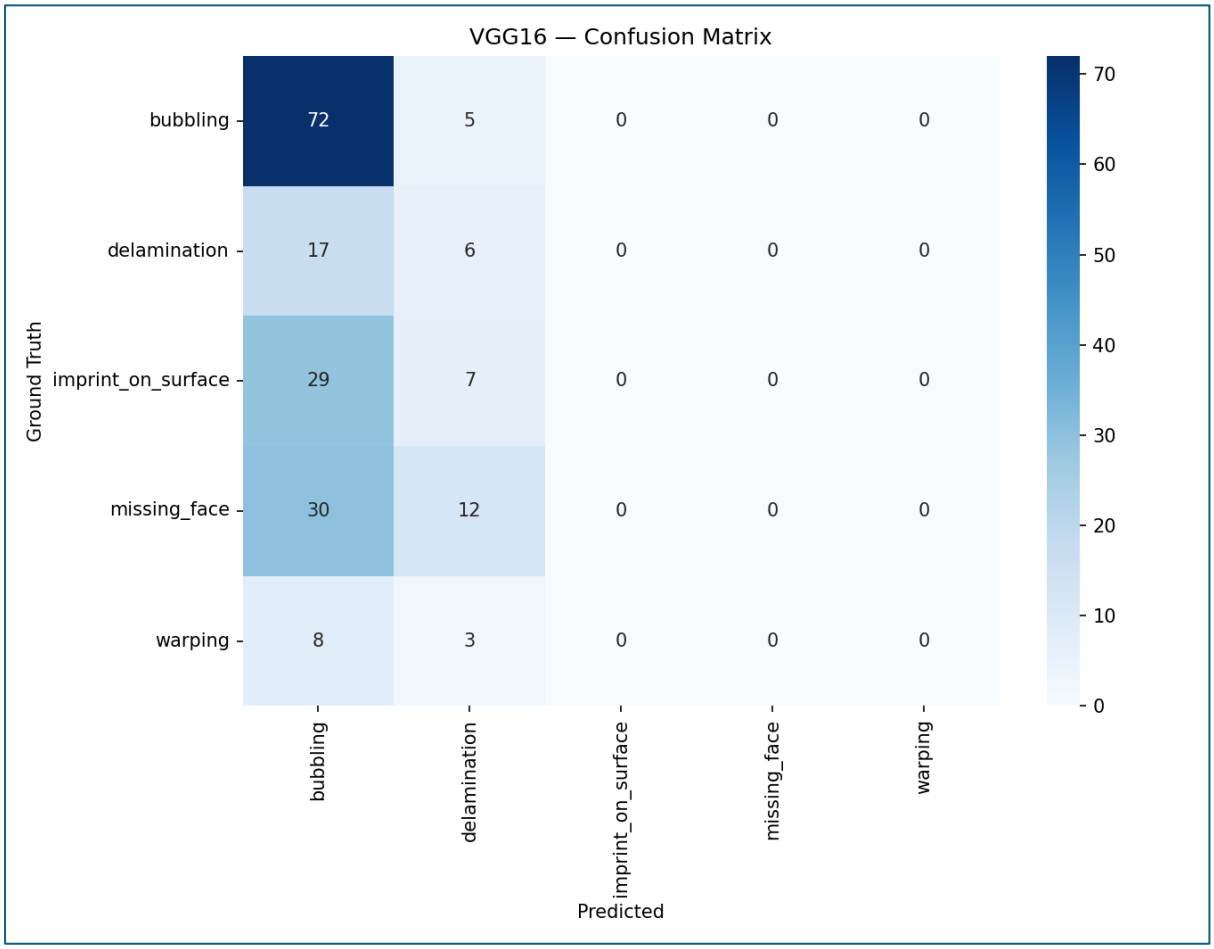


Figure 38 Confusion matrix- vgg16

```
Model          : vgg16
Threshold      : 0.5
Eval samples   : 189

Hamming Loss   : 0.6063
Subset Accuracy: 0.0053
Macro F1       : 0.4874
Weighted F1    : 0.5293
Top-2 Img Acc  : 85.71%

Per-class:
  bubbling                : P=0.3205  R=0.9615  F1=0.4808
  delamination            : P=0.1685  R=0.9688  F1=0.2870
  imprint_on_surface      : P=0.3165  R=0.9615  F1=0.4762
  missing_face            : P=0.5200  R=0.9579  F1=0.6741
  warping                 : P=0.3720  R=0.8592  F1=0.5191
```

Figure 39 perclass f1-scores of vgg16

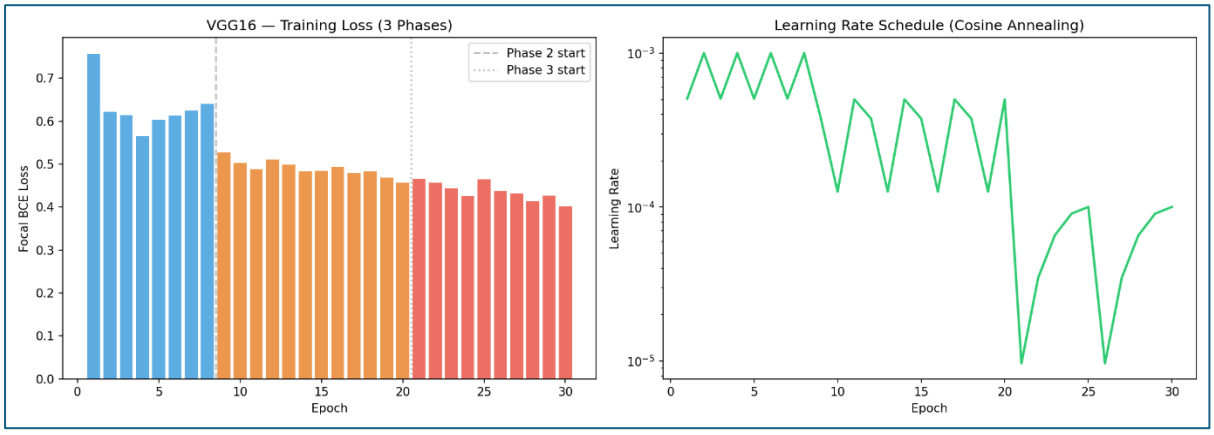


Figure 40 Training loss and learning rate schedule -vgg16

E.5 EfficientNet B2 model metrices

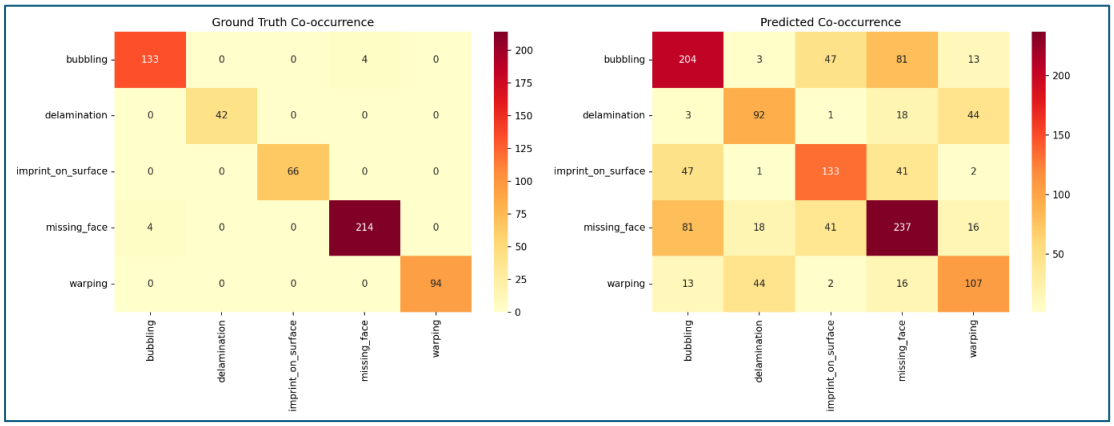


Figure 41 ground truth co-occurent -efficientNet B2

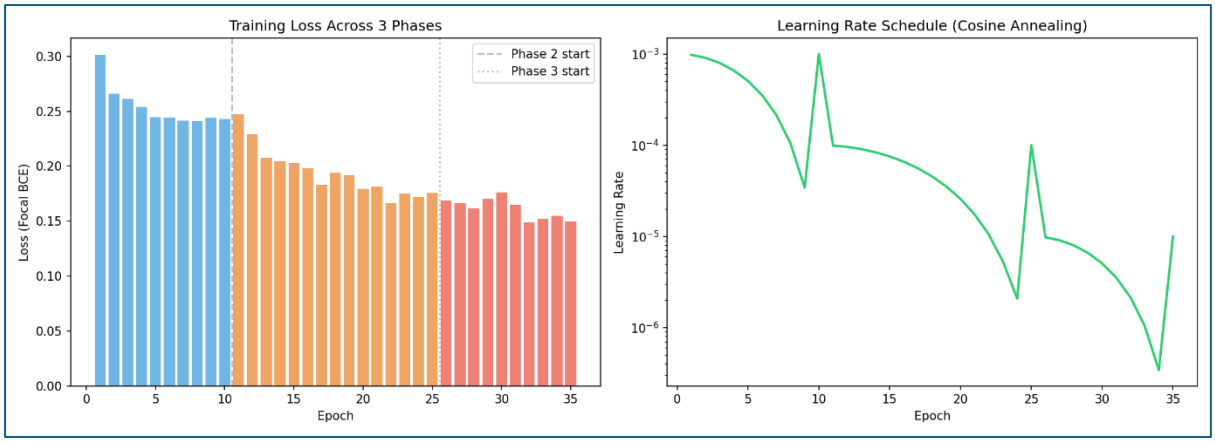


Figure 42 Training loss and Learning rate schedule -EfficientNetB2

```
===== MULTI-LABEL METRICS =====
Hamming Loss: 0.1901
Subset Accuracy: 0.3835
Macro F1: 0.5745
Weighted F1: 0.6205

===== SINGLE-LABEL EQUIVALENT (argmax) =====
                    precision    recall  f1-score   support

           bubbling     0.5929    0.6241    0.6081       133
      delamination      0.3731    0.5952    0.4587        42
imprint_on_surface      0.4271    0.6212    0.5062        66
      missing_face      0.7544    0.6143    0.6772       210
           warping      0.6901    0.5213    0.5939        94

          accuracy                         0.6000       545
         macro avg      0.5675    0.5952    0.5688       545
      weighted avg      0.6349    0.6000    0.6084       545
```

Figure 43 Perclass F1-scores -EfficientNetB2

```
Image-level Top-2 defect presence accuracy: 96.34%
Total images evaluated: 164
Correctly identified: 158
```

Figure 44 Top 2 level defect presence accuracy

E.5 AlexNet model metrices

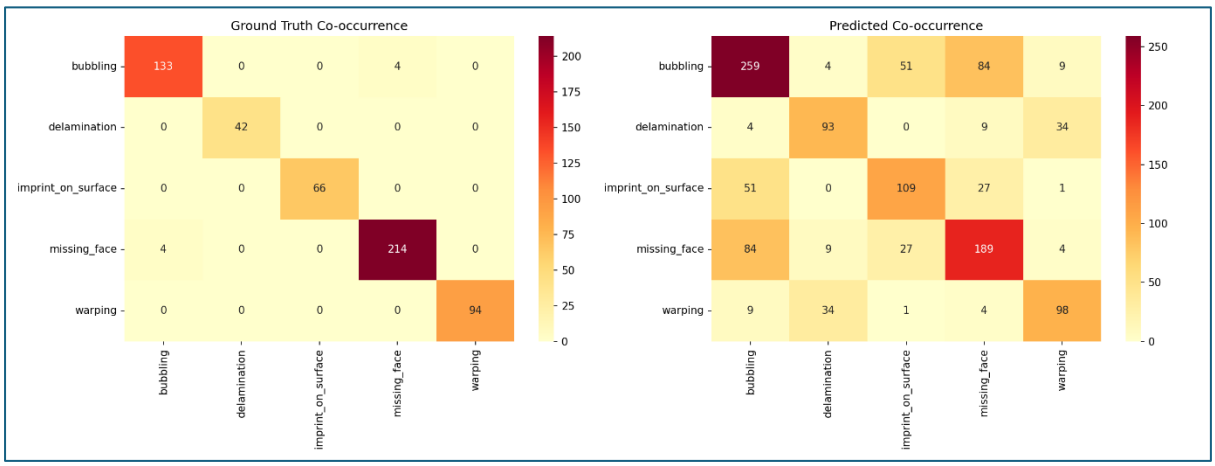


Figure 45 Ground truth co-occurence -Alexnet

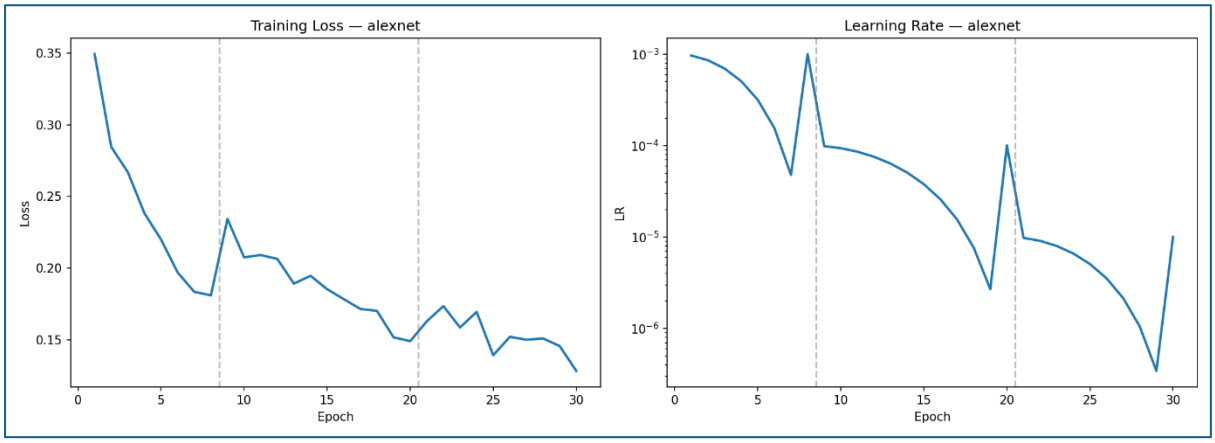


Figure 46 Training loss and learning rate- Alexnet

```
===== MULTI-LABEL METRICS =====
Hamming Loss: 0.1897
Subset Accuracy: 0.4128
Macro F1: 0.5723
Weighted F1: 0.6160

===== SINGLE-LABEL EQUIVALENT =====
                   precision    recall  f1-score   support

          bubbling     0.4921    0.6992    0.5776       133
      delamination     0.3038    0.5714    0.3967        42
imprint_on_surface     0.5316    0.6364    0.5793        66
      missing_face     0.8140    0.5000    0.6195       210
           warping     0.6087    0.4468    0.5153        94

          accuracy                         0.5615       545
         macro avg     0.5500    0.5708    0.5377       545
      weighted avg     0.6265    0.5615    0.5693       545
```

Figure 47 per-class F1 scores

```
Image-level Top-2 defect presence accuracy: 93.90%
Total images evaluated: 164
Correctly identified: 154
```

Figure 48 top 2 defect presence Accuracy -alexnet

E.6 Metrics ensemble

Metric	EfficientNet-B2	AlexNet	Ensemble (Best)
Macro F1	0.5745	0.5723	0.6262 *
Weighted F1	0.6205	0.6160	0.6640 *
Hamming Loss	0.1901	0.1897	0.1622 *
Subset Accuracy	0.3835	0.4128	0.4771 *

Top-2 Image Acc %	96.34% *	93.90%	95.12%
Strict Image Acc %	95.12% *	88.41%	91.46%

Table 49 ensemble metrics

E7. Threshold sweep analysis of ensemble model

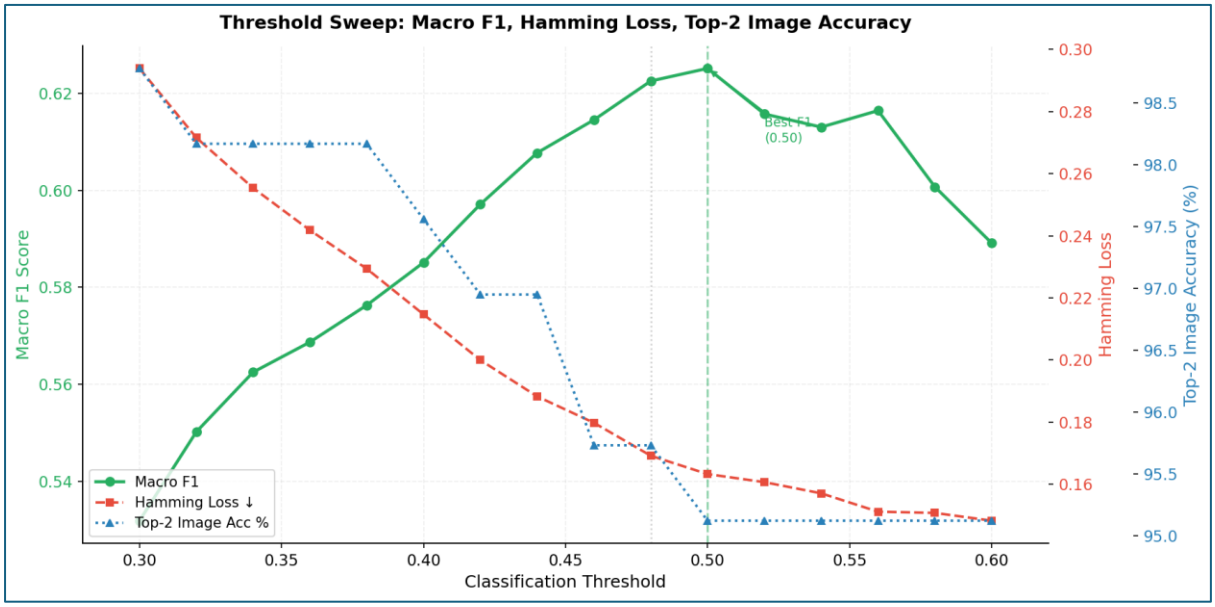


Figure 49 Threshold sweep analysis of ensemble model

Appendix F – Functional testing and nonfunctional testing results

F.1 Functional testing screenshots

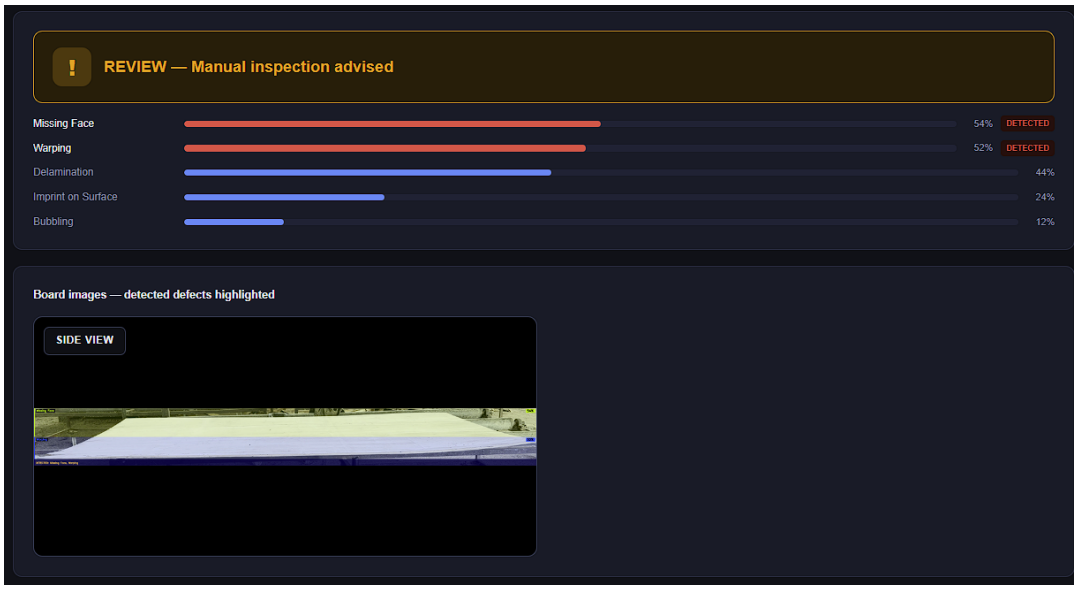


Figure 50 warping defect tested (self composed)



Figure 51 Delamination defect tested (self composed)

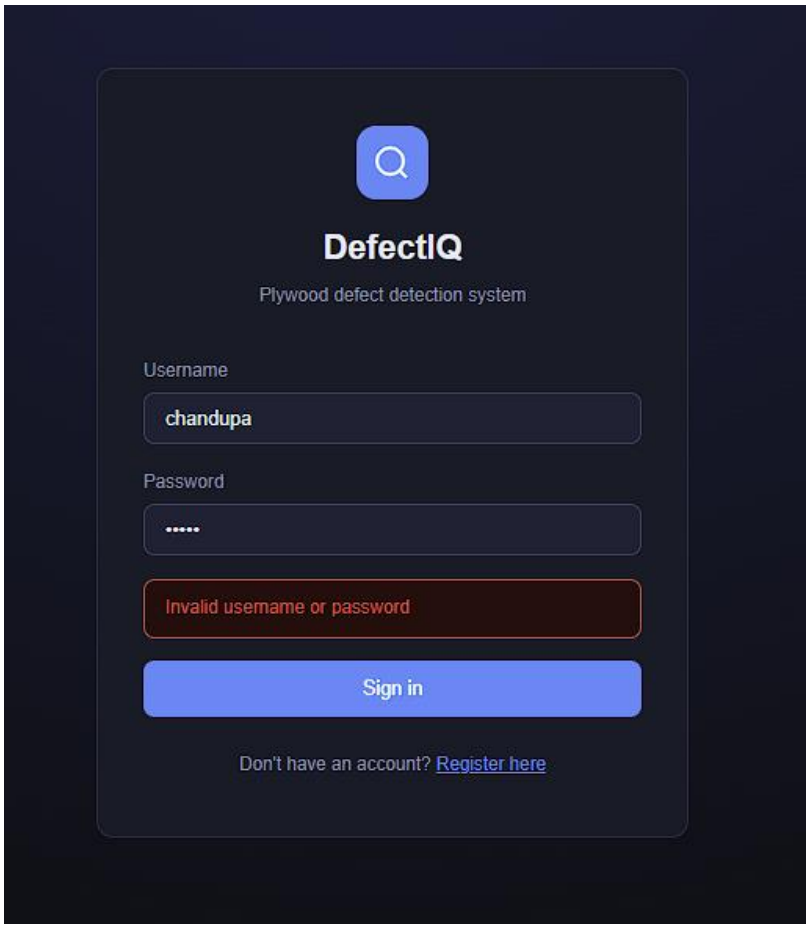


Figure 52 Invalid login credentials handled (self composed)

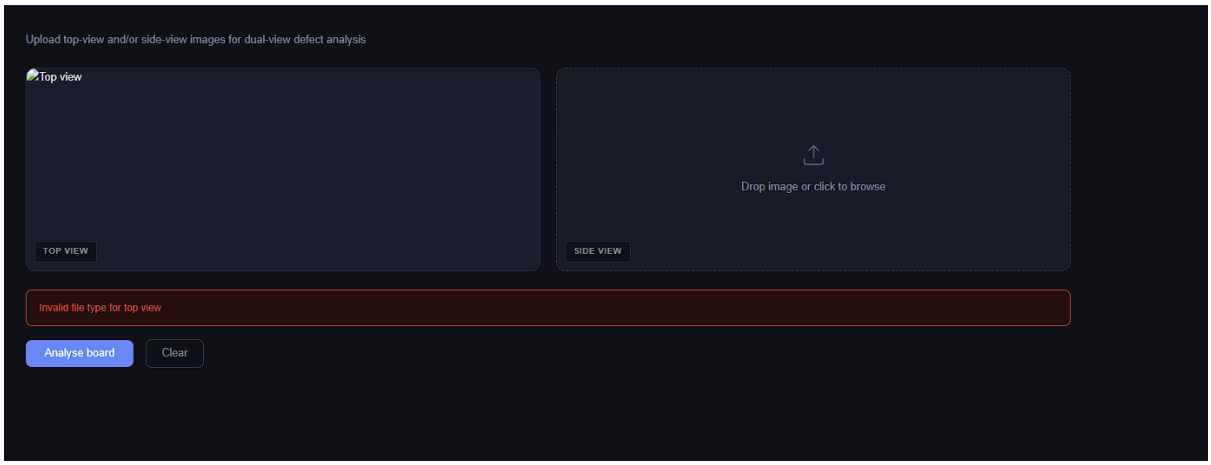


Figure 53 unsupported type uploaded (self composed)

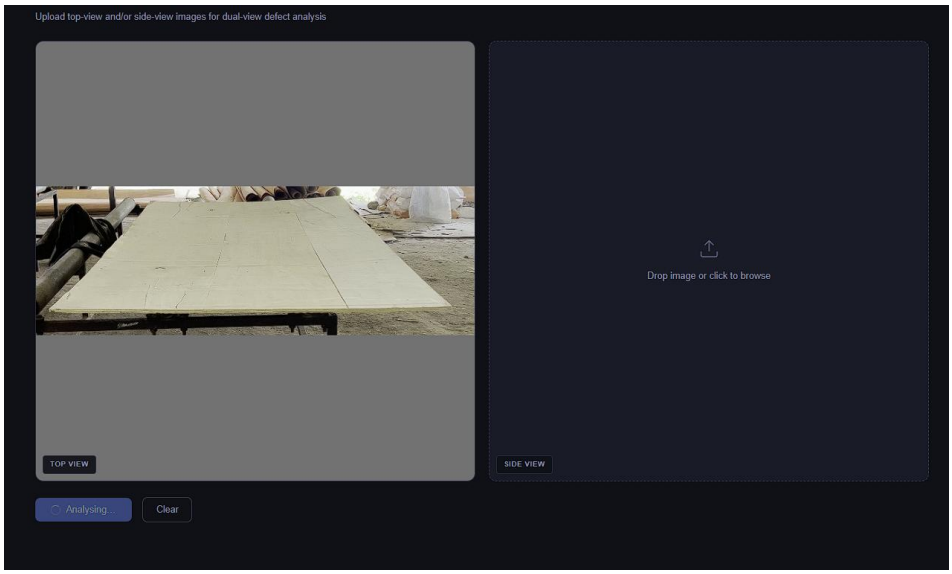


Figure 54 single view analysis without breaking(self composed)



Figure 55 single voew inspection results (self composed)

F.2 Nonfunctional testing

F.2.1 Response times

Response times were measured by submitting four inspection requests through the live system and recording the API response time for each /detect fetch request using the browser network tab (Chrome Dev tools). Table below records the results

Test	**Defect Scenario**	**API Response times**	**Result**
1	Defective board	2.71s	Pass
2	Defective board	3.10s	Pass
3	Defective board	1.99s	Pass
4	Defective board	1.37s	Pass
Average		**2.29s**	pass

Table 50 response test check - nonfunctionaltesting (self composed)

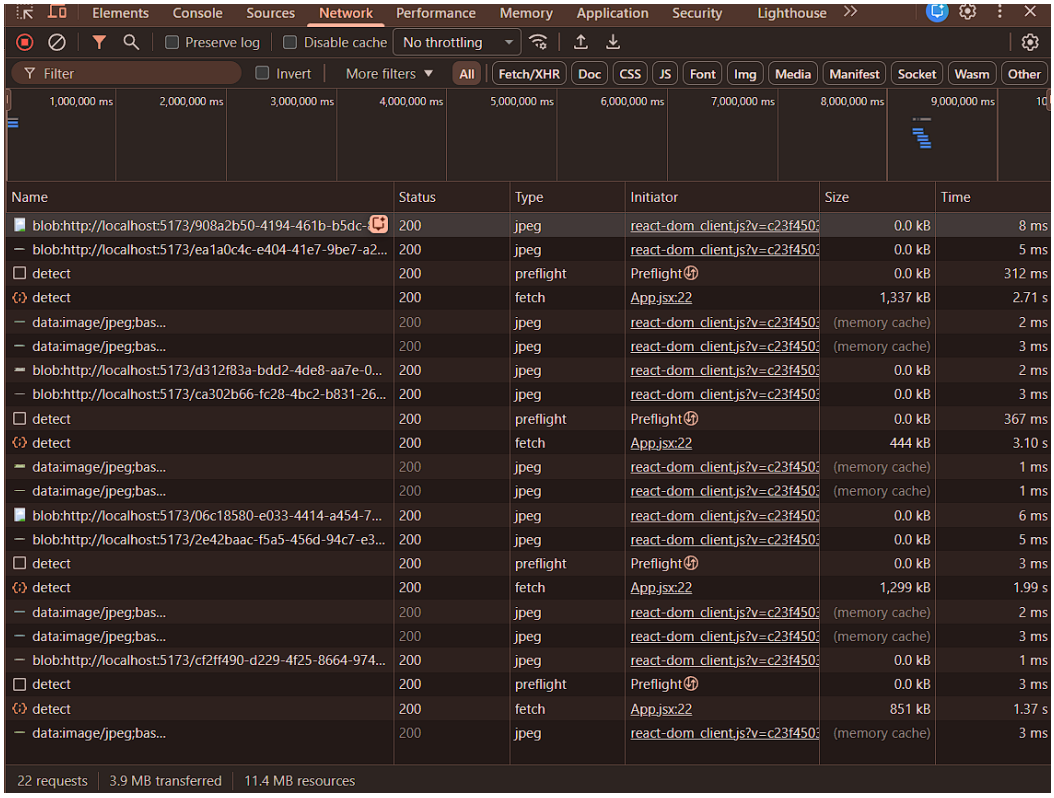


Figure 56 response times provided (self composed)

F.2.2 Security

Attempting to access the protected /inspect route without authentication redirects the user to the Login page

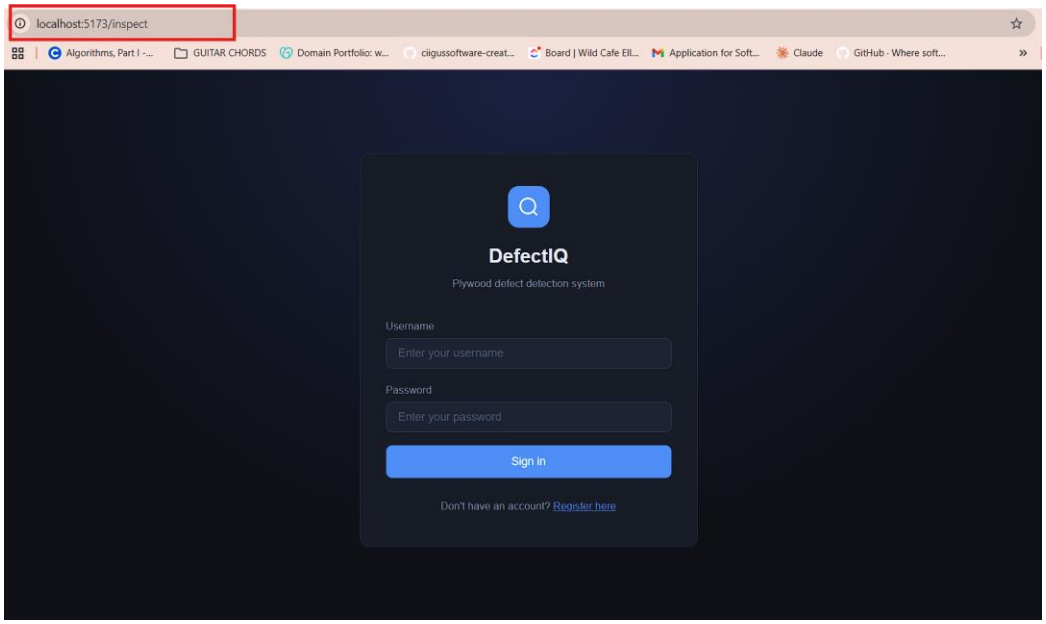


Figure 57 authentication check (self composed)

Route protection was verified by navigating directly to the protected /inspect URL while unauthenticated. The system redirected the browser to the Login page without exposing any inspection functionality, as shown in Appendix F, Figure F.X. This confirms that the React sessionStorage JWT guard correctly prevents unauthenticated access to protected views.

F.2.3 POST request to /detect endpoint without JWT token returns 401 Unauthorized with error message

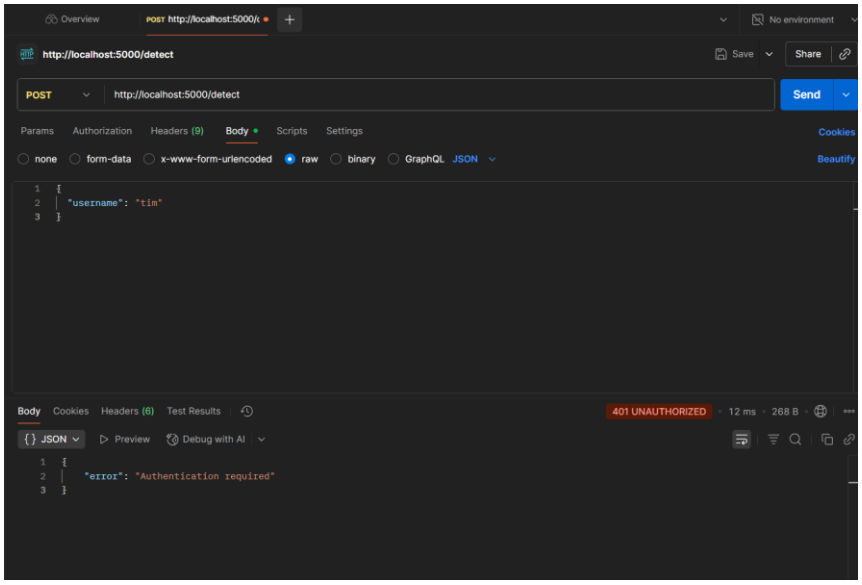


Figure 58 authentication check -post request (self composed)

API-level security was verified using Postman by sending a POST request directly to the /detect endpoint without an Authorization header. The Flask backend returned a 401 Unauthorized response with the message 'Authentication required', confirming that the JWT guard is enforced at the API layer independently of the frontend routing protection.

Appendix G – Evaluation

G.1 Model evaluators

ID	Name	Role	Experience
DE1	Samantha Marapana	Factory General Manager	35+ years in plywood manufacturing, – DCH Plywood manufacturers, Wood lanka (PVT) ltd.
DE2	Jaliya Wellawa	Factory Owner,	15 years in plywood manufacturing, Plywood house , Plymarc Enterprises
DE3	Dulanjana Marapana	Factory Owner,	6 years in plywood manufacturing, DCH plywood manufacturers
DE4	Thilina Samarathunga	Operational Manager	4 years in factory operations, , DCH Plywood manufacturers
DE5	Rohini Arukmulla	Factory Worker,	10+ years in production floor, DCH Plwyood manfacturers

Table 51 domain module evaluators (self composed)

ID	Name	Role	Experience
TE1	Sandaroo Wijewardana	Associate Technical Lead, Cambio Software Engineering	Product engineering, backend, API design
TE2	Sakila Dissanayake	Associate Software Architect, Cambio	System architecture, database design, modularity

		Software Engineering	
TE3	Nandun Perera	Software Engineer (AI experience), Cambio Software Engineering	Application development, ML model integration
TE4	Udeesha Kularatne	Machine Learning Expert, Sustainable Education Foundation, ICT Option.	ML models, dataset design, class imbalance
TE5	Inupa Bandara	Associate AI Engineer (AI student), Brown Company and Ltd.	AI/ML engineering, defect class performance

Table 52 Technical domain evaluators (self composed)

G.2 Questionnaires for evaluators

https://docs.google.com/document/d/1LekblobSd3W6Xb_ZrdIybcDdLk-Tcb3J/edit?usp=drive_link&ouid=111940683325705554197&rtpof=true&sd=true.
https://docs.google.com/document/d/1neTeYgyPkz_L8HkG9MAc3CNfrgeMqGie/edit?usp=drive_link&ouid=111940683325705554197&rtpof=true&sd=true.

G.3 Usability statements

Usability Statement	**Mean Score (/5)**
The interface is clear and easy to navigate	4.6
The PASS/REVIEW verdict is easy to understand	4.8
The defect recommendations are useful and actionable	4.4
The system is suitable for use in a factory environment	4.2

I would recommend this system for factory quality inspection	4.3

Table 53 Usability statements(self composed)

G.4 Self Evaluation table

Theme	Self-Evaluation
Research novelty and gap	DefectIQ fills a clear gap in the research literature, as no previous system was found to integrate post-manufactured plywood inspection, dual-view camera fusion, multi-label classification, and knowledge-based reasoning within a single pipeline. The shift from the single-label, patch-based classification used at the IPD stage to the multi-label, whole-board classification used in the final system is a significant methodological improvement. This multi-label design was adopted not only for technical reasons, but also to better represent the physical behaviour of real plywood defects. For example, delamination and warping can co-occur due to the same underlying manufacturing issue, particularly glue bond failure, meaning a single-label approach would not capture this relationship accurately. Likewise, the dual-view fusion strategy was designed around how defects physically appear on plywood boards, with surface defects more visible in the top view and structural defects more visible in the side view. Taken together, these contributions support the claim that DefectIQ is a novel system with strong practical relevance to the plywood manufacturing domain.
Scope and depth	The DefectIQ system incorporates a four-stage detection pipeline, a 2-model ensemble using weighted late fusion, YOLOv8-seg board detection, Test-Time Augmentation, JWT authentication, Supabase postgresql integration and a React frontend all in one undergraduate project timeline. The project also changed considerably from the IPD prototype, resulting in 14 changes in its architecture, resulting in a 49% relative improvement in Macro F1, and a 15-percentage point improvement in Top 2 Image Accuracy. This iterative development path continued technical effort and engineering judgement as opposed to a one-pass implementation.

System design and architecture	The modular architecture separating board detection, ensemble inference, dual-view fusion, and knowledge-based reasoning into distinct, independently testable components reflects sound software engineering principles. Each stage can be improved independently without breaking the pipeline. The Supabase PostgreSQL backend with JWT authentication provides a production-grade data layer appropriate for a multi-user factory deployment scenario. The React frontend with user-scoped inspection history and date filtering meets the functional requirements defined in Chapter 4. The design adheres to the principle of separation of concerns and demonstrates awareness of scalability considerations beyond the prototype stage.
Result quality	The final ensemble achieves a Macro F1 of 0.6262 and a Top-2 Image Accuracy of 95.12% on a genuinely challenging five-class multi-label problem with inherent class imbalance. Empirical benchmarking against ResNet-18 (F1=0.483), MobileNetV2 (F1=0.478), and VGG16 (F1=0.487) confirms the ensemble outperforms all standard baselines by a margin of 0.13–0.15 Macro F1 points. The delamination class (F1=0.524) remains the weakest, attributable to its limited training representation (115 samples) rather than a fundamental architectural limitation. The YOLOv8-seg board detector achieves an mAP50 of 0.979, confirming reliable Stage 1 performance. These results are considered meaningful given the dataset constraints and represent a solid foundation for future work with an expanded dataset.
Limitations acknowledged	The primary limitations are the dataset size (619 images from a single factory environment), the absence of real-time camera testing, and the partial implementation of FR02 (image validation). The delamination class performance gap is acknowledged as a direct consequence of dataset imbalance. Future work should prioritise dataset expansion to at least 300 samples per class, real-time camera integration (FR08), and Sinhala language support for the user interface (FR09) to improve accessibility for factory workers.

Table 54 self evaluation (self composed)

Appendix I – Functional and nonfunctional requirement evaluation

FR	Requirement	Status	Expert Verdict
FR01	Dual image upload	Implemented	Confirmed. Dual-view is essential for detecting both surface and structural defects simultaneously.
FR02	Image validation file type and size	Partial	TE5: file type validation functional; size/resolution validation absent. Known scope boundary.
FR03	Defect detection with confidence scores	Implemented	Confirmed. TE4 noted confidence display appropriate for technical users.
FR04	Surface and structural defect classification	Implemented	DE1, DE2 confirmed 5 classes cover commercially significant post-manufacturing defects.
FR05	Knowledge-based reasoning	Implemented	DE4: recommendations aligned with standard corrective procedures. High practical value.
FR06	User accounts and inspection history	Implemented	TE1, TE2: JWT auth and user-scoped history assessed as production-appropriate.
FR07	Date filtering on dashboard	Implemented	DE4: useful for tracking defect trends across production shifts.
FR08	Real-time camera integration	Future Work	All domain experts: highest priority for practical deployment.
FR09	Sinhala language support	Future Work	DE1: essential for accessibility in Sri Lankan factory context.

Table 55 functional and evaluation (self composed)

NFR	Requirement	Status	Expert Verdict
NFR01	Acceptable inference response time	Acceptable for static workflow	TE1: adequate for post-manufacturing inspection. Real-time deployment requires GPU optimization.
NFR02	Graceful degradation on edge inputs	Confirmed	TE2: Fallback to full image when board not detected is a sound reliability design.
NFR03	JWT + bcrypt security	Confirmed	TE2: industry-standard implementation suitable for factory intranet deployment.
NFR04	Non-technical user accessibility	Largely confirmed	DE4: navigable without training. DE1: Sinhala language required for full accessibility.
NFR05	Multi-user scalability	Confirmed	TE1: Supabase PostgreSQL supports concurrent multi-user access.

Table 56 non functional requirement evaluation(self composed)

Appendix J – Algorithm diagrams

Appendix j.1 – high level algorithm architecture flowcharts

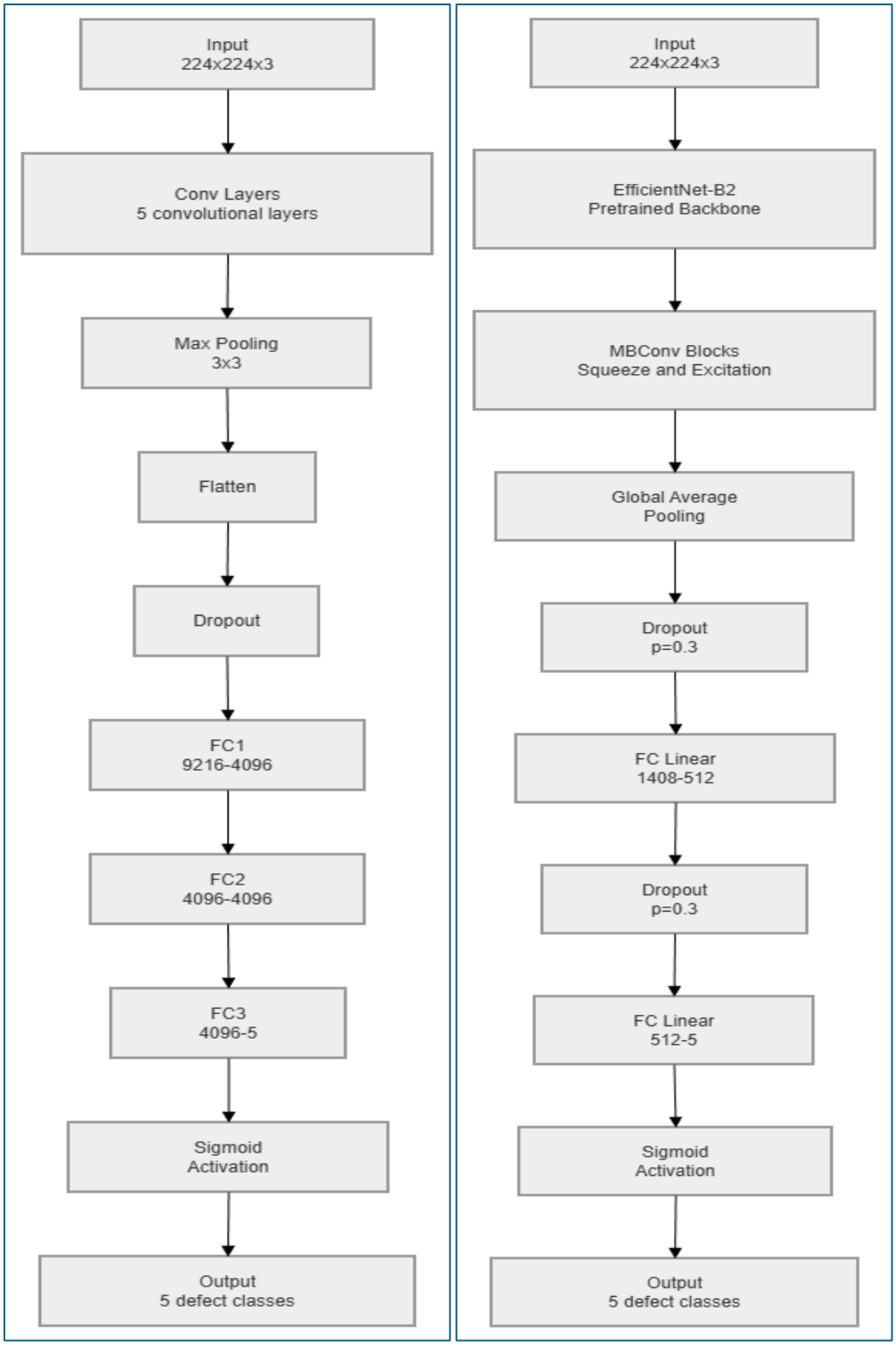


Figure 59 Algorithm architecture alexnet(left), efficientNetB2(right)

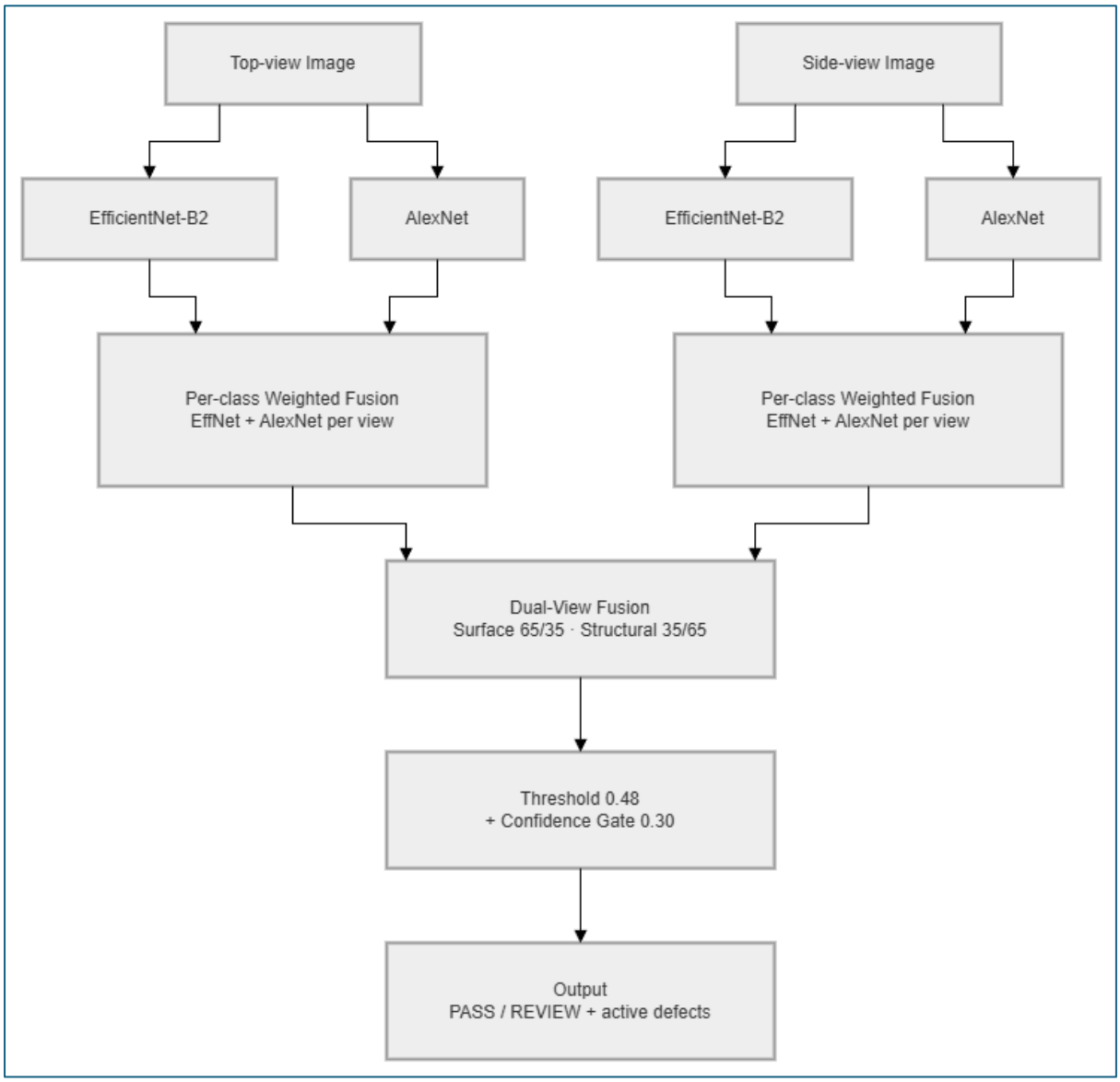


Figure 60 Ensemble algorithm architecture

Appendix j.2 Low fidelity wireframes

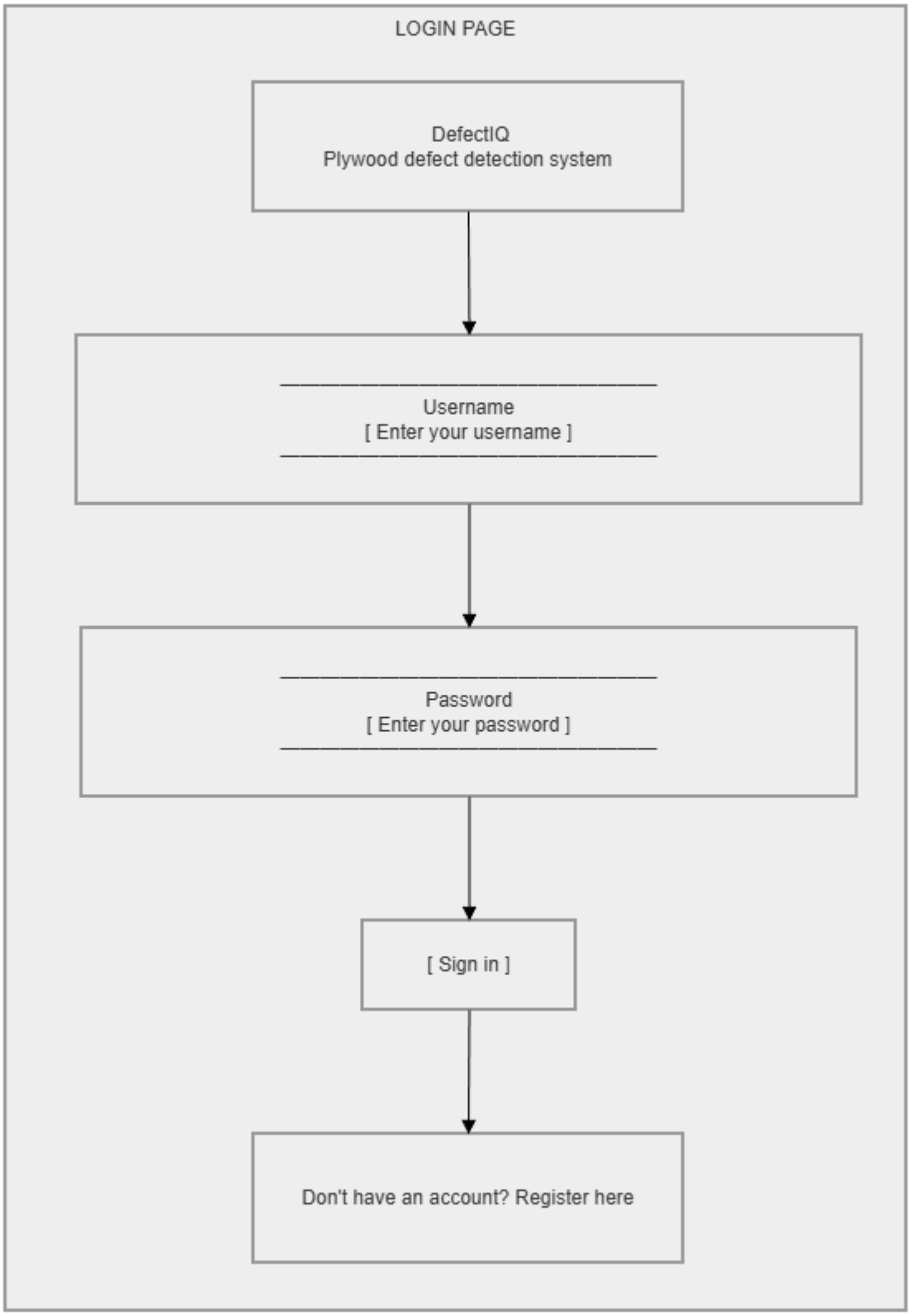


Figure 61 low fidelity wireframes login page (self composed)

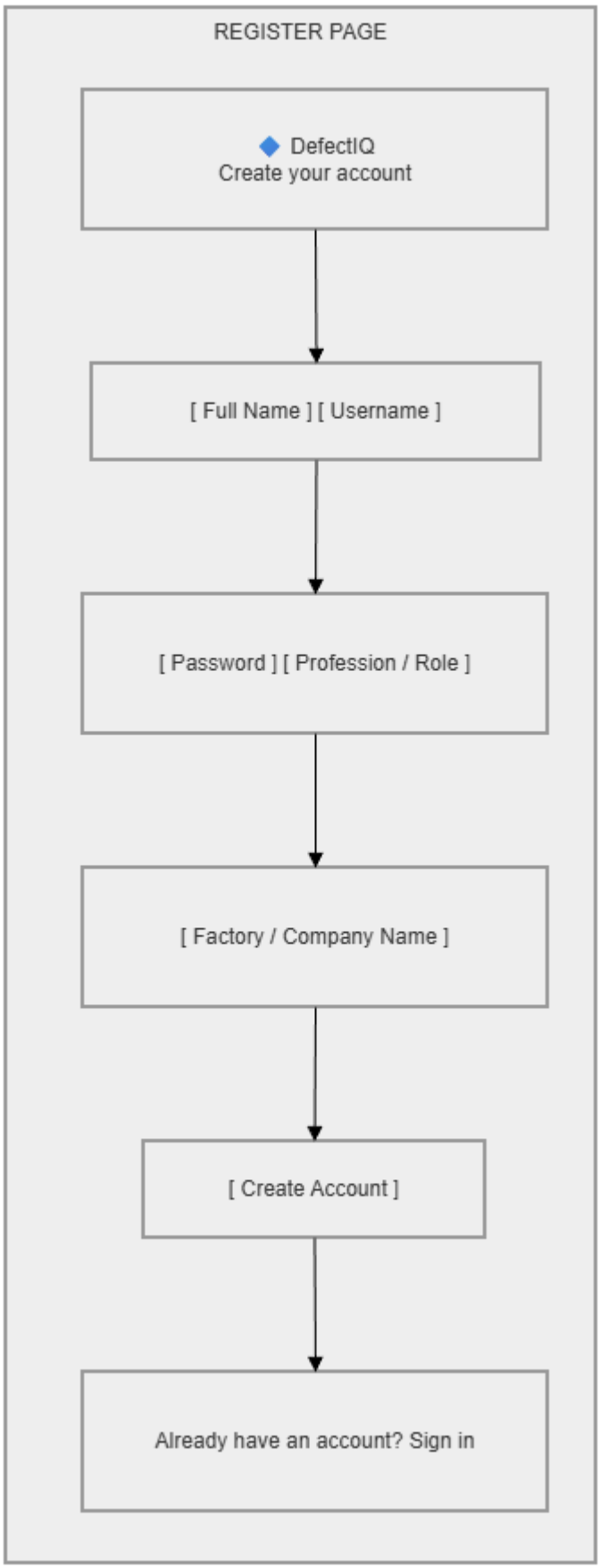


Figure 62 low fidelity wireframes -register page(self composed)

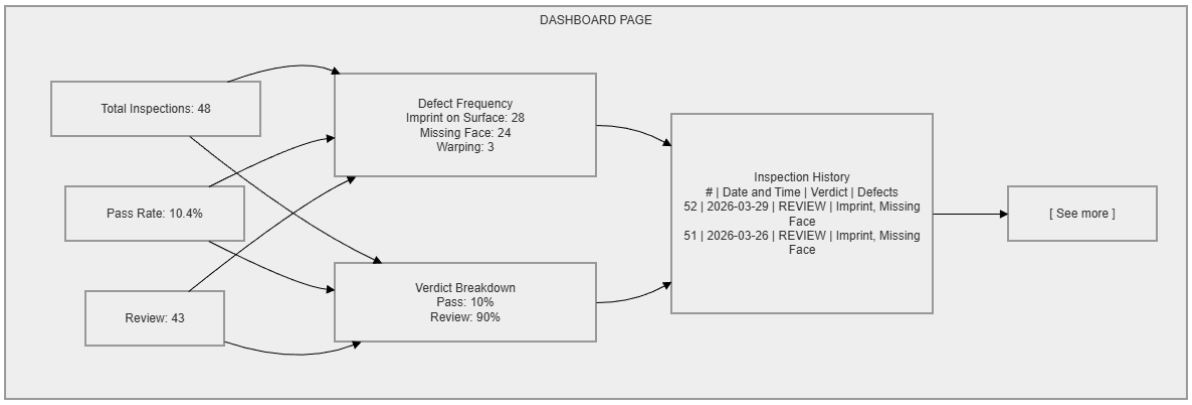


Figure 63 low fidelity wireframes - dashboard page(self composed)

Appendix K – ML pipeline technical detail

K.1 Training hyperparameters

The table documents the full hyperparameters configurations used for both EfficientNet-B2 and AlexNet training.

Parameter	EfficientNet-B2	AlexNet
Image size	224 × 224	224 × 224
Batch size	16	16
Random seed	42	42
Loss function	FocalBCEWithLogitsLoss (γ=2.0)	FocalBCEWithLogitsLoss (γ=2.0)
Label smoothing	0.05	0.05
Optimizer	AdamW (weight decay=0.01)	AdamW (weight decay=0.01)
LR scheduler	CosineAnnealingWarmRestarts	CosineAnnealingWarmRestarts
Dropout (classifier head)	0.4	0.0 (not added)
Gradient clipping	max_norm = 1.0	max_norm = 1.0
Multi-label IoU threshold	0.30	0.30
Inference threshold	0.50	0.50

Table 57 model training hyperparameters (self composed)

K.2 Three-phase fine-tuning schedule

The precise phase setups for both models are shown in Table K.2 including the unfrozen layers, the learning rates used, and whether or not CutMix/MixUp batch augmentation was enabled. Cosine Annealing with Warm Restarts was reset at the start of each phase.

Phase	Epochs	Layers unfrozen	Learning rates	CutMix/MixUp

Phase 1	EffNet: 10 | AlexNet: 8	Backbone frozen - classifier head only	Head LR = 1e-3	OFF
Phase 2	EffNet: 15 | AlexNet: 12	EffNet: last 3 backbone blocks + head | AlexNet: features[8:] + classifier	Backbone LR = 1e-4, Head LR = 5e-4	ON (prob=0.5)
Phase 3	EffNet: 10 | AlexNet: 10	EffNet: last 5 blocks + head (3 LR groups) | AlexNet: all features + classifier (3 LR groups)	EffNet: 1e-5 / 5e-5 / 1e-4 | AlexNet: 1e-5 / 5e-5 / 1e-4	ON (prob=0.5)

Table 58 Threephase finetuning schedule (self composed)

K.3 Training augmentation transforms

The Table lists the full online augmentation pipeline per crop before batching. CutMix and MixUp are applied after batch collation by DataLoader and therefore listed at the bottom.

Transform	**Parameter**	**Justification**
RandomResizedCrop	scale=(0.75, 1.0)	Plywood defects appear at varying scales depending on camera distance
RandomHorizontalFlip	p=0.5	Boards are laterally symmetric- flip is a valid augmentation
RandomVerticalFlip	p=0.3	Boards can be captured in either orientation
RandomRotation	±15°	Simulates minor camera angle variation during capture
ColorJitter	brightness=0.3, contrast=0.3,	Factory lighting varies significantly across shifts and locations

	saturation=0.2, hue=0.05	
RandomGrayscale	p=0.05	Encourages texture-based rather than colour-based discrimination
GaussianBlur	kernel=3, σ=(0.1, 1.0)	Simulates camera focus variation and motion blur in factory conditions
RandomErasing	p=0.2, scale=(0.02, 0.15)	Coarse dropout - simulates partial board occlusion and shadow artefacts
CutMix / MixUp	α=1.0/0.2, prob=0.5, Phase 2–3 only	Batch-level regularisation preventing memorisation on 619-image dataset

Table 59 Training augmentation transforms (self composed)

K.4 Test-time augmentation views

Five TTA views are used at inference, as in the Table. The output of sigmoids for the five views is averaged element-wise for ensemble.

View	Transform	Justification
1	Resize(224) + Normalise	Baseline - standard evaluation transform
2	HorizontalFlip(p=1.0) + Normalise	Boards are laterally symmetric; flip is a valid inference augmentation
3	VerticalFlip(p=1.0) + Normalise	Boards can be placed in either vertical orientation
4	ColorJitter(brightness=0.15) + Normalise	Simulates factory lighting variation at inference time
5	RandomResizedCrop(224, scale=(0.9, 1.0)) + Normalise	Slight scale variation simulates camera distance differences

Table 60 Test time augmentation views (self composed)

K.5 Per-class ensemble fusion weights

The final weights for ensemble fusion for each class are reported in Table K.5, as found on the validation set. The sum of the EfficientNet-B2 and AlexNet weights for each class is 1.0.

Defect Class	**EfficientNet-B2 weight**	**AlexNet weight**	**Rationale**	**Category**
bubbling	0.55	0.45	Texture-dominant; slight EffNet advantage	Surface
delamination	0.60	0.40	Fine-grained texture defect; EffNet SE blocks effective	Structural
imprint_on_surface	0.60	0.40	Embossed pattern detection; EffNet texture sensitivity	Surface
missing_face	0.50	0.50	Equal weighting - validation F1 maximised at parity	Surface
warping	0.40	0.60	Coarse spatial deformation; AlexNet large kernels effective	Structural

Table 61 per class ensemble fusion(self composed)

Appendix L – Methodology

L.1 Gantt chart

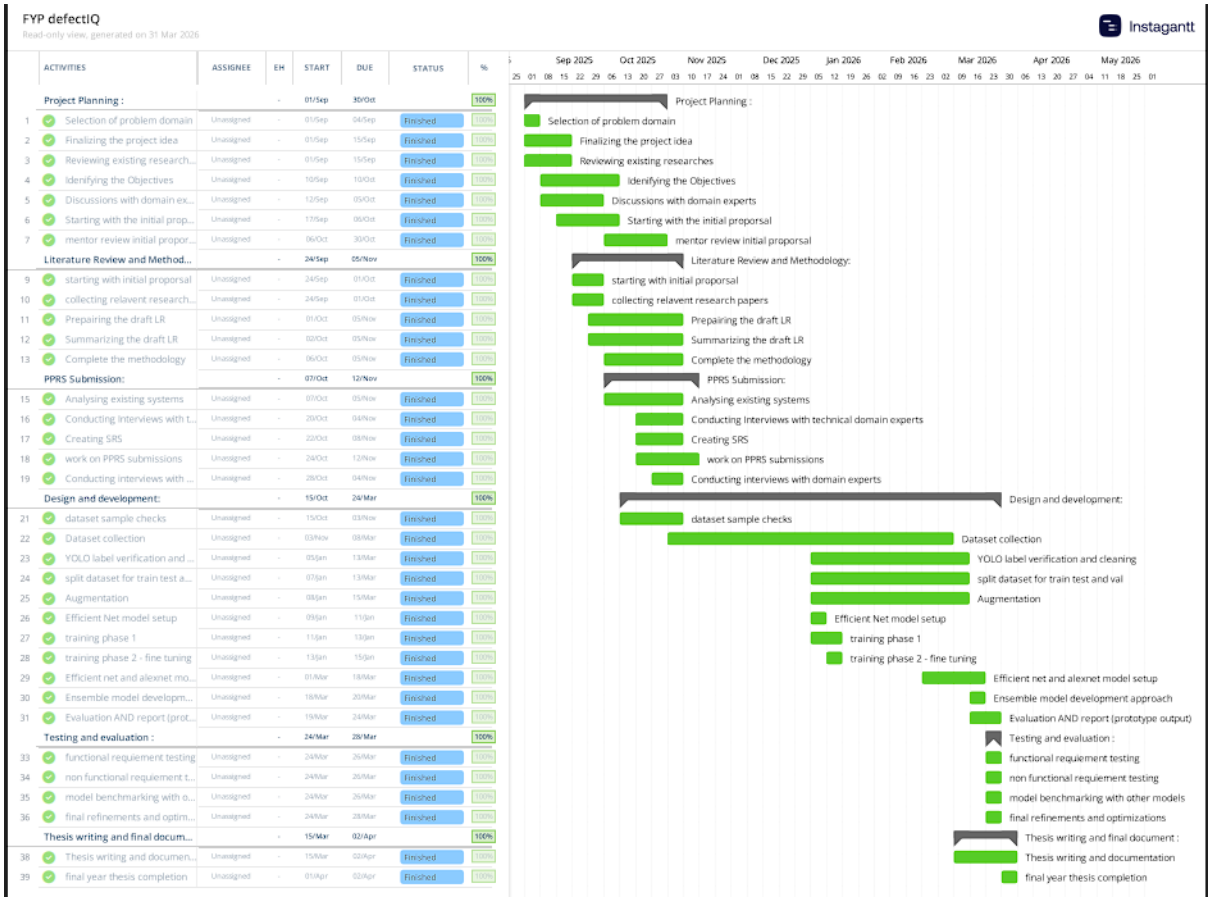


Figure 64 Gantt chart -final (self composed)

L.2 Project management sprint table

Sprint	Phase
Sprint 1	Dataset and Board Detection
Sprint 2	Ensemble Model Training
Sprint 3	Backend and Fusion
Sprint 4	Frontend and Evaluation

Table 62 project managament sprint table (self composed)